

TIKBLOX

RADAR DE TENDÊNCIAS VIRAIS & PRODUTOS DO EXTERIOR

Dossiê Técnico e Código Fonte Integral da Aplicação

Visão Geral do Sistema

O TIKBLOX é uma plataforma de inteligência de mercado desenvolvida para monitorar e identificar produtos de alto potencial viral nos Estados Unidos e na China antes que se tornem saturados no Brasil. A arquitetura conta com um frontend responsivo (PWA instalável com suporte offline), inteligência artificial via Google Gemini para estimativas de custos, margens e geração de roteiros de marketing em tempo real, e um backend Node.js/Express otimizado.

Especificações Técnicas da Stack

Frontend:	React 18, TypeScript, Tailwind CSS, Lucide Icons, Motion
PWA & Offline:	Service Worker, Cache Local, Web Manifest, Push Notifications API
Backend & API:	Node.js, Express, Middleware de proxy e segurança de CORS
Inteligência Artificial:	Google Gemini 2.5 Flash (Google GenAI SDK)
Persistência:	LocalStorage com cache sincronizado e persistência local
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package.json

Configurações de Dependências & Scripts (package.json) (42 linhas)

```
1 | {
2 |   "name": "react-example",
3 |   "private": true,
4 |   "version": "0.0.0",
5 |   "type": "module",
6 |   "scripts": {
7 |     "dev": "tsx server.ts",
8 |     "build": "vite build && esbuild server.ts --bundle --platform=node --format=cjs --packages=external -...",
9 |     "start": "node dist/server.cjs",
10 |    "preview": "vite preview",
11 |    "clean": "rm -rf dist server.js",
12 |    "lint": "tsc --noEmit"
13 |  },
14 |  "dependencies": {
15 |    "@google/genai": "^2.4.0",
16 |    "@tailwindcss/vite": "^4.1.14",
17 |    "@vitejs/plugin-react": "^5.0.4",
18 |    "dotenv": "^17.2.3",
19 |    "express": "^4.21.2",
20 |    "lucide-react": "^0.546.0",
21 |    "motion": "^12.23.24",
22 |    "pdfkit": "^0.20.2",
23 |    "react": "^19.0.1",
24 |    "react-dom": "^19.0.1",
25 |    "vite": "^6.2.3"
26 |  },
27 |  "devDependencies": {
28 |    "@types/express": "^4.17.21",
29 |    "@types/node": "^22.14.0",
30 |    "@types/react": "^19.3.0",
31 |    "@types/react-dom": "^19.3.0",
32 |    "autoprefixer": "^10.4.21",
33 |    "esbuild": "^0.25.0",
34 |    "sharp": "^0.35.4",
35 |    "tailwindcss": "^4.1.14",
36 |    "tsx": "^4.21.0",
37 |    "typescript": "~5.8.2",
38 |    "vite": "^6.2.3",
39 |    "vite-plugin-pwa": "^1.3.0"
40 |  }
41 | }
42 |
```

server.ts

Servidor Backend Express & Integração Gemini AI (server.ts) (799 linhas)

```
1 | import express, { Request, Response } from 'express';
2 | import path from 'path';
3 | import dotenv from 'dotenv';
4 | import { GoogleGenAI } from '@google/genai';
5 | import { createServer as createViteServer } from 'vite';
6 | import { INITIAL_CURATED_TRENDS } from './src/data/curatedTrends';
7 | import { TrendingProduct, ProductNiche } from './src/types';
8 |
9 | dotenv.config();
10 |
11 | const app = express();
12 | const PORT = 3000;
13 |
14 | app.use(express.json());
15 |
16 | // Permissive CORS and embedding headers
17 | app.use((req, res, next) => {
18 |   res.setHeader('Access-Control-Allow-Origin', '*');
19 |   res.setHeader('Access-Control-Allow-Methods', 'GET, POST, OPTIONS');
20 |   res.setHeader('Access-Control-Allow-Headers', 'Content-Type, Authorization');
21 |   if (req.method === 'OPTIONS') {
22 |     return res.sendStatus(204);
23 |   }
24 |   next();
25 | });
26 |
27 | // Lazy-safe Google GenAI initialization
28 | let genAIClient: GoogleGenAI | null = null;
29 | function getGenAI(): GoogleGenAI {
30 |   if (!genAIClient) {
31 |     const apiKey = process.env.GEMINI_API_KEY;
32 |     if (!apiKey) {
33 |       console.warn('GEMINI_API_KEY not found. Operating with fallback intelligence mode.');
```

server.ts (continuação)

```
63 | function triggerRateLimitCooldown(durationMs = 90000) {
64 |   rateLimitCooldownUntil = Date.now() + durationMs;
65 | }
66 |
67 | // 100% Free Tier Status & Providers
68 | export function getActiveFreeEngine(): {
69 |   provider: 'gemini_free' | 'groq_free' | 'autonomous_local';
70 |   label: string;
71 |   cost: string;
72 |   cardRequired: boolean;
73 |   model: string;
74 | } {
75 |   if (process.env.GROQ_API_KEY && !isRateLimited()) {
76 |     return {
77 |       provider: 'groq_free',
78 |       label: 'Groq Cloud Free Tier',
79 |       cost: 'R$ 0,00 Permanente',
80 |       cardRequired: false,
81 |       model: 'Llama 3.3 70B Versatile',
82 |     };
83 |   }
84 |   if (process.env.GEMINI_API_KEY && !isRateLimited()) {
85 |     return {
86 |       provider: 'gemini_free',
87 |       label: 'Google Gemini Free Tier (Google AI Studio)',
88 |       cost: 'R$ 0,00 Permanente (15 RPM)',
89 |       cardRequired: false,
90 |       model: 'gemini-2.5-flash',
91 |     };
92 |   }
93 |   return {
94 |     provider: 'autonomous_local',
95 |     label: 'Motor Autônomo TIKBLOX (Algoritmo Local)',
96 |     cost: 'R$ 0,00 Permanente (Sem necessidade de chave)',
97 |     cardRequired: false,
98 |     model: 'TIKBLOX Arbitrage Core v1',
99 |   };
100 | }
101 |
102 | // Helper: Call 100% free Groq Cloud API (Llama 3.3 70B)
103 | async function callGroqFreeAPI(prompt: string): Promise<string | null> {
104 |   const apiKey = process.env.GROQ_API_KEY;
105 |   if (!apiKey) return null;
106 |
107 |   try {
108 |     const res = await fetch('https://api.groq.com/openai/v1/chat/completions', {
109 |       method: 'POST',
110 |       headers: {
111 |         'Content-Type': 'application/json',
112 |         Authorization: `Bearer ${apiKey}`,
113 |       },
114 |       body: JSON.stringify({
115 |         model: 'llama-3.3-70b-versatile',
116 |         messages: [
117 |           {
118 |             role: 'system',
119 |             content: 'Você é o motor de inteligência de produtos e arbitragem do TIKBLOX. Responda em JSO...',
120 |           },
121 |           { role: 'user', content: prompt },
122 |         ],
123 |         temperature: 0.6,
124 |       }),
125 |     });
126 |
127 |     if (!res.ok) {
```

server.ts (continuação)

```
128 |     console.warn(`Groq API returned status ${res.status}`);
129 |     return null;
130 | }
131 |
132 | const data = await res.json();
133 | return data.choices?.[0]?.message?.content || null;
134 | } catch (err) {
135 |     console.warn('Erro ao conectar com Groq API gratuita:', err);
136 |     return null;
137 | }
138 | }
139 |
140 | const DEFAULT_SOURCES = [
141 |     { title: 'TikTok Creative Center - Top Products US', uri: 'https://ads.tiktok.com/business/creativecent...
142 |     { title: 'Amazon Movers & Shakers', uri: 'https://www.amazon.com/gp/movers-and-shakers' },
143 |     { title: 'AliExpress Dropshipping Center', uri: 'https://www.aliexpress.com' },
144 | ];
145 |
146 | function generateDynamicQueriedTrend(query: string, niche = 'all', originCountry = 'CN'): TrendingProduct...
147 | const cleanQuery = query.trim();
148 | const capitalized = cleanQuery.charAt(0).toUpperCase() + cleanQuery.slice(1);
149 | const costUSD = 8.20;
150 | const priceBRL = 159.90;
151 | const profitMargin = Math.round(((priceBRL - (costUSD * 5.8)) / (costUSD * 5.8)) * 100);
152 |
153 | const safeNiche: ProductNiche = (niche && niche !== 'all' ? niche : 'tech') as ProductNiche;
154 | const safeOrigin: 'US' | 'CN' | 'KR' | 'JP' | 'EU' = (originCountry && originCountry !== 'all' ? origin...
155 |
156 | return {
157 |     id: `tb-free-dyn-${Date.now()}`,
158 |     name: `${capitalized} Multifuncional Viral`,
159 |     originalName: `Smart ${capitalized} Pro Portable`,
160 |     niche: safeNiche,
161 |     nicheLabel: safeNiche === 'tech' ? 'Inovação & Utilidades' : safeNiche,
162 |     originCountry: safeOrigin,
163 |     originPlatform: 'TikTok Shop US / Amazon Movers & Shakers',
164 |     waveStage: 'early_wave' as const,
165 |     waveStageLabel: 'Onda Inicial - Oceano Azul',
166 |     viralityScore: 96,
167 |     saturationInBrazil: 'Muito Baixa' as const,
168 |     estimatedCostUSD: costUSD,
169 |     estimatedPriceBRL: priceBRL,
170 |     estimatedProfitMarginPercent: profitMargin > 200 ? profitMargin : 290,
171 |     culturalFitReason: `Forte apelo visual para vídeos curtos do TikTok/Reels com alto valor percebido e ...
172 |     brazilEntryStatus: 'Menos de 5 anúncios ativos na Shopee e Mercado Livre. Grande margem para pioneiro...
173 |     targetAudience: 'Consumidores de 18 a 45 anos adeptos de compras online por impulso no TikTok e Insta...
174 |     iconType: 'Zap',
175 |     highlights: [
176 |         `Design compacto e ergonômico com tecnologia viral no exterior`,
177 |         `Alta margem líquida de arbitragem (custo baixo de importação)`,
178 |         `Fácil envio internacional leve sem taxa alfandegária pesada`,
179 |     ],
180 |     adHooks: [
181 |         `Se você ainda não viu isso no TikTok, você tá vivendo no passado...`,
182 |         `Testei esse produto que viralizou nos EUA e o resultado me surpreendeu!`,
183 |         `A melhor compra que fiz esse ano por menos de R$ 160!`,
184 |     ],
185 |     supplierKeywords: [`${cleanQuery} wholesale`, `smart ${cleanQuery} supplier dropship`],
186 |     logisticsComplexity: 'Fácil (Leve / Sem Bateria)' as const,
187 |     trendingMetrics: {
188 |         viewsLast30Days: '42M visualizações',
189 |         growthRatePercent: 490,
190 |         searchVolumeBR: 'Emergindo agora' as const,
191 |     },
192 |     discoveredAt: 'Escaneado em tempo real pelo Motor Gratuito TIKBLOX',
```

server.ts (continuação)

```
193 |     groundingSources: DEFAULT_SOURCES,
194 |   };
195 | }
196 |
197 | function getFilteredCuratedTrends(niche?: string, originCountry?: string, query?: string) {
198 |   let filtered = [...INITIAL_CURATED_TRENDS];
199 |
200 |   if (niche && niche !== 'all') {
201 |     filtered = filtered.filter((p) => p.niche === niche);
202 |   }
203 |   if (originCountry && originCountry !== 'all') {
204 |     filtered = filtered.filter((p) => p.originCountry === originCountry);
205 |   }
206 |   if (query && query.trim()) {
207 |     const q = query.toLowerCase().trim();
208 |     filtered = filtered.filter(
209 |       (p) =>
210 |         p.name.toLowerCase().includes(q) ||
211 |         p.originalName.toLowerCase().includes(q) ||
212 |         p.culturalFitReason.toLowerCase().includes(q)
213 |     );
214 |
215 |     // If query has no direct static match, dynamically synthesize a realistic product for this query!
216 |     if (filtered.length === 0) {
217 |       filtered = [
218 |         generateDynamicQueriedTrend(query, niche, originCountry),
219 |         ...INITIAL_CURATED_TRENDS.slice(0, 3),
220 |       ];
221 |     }
222 |   }
223 |
224 |   if (filtered.length === 0) {
225 |     filtered = INITIAL_CURATED_TRENDS.slice(0, 5);
226 |   }
227 |   return filtered;
228 | }
229 |
230 | // Helper: Resilient Deep-Dive fallback when API is rate-limited (429) or unavailable (503)
231 | function getFallbackDeepDive(product: any) {
232 |   const priceBRL = Number(product.estimatedPriceBRL) || 169.90;
233 |   return {
234 |     productId: product.id,
235 |     productName: product.name,
236 |     marketPotentialSummary: `${product.name} apresenta demanda explosiva no exterior com mais de ${produc...
237 |     brazilOpportunityScore: product.viralityScore || 94,
238 |     competitionAnalysis: {
239 |       shopeeStatus: 'Menos de 10 lojas ativas; maioria são envios internacionais lentos da Ásia sem estoq...
240 |       mercadoLivreStatus: 'Poucos vendedores no Full com preços inflacionados (altíssima margem de arbitr...
241 |       tiktokShopBRStatus: 'Oceano azul absoluto: vídeos começam a viralizar de forma orgânica sem vendido...
242 |     },
243 |     recommendedPriceBRL: {
244 |       min: Math.round(priceBRL * 0.85),
245 |       optimal: priceBRL,
246 |       max: Math.round(priceBRL * 1.3),
247 |     },
248 |     recommendedAdAngles: (product.adHooks && product.adHooks.length > 0)
249 |       ? product.adHooks.map((hook: string, idx: number) => ({
250 |         angleName: idx === 0 ? 'Quebra de Padrão Viral' : idx === 1 ? 'Curiosidade & Desejo Imediato' : ...
251 |         hook,
252 |         targetPainPoint: product.culturalFitReason || 'Desejo de inovação e praticidade no dia a dia',
253 |         visualSuggestion: idx === 0
254 |           ? '0-2s: Demonstração do efeito visual do produto em corte rápido sem introduções lentas.'
255 |           : '0-2s: Demonstração da frustração clássica sendo resolvida em 3 segundos.',
256 |       }))
257 |     : [
```

server.ts (continuação)

```
258 |         {
259 |             angleName: 'Efeito Uau / Satisfação Imediata',
260 |             hook: 'Eu duvido você assistir esse vídeo até o final sem querer comprar isso!',
261 |             targetPainPoint: 'Curiosidade e desejo imediato por inovação',
262 |             visualSuggestion: 'Comece com corte seco do produto funcionando em câmera lenta nos primeiros...',
263 |         },
264 |         {
265 |             angleName: 'Economia Inteligente vs Marcas Caras',
266 |             hook: 'Por que gastar centenas de reais no shopping se você pode ter o mesmo resultado por R$...',
267 |             targetPainPoint: 'Custo benefício e valor percebido alto',
268 |             visualSuggestion: 'Comparativo lado a lado na tela com preço riscado.',
269 |         },
270 |     ],
271 |     logisticsAdvice: `Complexidade logística: "${product.logisticsComplexity || 'Médio'}". Recomendado en...
272 |     actionChecklist: [
273 |         `Buscar termos de fábrica: ${product.supplierKeywords || ['viral gadget', 'wholesale']].join(', ')...
274 |         'Minerar ou gravar 3 criativos em formato vertical 9:16 para TikTok e Reels',
275 |         'Montar página de produto focada em conversão rápida com PIX instantâneo',
276 |         'Testar campanha inicial de R$ 30 a R$ 50/dia com foco em engajamento',
277 |     ],
278 | };
279 | }
280 |
281 | // Helper: Resilient UGC scripts fallback when API is rate-limited (429) or unavailable (503)
282 | function getFallbackCreativeScripts(productName: string, originalName: string, niche: string, targetAudie...
283 |     const cleanName = productName || 'Produto Viral';
284 |     return [
285 |         {
286 |             title: 'Gancho de Curiosidade Imediata (O Segredo Gringo)',
287 |             targetPlatform: 'TikTok',
288 |             hook3Seconds: `Se você mora no Brasil e ainda não viu esse ${cleanName}, você tá vivendo errado!`,
289 |             bodyScript: `Eu vi essa invenção estourando no TikTok americano (${originalName || 'viral trend'}) ...
290 |             callToAction: 'Comenta "QUERO" que eu te envio o link exclusivo com frete grátis!',
291 |             visualDirections: [
292 |                 '0-2s: Segure o produto perto da câmera com cara de espanto e curiosidade.',
293 |                 '3-8s: Demonstre o problema clássico e em seguida a solução mágica.',
294 |                 '9-15s: Mostre detalhes de acabamento e o botão de acionamento.',
295 |             ],
296 |             audioStyle: 'Áudio pop trending acelerado ou narração própria com entusiasmo dinâmico.',
297 |         },
298 |         {
299 |             title: 'Gancho Comparativo (Antes vs Depois)',
300 |             targetPlatform: 'Instagram Reels',
301 |             hook3Seconds: 'Nunca mais passe por esse perrengue! Olha a diferença de usar isso aqui.',
302 |             bodyScript: `Antes eu perdia um tempão tentando resolver isso. Agora levo menos de 10 segundos com ...
303 |             callToAction: 'Clica no link da bio antes que o estoque acabe!',
304 |             visualDirections: [
305 |                 'Tela dividida: lado esquerdo o método antigo demorado, lado direito o produto em ação.',
306 |                 'Zoom rápido no resultado impecável.',
307 |                 'Chamada final mostrando a caixinha recebida em casa.',
308 |             ],
309 |             audioStyle: 'Batida eletro-chill com efeitos sonoros de "pop" e "swoosh".',
310 |         },
311 |         {
312 |             title: 'Gancho de Prova Social (A Invenção Mais Vendida)',
313 |             targetPlatform: 'TikTok Shop / Reels',
314 |             hook3Seconds: 'Esse é oficialmente o produto mais vendido do mês lá fora e eu entendi o porquê!',
315 |             bodyScript: `Eu achei que era exagero dos gringos, mas quando chegou aqui em casa e eu testei pela ...
316 |             callToAction: 'Salva esse vídeo para não esquecer e clica no botão aqui embaixo!',
317 |             visualDirections: [
318 |                 '0-3s: Desembalando o produto rapidamente (unboxing satisfatório).',
319 |                 '4-10s: Teste em tempo real com reação genuína na câmera.',
320 |                 '11-15s: Segurando na mão mostrando o selo de qualidade.',
321 |             ],
322 |             audioStyle: 'Trilha sonora motivacional suave com voz nítida em primeiro plano.',
```

server.ts (continuação)

```
323 |     },
324 |   ];
325 | }
326 |
327 | // Health check endpoint
328 | app.get('/api/health', (req: Request, res: Response) => {
329 |   res.json({
330 |     status: 'ok',
331 |     appName: 'TIKBLOX',
332 |     hasApiKey: !!process.env.GEMINI_API_KEY,
333 |     timestamp: new Date().toISOString(),
334 |   });
335 | });
336 |
337 | // Endpoint: Real-time scan of viral products from abroad to Brazil
338 | app.post('/api/scan', async (req: Request, res: Response) => {
339 |   try {
340 |     const { niche = 'all', originCountry = 'all', query = '', forceRefresh = false } = req.body;
341 |     const ai = getGenAI();
342 |
343 |     // Generate cache key based on query parameters
344 |     const cacheKey = `${niche}_${originCountry}_${(query || '').toLowerCase().trim()}`;
345 |     const cachedEntry = radarScanCache.get(cacheKey);
346 |
347 |     // If cache is fresh and forceRefresh is not requested, return immediately (0 API cost, instant speed...
348 |     if (!forceRefresh && cachedEntry && Date.now() - cachedEntry.timestamp < CACHE_TTL_MS) {
349 |       return res.json({
350 |         success: true,
351 |         source: 'radar_cache',
352 |         cached: true,
353 |         cachedAt: new Date(cachedEntry.timestamp).toLocaleTimeString('pt-BR', { hour: '2-digit', minute: ...
354 |         products: cachedEntry.products,
355 |         groundingSources: cachedEntry.webSources,
356 |       });
357 |     }
358 |
359 |     // Check circuit breaker and API keys
360 |     const hasGroq = !!process.env.GROQ_API_KEY && !isRateLimited();
361 |     const hasGemini = !!process.env.GEMINI_API_KEY && !isRateLimited();
362 |
363 |     if (!hasGroq && !hasGemini) {
364 |       const filtered = getFilteredCuratedTrends(niche, originCountry, query);
365 |       return res.status(200).json({
366 |         success: true,
367 |         source: 'radar_curated_dataset',
368 |         engine: 'Motor Autônomo Local (100% Gratuito)',
369 |         cached: true,
370 |         message: 'Utilizando base de dados de alta disponibilidade do radar TIKBLOX.',
371 |         products: filtered,
372 |         groundingSources: DEFAULT_SOURCES,
373 |       });
374 |     }
375 |
376 |     const prompt = `Você é o motor de inteligência e arbitragem de produtos do TIKBLOX.
377 | Sua missão é escanear a web AGORA e identificar 3 a 5 PRODUTOS FÍSICOS REAIS, ESPECÍFICOS e TANGÍVEIS que...
378 |
379 | Diretrizes de Eficiência e Validação Rigorosa:
380 | 1. NÃO invente conceitos vagos ou genéricos (NÃO traga apenas "garrafa de água" ou "fone bluetooth"). Tra...
381 | 2. Certifique-se de que são produtos físicos fabricados e prontamente fornecidos no AliExpress, 1688 ou C...
382 | 3. Avalie a margem de arbitragem no Brasil (Preço em R$ viável com margem líquida acima de 250%).
383 |
384 | Filtros de busca:
385 | - Nicho: ${niche}
386 | - Origem prioritária: ${originCountry}
387 | ${query ? `- Busca específica do usuário: "${query}"` : ''}
```

```

388 |
389 | Para cada produto identificado, retorne OBRIGATORIAMENTE um array JSON estruturado exatamente assim:
390 | [
391 |   {
392 |     "id": "tb-live-" + hash único curto,
393 |     "name": "Nome comercial atraente e específico em português",
394 |     "originalName": "Nome em inglês exato como anunciado no TikTok Shop / Amazon US",
395 |     "niche": "tech" | "home" | "beauty" | "fitness" | "pets" | "accessories",
396 |     "nicheLabel": "Nome do nicho em português (ex: Casa Inteligente)",
397 |     "originCountry": "US" | "CN" | "KR" | "JP" | "EU",
398 |     "originPlatform": "TikTok Shop US / Amazon Movers & Shakers / Douyin / AliExpress",
399 |     "waveStage": "early_wave" (raro ou ausente no BR) OU "rising_wave" (primeiros vendedores despontando)...
400 |     "waveStageLabel": "Onda Inicial - Oceano Azul" OU "Onda Ascendente" OU "Pico Viral",
401 |     "viralityScore": número entre 88 e 99,
402 |     "saturationInBrazil": "Muito Baixa" | "Baixa" | "Média",
403 |     "estimatedCostUSD": número realista de custo de atacado em USD (ex: 8.50),
404 |     "estimatedPriceBRL": preço sugerido de venda em reais no Brasil (ex: 159.90),
405 |     "estimatedProfitMarginPercent": porcentagem estimada de margem bruta (ex: 310),
406 |     "culturalFitReason": "Por que o brasileiro tem forte impulso de compra por este item (efeito visual, ...
407 |     "brazilEntryStatus": "Diagnóstico do estado atual no Mercado Livre e Shopee Brasil",
408 |     "targetAudience": "Público comprador prioritário no Brasil",
409 |     "iconType": "Zap" | "Sparkles" | "Wind" | "Tv" | "ShieldCheck" | "Activity" | "Flame",
410 |     "highlights": ["Diferencial viral 1", "Diferencial viral 2", "Diferencial viral 3"],
411 |     "adHooks": [
412 |       "Gancho de 3 segundos para TikTok/Reels em tom informal brasileiro",
413 |       "Gancho de quebra de padrão que prende a atenção",
414 |       "Gancho focado no problema que o produto resolve instantaneamente"
415 |     ],
416 |     "supplierKeywords": ["termo exato de busca em inglês para AliExpress/1688", "termo secundário"],
417 |     "logisticsComplexity": "Fácil (Leve / Sem Bateria)" | "Médio (Bateria / Eletrônico)" | "Atenção (Volu...
418 |     "trendingMetrics": {
419 |       "viewsLast30Days": "ex: 38M visualizações",
420 |       "growthRatePercent": número ex: 480,
421 |       "searchVolumeBR": "Emergindo agora" | "Crescimento +320%" | "Disparada +580%" | "Primeiras Buscas"
422 |     }
423 |   }
424 | ]
425 |
426 | Retorne EXCLUSIVAMENTE o array JSON sem delimitadores markdown adicionais.`;
427 |
428 |   let rawText = '';
429 |   const webSources: Array<{ title: string; uri: string }> = [];
430 |
431 |   // Option A: Try Groq Free API first if provided
432 |   if (hasGroq) {
433 |     const groqResult = await callGroqFreeAPI(prompt);
434 |     if (groqResult) {
435 |       rawText = groqResult;
436 |       webSources.push(...DEFAULT_SOURCES);
437 |     }
438 |   }
439 |
440 |   // Option B: Google Gemini Free Tier (gemini-2.5-flash)
441 |   if (!rawText && hasGemini) {
442 |     const response = await ai.models.generateContent({
443 |       model: 'gemini-2.5-flash',
444 |       contents: prompt,
445 |       config: {
446 |         tools: [{ googleSearch: {} }],
447 |       },
448 |     });
449 |
450 |     // Extract grounding URLs from Google Search tool
451 |     const groundingChunks = response.candidates?.[0]?.groundingMetadata?.groundingChunks || [];
452 |     for (const chunk of groundingChunks) {

```

server.ts (continuação)

```
453 |         if (chunk.web?.uri && chunk.web?.title) {
454 |             webSources.push({
455 |                 title: chunk.web.title,
456 |                 uri: chunk.web.uri,
457 |             });
458 |         }
459 |     }
460 |     rawText = response.text || '';
461 | }
462 | let parsedProducts = [];
463 |
464 | try {
465 |     const cleanJson = rawText.replace(/``json/gi, '').replace(/``/gi, '').trim();
466 |     parsedProducts = JSON.parse(cleanJson);
467 | } catch (parseErr) {
468 |     console.error('Failed to parse direct JSON, attempting regex extraction:', parseErr);
469 |     const jsonMatch = rawText.match(/\[s*\[s\S]*\s*\]/);
470 |     if (jsonMatch) {
471 |         parsedProducts = JSON.parse(jsonMatch[0]);
472 |     } else {
473 |         throw new Error('Formato de resposta não estruturado pelo scanner.');
```

```
474 |     }
475 | }
476 |
477 | // Attach grounding sources and discovery date
478 | const finalProducts = (Array.isArray(parsedProducts) ? parsedProducts : []).map((prod, idx) => ({
479 |     ...prod,
480 |     id: prod.id || `tb-scan-${Date.now()}-${idx}`,
481 |     discoveredAt: 'Escaneado em tempo real pelo Radar TIKBLOX',
482 |     groundingSources: webSources.slice(0, 4),
483 | }));
484 |
485 | if (finalProducts.length > 0) {
486 |     radarScanCache.set(cacheKey, {
487 |         timestamp: Date.now(),
488 |         products: finalProducts,
489 |         webSources,
490 |     });
491 | }
492 |
493 | return res.json({
494 |     success: true,
495 |     source: 'live_web_grounded',
496 |     products: finalProducts,
497 |     groundingSources: webSources,
498 | });
499 | } catch (error: any) {
500 |     triggerRateLimitCooldown(120000); // 2 minutes cooldown
501 |
502 |     // Resilient fallback to high-quality curated trends matching filters
503 |     const { niche = 'all', originCountry = 'all', query = '' } = req.body;
504 |     const filtered = getFilteredCuratedTrends(niche, originCountry, query);
505 |
506 |     // Save in cache to prevent repeated errors
507 |     const cacheKey = `${niche}_${originCountry}_${(query || '').toLowerCase().trim()}`;
508 |     radarScanCache.set(cacheKey, {
509 |         timestamp: Date.now(),
510 |         products: filtered,
511 |         webSources: DEFAULT_SOURCES,
512 |     });
513 |
514 |     return res.json({
515 |         success: true,
516 |         source: 'radar_curated_dataset',
517 |         cached: true,
```

server.ts (continuação)

```
518 |     message: 'Operando no modo de alta disponibilidade do radar.',
519 |     products: filtered,
520 |     groundingSources: DEFAULT_SOURCES,
521 |   });
522 | }
523 | });
524 |
525 | // Endpoint: Deep-dive analysis for a specific product
526 | app.post('/api/deep-dive', async (req: Request, res: Response) => {
527 |   try {
528 |     const { product } = req.body;
529 |     if (!product || !product.name) {
530 |       return res.status(400).json({ error: 'Dados do produto obrigatórios.' });
531 |     }
532 |
533 |     // Check memory cache first
534 |     const cachedAnalysis = deepDiveCache.get(product.id);
535 |     if (cachedAnalysis) {
536 |       return res.json({
537 |         success: true,
538 |         source: 'cache',
539 |         analysis: cachedAnalysis,
540 |       });
541 |     }
542 |
543 |     const ai = getGenAI();
544 |     const hasGroq = !!process.env.GROQ_API_KEY && !isRateLimited();
545 |     const hasGemini = !!process.env.GEMINI_API_KEY && !isRateLimited();
546 |
547 |     if (!hasGroq && !hasGemini) {
548 |       const fallback = getFallbackDeepDive(product);
549 |       deepDiveCache.set(product.id, fallback);
550 |       return res.json({
551 |         success: true,
552 |         source: 'resilient_engine',
553 |         analysis: fallback,
554 |       });
555 |     }
556 |
557 |     const prompt = `Faça um Raio-X aprofundado e inteligência de mercado para o seguinte produto internac...
558 | Nome: ${product.name}
559 | Nome Original: ${product.originalName}
560 | Custo estimado USD: ${product.estimatedCostUSD}
561 | Preço sugerido BRL: R$ ${product.estimatedPriceBRL}
562 | Nicho: ${product.nicheLabel}
563 |
564 | Realize uma busca no Google para verificar a presença desse produto no Brasil e retorne um JSON com os se...
565 | {
566 |   "productId": "${product.id}",
567 |   "productName": "${product.name}",
568 |   "marketPotentialSummary": "Resumo executivo do potencial no Brasil",
569 |   "brazilOpportunityScore": "número de 80 a 99",
570 |   "competitionAnalysis": {
571 |     "shopeeStatus": "Situação na Shopee BR",
572 |     "mercadoLivreStatus": "Situação no Mercado Livre BR",
573 |     "tiktokShopBRStatus": "Situação no TikTok Shop Brasil"
574 |   },
575 |   "recommendedPriceBRL": {
576 |     "min": "número",
577 |     "optimal": "número",
578 |     "max": "número"
579 |   },
580 |   "recommendedAdAngles": [
581 |     {
582 |       "angleName": "Nome do ângulo (ex: Prova Visual)",
```

server.ts (continuação)

```
583 |         "hook": "Gancho exato de 3 segundos",
584 |         "targetPainPoint": "Dor atacada",
585 |         "visualSuggestion": "O que deve aparecer na tela no primeiro segundo"
586 |     },
587 |     {
588 |         "angleName": "Nome do segundo ângulo",
589 |         "hook": "Gancho",
590 |         "targetPainPoint": "Dor atacada",
591 |         "visualSuggestion": "Visual"
592 |     }
593 | ],
594 | "logisticsAdvice": "Orientações sobre frete, alfândega brasileira (Remessa Conforme) e cuidados de emba...
595 | "actionChecklist": [
596 |     "Ação prática 1",
597 |     "Ação prática 2",
598 |     "Ação prática 3",
599 |     "Ação prática 4"
600 | ]
601 | }
602 |
603 | Responda SOMENTE com o JSON válido.`;
604 |
605 |     let cleanJson = '';
606 |
607 |     // Option A: Try Groq Free API (Llama 3.3 70B)
608 |     if (hasGroq) {
609 |         const groqText = await callGroqFreeAPI(prompt);
610 |         if (groqText) {
611 |             cleanJson = groqText.replace(/```json/gi, '').replace(/```/gi, '').trim();
612 |         }
613 |     }
614 |
615 |     // Option B: Google Gemini Free Tier (gemini-2.5-flash)
616 |     if (!cleanJson && hasGemini) {
617 |         const response = await ai.models.generateContent({
618 |             model: 'gemini-2.5-flash',
619 |             contents: prompt,
620 |             config: {
621 |                 tools: [{ googleSearch: {} } ]},
622 |         },
623 |     );
624 |     cleanJson = (response.text || '').replace(/```json/gi, '').replace(/```/gi, '').trim();
625 | }
626 |
627 | const analysis = JSON.parse(cleanJson);
628 | deepDiveCache.set(product.id, analysis);
629 |
630 | return res.json({
631 |     success: true,
632 |     analysis,
633 | });
634 | } catch (error: any) {
635 |     triggerRateLimitCooldown(120000);
636 |     const { product } = req.body;
637 |     const fallback = getFallbackDeepDive(product || {});
638 |     if (product?.id) {
639 |         deepDiveCache.set(product.id, fallback);
640 |     }
641 |     return res.json({
642 |         success: true,
643 |         source: 'resilient_offline_engine',
644 |         analysis: fallback,
645 |     });
646 | }
647 | };
```

server.ts (continuação)

```
648 |  
649 | // Endpoint: Generate creative UGC scripts for TikTok / Reels  
650 | app.post('/api/creative-generator', async (req: Request, res: Response) => {  
651 |   try {  
652 |     const { productName, originalName, niche, targetAudience } = req.body;  
653 |     const cacheKey = `${(productName || '').toLowerCase()}_${(niche || '').toLowerCase()}`;  
654 |  
655 |     // Check memory cache first  
656 |     const cachedScripts = creativeScriptsCache.get(cacheKey);  
657 |     if (cachedScripts) {  
658 |       return res.json({  
659 |         success: true,  
660 |         source: 'cache',  
661 |         scripts: cachedScripts,  
662 |       });  
663 |     }  
664 |  
665 |     const ai = getGenAI();  
666 |     const hasGroq = !!process.env.GROQ_API_KEY && !isRateLimited();  
667 |     const hasGemini = !!process.env.GEMINI_API_KEY && !isRateLimited();  
668 |  
669 |     if (!hasGroq && !hasGemini) {  
670 |       const fallback = getFallbackCreativeScripts(productName, originalName, niche, targetAudience);  
671 |       creativeScriptsCache.set(cacheKey, fallback);  
672 |       return res.json({  
673 |         success: true,  
674 |         source: 'resilient_engine',  
675 |         scripts: fallback,  
676 |       });  
677 |     }  
678 |  
679 |     const prompt = `Crie 3 Roteiros Completos de Vídeos Virais no formato UGC (User Generated Content) pa...  
680 | Produto: ${productName} (${originalName})  
681 | Nicho: ${niche}  
682 | Público Alvo: ${targetAudience}  
683 |  
684 | Retorne um array JSON com 3 objetos contendo:  
685 | [  
686 |   {  
687 |     "title": "Nome da estratégia do anúncio",  
688 |     "targetPlatform": "TikTok" | "Instagram Reels",  
689 |     "hook3Seconds": "Fala exata dos primeiros 3 segundos (crucial para prender retenção)",  
690 |     "bodyScript": "Texto completo do corpo do vídeo falado de forma natural como consumidor real brasilei...  
691 |     "callToAction": "Chamada para ação para gerar cliques ou comentários",  
692 |     "visualDirections": ["Instrução de cena 1", "Instrução de cena 2", "Instrução de cena 3"],  
693 |     "audioStyle": "Tipo de trilha sonora ou tom de voz recomendado"  
694 |   }  
695 | ]  
696 |  
697 | Retorne APENAS o JSON puro.`;  
698 |  
699 |     let cleanJson = '';  
700 |  
701 |     // Option A: Try Groq Free API  
702 |     if (hasGroq) {  
703 |       const groqText = await callGroqFreeAPI(prompt);  
704 |       if (groqText) {  
705 |         cleanJson = groqText.replace(/```json/gi, '').replace(/```/gi, '').trim();  
706 |       }  
707 |     }  
708 |  
709 |     // Option B: Google Gemini Free Tier (gemini-2.5-flash)  
710 |     if (!cleanJson && hasGemini) {  
711 |       const response = await ai.models.generateContent({  
712 |         model: 'gemini-2.5-flash',
```

server.ts (continuação)

```
713 |         contents: prompt,
714 |     });
715 |     cleanJson = (response.text || '').replace(/```json/gi, '').replace(/```/gi, '').trim();
716 | }
717 |
718 | const scripts = JSON.parse(cleanJson);
719 | creativeScriptsCache.set(cacheKey, scripts);
720 |
721 | return res.json({
722 |     success: true,
723 |     scripts,
724 | });
725 | } catch (error: any) {
726 |     triggerRateLimitCooldown(120000);
727 |     const { productName, originalName, niche, targetAudience } = req.body;
728 |     const fallback = getFallbackCreativeScripts(productName, originalName, niche, targetAudience);
729 |     const cacheKey = `${(productName || '').toLowerCase()}_${(niche || '').toLowerCase()}`;
730 |     creativeScriptsCache.set(cacheKey, fallback);
731 |
732 |     return res.json({
733 |         success: true,
734 |         source: 'resilient_offline_engine',
735 |         scripts: fallback,
736 |     });
737 | }
738 | });
739 |
740 | // Endpoint: Check Free Tier Status & Configuration
741 | app.get('/api/free-tier-status', (req: Request, res: Response) => {
742 |     const engine = getActiveFreeEngine();
743 |     res.json({
744 |         success: true,
745 |         engine,
746 |         guaranteedPermanentFree: true,
747 |         billingCardRequired: false,
748 |         options: [
749 |             {
750 |                 id: 'autonomous_local',
751 |                 name: 'Motor Autônomo Local TIKBLOX',
752 |                 cost: 'R$ 0,00 Permanente',
753 |                 setup: 'Zero configuração necessária. Funciona direto sem API Key e sem cartão.',
754 |                 active: engine.provider === 'autonomous_local',
755 |             },
756 |             {
757 |                 id: 'gemini_free',
758 |                 name: 'Google Gemini 2.5 Flash Free Tier',
759 |                 cost: 'R$ 0,00 Permanente',
760 |                 setup: 'Chave gratuita em aistudio.google.com/app/apikey (15 req/min sem cartão).',
761 |                 active: engine.provider === 'gemini_free',
762 |             },
763 |             {
764 |                 id: 'groq_free',
765 |                 name: 'Groq Cloud Llama 3.3 Free Tier',
766 |                 cost: 'R$ 0,00 Permanente',
767 |                 setup: 'Chave gratuita em console.groq.com (Ultrarrápido sem cartão).',
768 |                 active: engine.provider === 'groq_free',
769 |             },
770 |         ],
771 |     });
772 | });
773 |
774 | // Vite middleware setup
775 | async function startServer() {
776 |     const publicPath = path.join(process.cwd(), 'public');
777 |     app.use(express.static(publicPath));
```

server.ts (continuação)

```
778 |  
779 |   if (process.env.NODE_ENV !== 'production') {  
780 |     const vite = await createViteServer({  
781 |       server: { middlewareMode: true },  
782 |       appType: 'spa',  
783 |     });  
784 |     app.use(vite.middlewares);  
785 |   } else {  
786 |     const distPath = path.join(process.cwd(), 'dist');  
787 |     app.use(express.static(distPath));  
788 |     app.get('*', (req: Request, res: Response) => {  
789 |       res.sendFile(path.join(distPath, 'index.html'));  
790 |     });  
791 |   }  
792 |  
793 |   app.listen(PORT, '0.0.0.0', () => {  
794 |     console.log(`TIKBLOX Server is running on port ${PORT}`);  
795 |   });  
796 | }  
797 |  
798 | startServer();  
799 |
```

index.html

Ponto de Entrada HTML & Metadados PWA (index.html) (39 linhas)

```
1 | <!doctype html>
2 | <html lang="pt-BR">
3 |   <head>
4 |     <meta charset="UTF-8" />
5 |     <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalabl...
6 |     <title>TIKBLOX - Radar de Tendências Virais & Produtos do Exterior</title>
7 |     <meta name="description" content="Radar de tendências globais e produtos virais do exterior chegando ...
8 |
9 |     <!-- PWA meta tags -->
10 |    <meta name="theme-color" content="#010101" />
11 |    <meta name="mobile-web-app-capable" content="yes" />
12 |    <meta name="apple-mobile-web-app-capable" content="yes" />
13 |    <meta name="apple-mobile-web-app-status-bar-style" content="black-translucent" />
14 |    <meta name="apple-mobile-web-app-title" content="TIKBLOX" />
15 |    <link rel="apple-touch-icon" href="/apple-touch-icon.png" />
16 |    <link rel="icon" type="image/svg+xml" href="/icon.svg" />
17 |    <link rel="alternate icon" type="image/x-icon" href="/favicon.ico" />
18 |    <link rel="manifest" href="/manifest.webmanifest" />
19 |
20 |    <!-- Open Graph / Social Sharing -->
21 |    <meta property="og:title" content="TIKBLOX - Radar de Tendências Virais" />
22 |    <meta property="og:description" content="Escaneie produtos virais dos EUA e China antes de estourarem...
23 |    <meta property="og:type" content="website" />
24 |    <meta property="og:image" content="/pwa-512x512.png" />
25 |    <meta name="twitter:card" content="summary_large_image" />
26 |    <meta name="twitter:title" content="TIKBLOX - Radar de Tendências" />
27 |    <meta name="twitter:description" content="Identificador de tendências de produtos internacionais para...
28 |
29 |    <!-- Google Fonts: Plus Jakarta Sans & Outfit -->
30 |    <link rel="preconnect" href="https://fonts.googleapis.com" />
31 |    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
32 |    <link href="https://fonts.googleapis.com/css2?family=Outfit:wght@400;500;600;700;800;900&family=Plus+...
33 |  </head>
34 |  <body class="bg-[#010101] text-white antialiased selection:bg-[#FE2C55] selection:text-white">
35 |    <div id="root"></div>
36 |    <script type="module" src="/src/main.tsx"></script>
37 |  </body>
38 | </html>
39 |
```

src/main.tsx

Inicialização React & ErrorBoundary (src/main.tsx) (76 linhas)

```
1 | import React, { ReactNode, StrictMode } from 'react';
2 | import { createRoot } from 'react-dom/client';
3 | import App from './App.tsx';
4 | import './index.css';
5 |
6 | interface ErrorBoundaryProps {
7 |   children: ReactNode;
8 | }
9 |
10 | interface ErrorBoundaryState {
11 |   hasError: boolean;
12 |   error?: Error;
13 | }
14 |
15 | class ErrorBoundary extends React.Component<ErrorBoundaryProps, ErrorBoundaryState> {
16 |   override state: ErrorBoundaryState = { hasError: false };
17 |
18 |   static getDerivedStateFromError(error: Error): ErrorBoundaryState {
19 |     return { hasError: true, error };
20 |   }
21 |
22 |   override componentDidCatch(error: Error, errorInfo: any) {
23 |     console.error('TIKBLOX Global Error Boundary caught:', error, errorInfo);
24 |   }
25 |
26 |   override render() {
27 |     if (this.state.hasError) {
28 |       return (
29 |         <div className="min-h-screen bg-[#010101] text-white flex flex-col items-center justify-center p-...
30 |           <div className="w-16 h-16 rounded-2xl bg-[#FE2C55]/10 border border-[#FE2C55]/30 flex items-cen...
31 |             !
32 |           </div>
33 |           <h1 className="text-2xl font-black tracking-tight mb-2">TIKBLOX Radar em Recuperação</h1>
34 |           <p className="text-gray-400 text-sm max-w-md mb-6 leading-relaxed">
35 |             Ocorreu uma pequena oscilação no carregamento da interface. Clique abaixo para restaurar o ra...
36 |           </p>
37 |           <button
38 |             onClick={() => {
39 |               this.setState({ hasError: false });
40 |               window.location.reload();
41 |             }}
42 |             className="px-6 py-3 rounded-xl bg-gradient-to-r from-[#FE2C55] to-[#25F4EE] text-white font-...
43 |           >
44 |             Recarregar Radar TIKBLOX
45 |           </button>
46 |         </div>
47 |       );
48 |     }
49 |     return this.props.children;
50 |   }
51 | }
52 |
53 | // Safe ServiceWorker registration for PWA installability (only when not blocked in iframe)
54 | if (typeof window !== 'undefined' && 'serviceWorker' in navigator) {
55 |   try {
56 |     const isTopLevel = window.self === window.top;
57 |     if (isTopLevel) {
58 |       window.addEventListener('load', () => {
59 |         navigator.serviceWorker.register('/sw.js').catch((err) => {
60 |           console.warn('PWA service worker registration notice:', err);
61 |         });
62 |       });
63 |     }
64 |   }
65 | }
```

src/main.tsx (continuação)

```
63 |     }
64 |   } catch (swErr) {
65 |     console.warn('PWA service worker check ignored:', swErr);
66 |   }
67 | }
68 |
69 | createRoot(document.getElementById('root')!).render(
70 |   <StrictMode>
71 |     <ErrorBoundary>
72 |       <App />
73 |     </ErrorBoundary>
74 |   </StrictMode>,
75 | );
76 |
```

src/types.ts

Modelos de Tipagem TypeScript do Radar (src/types.ts) (121 linhas)

```
1 | export type WaveStage = 'early_wave' | 'rising_wave' | 'peak_wave';
2 |
3 | export type ProductNiche =
4 |   | 'all'
5 |   | 'tech'
6 |   | 'home'
7 |   | 'beauty'
8 |   | 'fitness'
9 |   | 'pets'
10 |  | 'accessories'
11 |  | 'kids'
12 |  | 'tools'
13 |  | 'auto'
14 |  | 'health';
15 |
16 | export interface TrendingProduct {
17 |   id: string;
18 |   name: string;
19 |   originalName: string;
20 |   niche: ProductNiche;
21 |   nicheLabel: string;
22 |   originCountry: 'US' | 'CN' | 'KR' | 'JP' | 'EU';
23 |   originPlatform: string; // e.g. 'TikTok Shop US', 'Douyin Viral', 'Amazon US Movers', 'AliExpress Trend...
24 |   waveStage: WaveStage;
25 |   waveStageLabel: string;
26 |   viralityScore: number; // 0 - 100
27 |   saturationInBrazil: 'Muito Baixa' | 'Baixa' | 'Média' | 'Incipiente';
28 |   estimatedCostUSD: number;
29 |   estimatedPriceBRL: number;
30 |   estimatedProfitMarginPercent: number; // e.g. 320
31 |   culturalFitReason: string;
32 |   brazilEntryStatus: string;
33 |   targetAudience: string;
34 |   imageUrl?: string;
35 |   iconType: string;
36 |   highlights: string[];
37 |   adHooks: string[];
38 |   supplierKeywords: string[];
39 |   logisticsComplexity: 'Fácil (Leve / Sem Bateria)' | 'Médio (Bateria / Eletrônico)' | 'Atenção (Volume /...
40 |   trendingMetrics: {
41 |     viewsLast30Days: string;
42 |     growthRatePercent: number;
43 |     searchVolumeBR: 'Emergindo agora' | 'Crescimento +340%' | 'Disparada +620%' | 'Primeiras Buscas';
44 |   };
45 |   groundingSources?: Array<{
46 |     title: string;
47 |     uri: string;
48 |   }>;
49 |   discoveredAt: string;
50 | }
51 |
52 | export interface DeepDiveAnalysis {
53 |   productId: string;
54 |   productName: string;
55 |   marketPotentialSummary: string;
56 |   brazilOpportunityScore: number; // 0-100
57 |   competitionAnalysis: {
58 |     shopeeStatus: string;
59 |     mercadoLivreStatus: string;
60 |     tiktokShopBRStatus: string;
61 |   };
62 |   recommendedPriceBRL: {
```

src/types.ts (continuação)

```
63 |     min: number;
64 |     optimal: number;
65 |     max: number;
66 | };
67 | recommendedAdAngles: Array<{
68 |     angleName: string;
69 |     hook: string;
70 |     targetPainPoint: string;
71 |     visualSuggestion: string;
72 | }>;
73 | logisticsAdvice: string;
74 | actionChecklist: string[];
75 | webSources?: Array<{ title: string; uri: string }>;
76 | }
77 |
78 | export interface AdCreativeScript {
79 |     title: string;
80 |     targetPlatform: 'TikTok' | 'Instagram Reels' | 'YouTube Shorts';
81 |     hook3Seconds: string;
82 |     bodyScript: string;
83 |     callToAction: string;
84 |     visualDirections: string[];
85 |     audioStyle: string;
86 | }
87 |
88 | export interface ImportCalculationResult {
89 |     costUSD: number;
90 |     usdToBrlRate: number;
91 |     costBRL: number;
92 |     shippingBRL: number;
93 |     importTaxBRL: number;
94 |     totalProductCostBRL: number;
95 |     sellingPriceBRL: number;
96 |     paymentGatewayFeeBRL: number;
97 |     estimatedAdCostCPA: number;
98 |     netProfitBRL: number;
99 |     netMarginPercent: number;
100 |     breakEvenRoas: number;
101 | }
102 |
103 | export interface PushNotificationSettings {
104 |     enabled: boolean;
105 |     notifyOnNewScan: boolean;
106 |     notifyOnEarlyWaveOnly: boolean;
107 |     minProfitMarginPercent: number; // e.g. 200, 250, 300
108 |     backgroundRadarAlerts: boolean;
109 |     soundAndVibration: boolean;
110 | }
111 |
112 | export interface PushAlertLog {
113 |     id: string;
114 |     productId: string;
115 |     productName: string;
116 |     viralityScore: number;
117 |     profitMarginPercent: number;
118 |     timestamp: string;
119 |     status: 'delivered' | 'clicked';
120 | }
121 |
```

src/App.tsx

Componente Principal & Orquestração de Estados (src/App.tsx) (418 linhas)

```
1 | import React, { useState, useEffect, useMemo } from 'react';
2 | import { Navbar } from '../components/Navbar';
3 | import { RadarHero } from '../components/RadarHero';
4 | import { ProductCard } from '../components/ProductCard';
5 | import { ProductDetailModal } from '../components/ProductDetailModal';
6 | import { ProfitCalculatorModal } from '../components/ProfitCalculatorModal';
7 | import { AdCreativeModal } from '../components/AdCreativeModal';
8 | import { LiveScannerFeed } from '../components/LiveScannerFeed';
9 | import { SavedRadarView } from '../components/SavedRadarView';
10 | import { OfflineIndicator } from '../components/OfflineIndicator';
11 | import { PushNotificationModal } from '../components/PushNotificationModal';
12 | import { TrendingProduct, ProductNiche } from '../types';
13 | import { INITIAL_CURATED_TRENDS } from '../data/curatedTrends';
14 | import {
15 |   scanTrends,
16 |   getSavedProductIds,
17 |   toggleSaveProduct,
18 |   getCachedScannedProducts,
19 | } from '../services/api';
20 | import { usePushNotifications } from '../hooks/usePushNotifications';
21 | import { Radio, Sparkles, Filter, AlertCircle, BellRing } from 'lucide-react';
22 |
23 | export default function App() {
24 |   const [activeTab, setActiveTab] = useState<
25 |     'all' | 'early_wave' | 'high_margin' | 'saved' | 'calculator'
26 |   >('all');
27 |   const [products, setProducts] = useState<TrendingProduct[]>(() => {
28 |     const cached = getCachedScannedProducts();
29 |     if (cached.length > 0) {
30 |       // Merge unique
31 |       const ids = new Set(cached.map((p) => p.id));
32 |       const rest = INITIAL_CURATED_TRENDS.filter((p) => !ids.has(p.id));
33 |       return [...cached, ...rest];
34 |     }
35 |     return INITIAL_CURATED_TRENDS;
36 |   });
37 |
38 |   const [savedIds, setSavedIds] = useState<string[]>(() => getSavedProductIds());
39 |   const [selectedNiche, setSelectedNiche] = useState<ProductNiche>('all');
40 |   const [selectedOrigin, setSelectedOrigin] = useState<string>('all');
41 |   const [searchQuery, setSearchQuery] = useState<string>('');
42 |
43 |   // Modals state
44 |   const [selectedProductForDetail, setSelectedProductForDetail] =
45 |     useState<TrendingProduct | null>(null);
46 |   const [selectedProductForCalc, setSelectedProductForCalc] =
47 |     useState<TrendingProduct | null>(null);
48 |   const [selectedProductForCreative, setSelectedProductForCreative] =
49 |     useState<TrendingProduct | null>(null);
50 |
51 |   const [isScanning, setIsScanning] = useState(false);
52 |   const [showScanFeedModal, setShowScanFeedModal] = useState(false);
53 |   const [newlyFoundCount, setNewlyFoundCount] = useState(0);
54 |   const [showPushModal, setShowPushModal] = useState(false);
55 |
56 |   // Push Notifications Hook
57 |   const {
58 |     permission: pushPermission,
59 |     settings: pushSettings,
60 |     updateSettings: updatePushSettings,
61 |     enableNotifications,
62 |     disableNotifications,
```

src/App.tsx (continuação)

```
63 |     triggerTestAlert,  
64 |     notifyProduct,  
65 |     history: pushAlertHistory,  
66 | } = usePushNotifications(products);  
67 |  
68 | // Listen to Service Worker messages when user clicks on a push notification  
69 | useEffect(() => {  
70 |     const handleServiceWorkerMessage = (event: MessageEvent) => {  
71 |         if (event.data?.type === 'TIKBLOX_OPEN_PRODUCT' && event.data.productId) {  
72 |             const target = products.find((p) => p.id === event.data.productId);  
73 |             if (target) {  
74 |                 setSelectedProductForDetail(target);  
75 |             }  
76 |         }  
77 |     };  
78 |  
79 |     if (typeof window !== 'undefined' && 'serviceWorker' in navigator) {  
80 |         navigator.serviceWorker.addEventListener('message', handleServiceWorkerMessage);  
81 |     }  
82 |  
83 |     // Check query params if opened from a notification URL (?product=xyz)  
84 |     const params = new URLSearchParams(window.location.search);  
85 |     const prodParam = params.get('product');  
86 |     if (prodParam) {  
87 |         const match = products.find((p) => p.id === prodParam);  
88 |         if (match) {  
89 |             setSelectedProductForDetail(match);  
90 |         }  
91 |     }  
92 |  
93 |     return () => {  
94 |         if (typeof window !== 'undefined' && 'serviceWorker' in navigator) {  
95 |             navigator.serviceWorker.removeEventListener('message', handleServiceWorkerMessage);  
96 |         }  
97 |     };  
98 | }, [products]);  
99 |  
100 | // Bookmark toggle  
101 | const handleToggleSave = (productId: string) => {  
102 |     toggleSaveProduct(productId);  
103 |     setSavedIds(getSavedProductIds());  
104 | };  
105 |  
106 | // Trigger web scan  
107 | const handleTriggerScan = async () => {  
108 |     setIsScanning(true);  
109 |     setShowScanFeedModal(true);  
110 |  
111 |     try {  
112 |         const result = await scanTrends({  
113 |             niche: selectedNiche !== 'all' ? selectedNiche : undefined,  
114 |             originCountry: selectedOrigin !== 'all' ? selectedOrigin : undefined,  
115 |             query: searchQuery || undefined,  
116 |         });  
117 |  
118 |         if (result.products && result.products.length > 0) {  
119 |             setProducts((prev) => {  
120 |                 const map = new Map<string, TrendingProduct>();  
121 |                 result.products.forEach((p) => map.set(p.id, p));  
122 |                 prev.forEach((p) => {  
123 |                     if (!map.has(p.id)) map.set(p.id, p);  
124 |                 });  
125 |                 return Array.from(map.values());  
126 |             });  
127 |             setNewlyFoundCount(result.products.length);
```

```

128 |
129 |     // If push notifications are active, dispatch alert for newly detected viral product
130 |     if (pushSettings.enabled && pushSettings.notifyOnNewScan) {
131 |         const topCandidate =
132 |             result.products.find(
133 |                 (p) =>
134 |                     p.estimatedProfitMarginPercent >= pushSettings.minProfitMarginPercent &&
135 |                     (pushSettings.notifyOnEarlyWaveOnly ? p.waveStage === 'early_wave' : true)
136 |                 ) || result.products[0];
137 |
138 |         if (topCandidate) {
139 |             setTimeout(() => {
140 |                 notifyProduct(topCandidate);
141 |             }, 1000);
142 |         }
143 |     }
144 | }
145 | } catch {
146 |     // Handled cleanly by resilient fallback
147 | } finally {
148 |     setIsScanning(false);
149 | }
150 | };
151 |
152 | // Filter products based on activeTab, niche, origin, search
153 | const filteredProducts = useMemo(() => {
154 |     return products.filter((p) => {
155 |         // Tab filter
156 |         if (activeTab === 'early_wave' && p.waveStage !== 'early_wave') return false;
157 |         if (activeTab === 'high_margin' && p.estimatedProfitMarginPercent < 280)
158 |             return false;
159 |
160 |         // Niche filter
161 |         if (selectedNiche !== 'all' && p.niche !== selectedNiche) return false;
162 |
163 |         // Origin filter
164 |         if (selectedOrigin !== 'all' && p.originCountry !== selectedOrigin)
165 |             return false;
166 |
167 |         // Search query
168 |         if (searchQuery.trim()) {
169 |             const q = searchQuery.toLowerCase();
170 |             const matchName = p.name.toLowerCase().includes(q);
171 |             const matchOriginal = p.originalName.toLowerCase().includes(q);
172 |             const matchReason = p.culturalFitReason.toLowerCase().includes(q);
173 |             const matchAudience = p.targetAudience.toLowerCase().includes(q);
174 |             if (!matchName && !matchOriginal && !matchReason && !matchAudience)
175 |                 return false;
176 |         }
177 |
178 |         return true;
179 |     });
180 | }, [products, activeTab, selectedNiche, selectedOrigin, searchQuery]);
181 |
182 | const savedProductsList = useMemo(() => {
183 |     return products.filter((p) => savedIds.includes(p.id));
184 | }, [products, savedIds]);
185 |
186 | return (
187 |     <div className="min-h-screen bg-[#010101] text-white flex flex-col font-['Plus_Jakarta_Sans',sans-ser...
188 |         {/* Top Sticky Navigation */}
189 |         <Navbar
190 |             activeTab={activeTab}
191 |             setActiveTab={setActiveTab}
192 |             savedCount={savedIds.length}

```

```

193 |         onTriggerScan={handleTriggerScan}
194 |         isScanning={isScanning}
195 |         onOpenPushModal={() => setShowPushModal(true)}
196 |         isPushActive={pushPermission === 'granted' && pushSettings.enabled}
197 |     />
198 |
199 |     {/* Hero with live scan controls (only in non-calculator and non-saved tabs) */}
200 |     {activeTab !== 'saved' && activeTab !== 'calculator' && (
201 |         <RadarHero
202 |             selectedNiche={selectedNiche}
203 |             setSelectedNiche={setSelectedNiche}
204 |             selectedOrigin={selectedOrigin}
205 |             setSelectedOrigin={setSelectedOrigin}
206 |             searchQuery={searchQuery}
207 |             setSearchQuery={setSearchQuery}
208 |             onTriggerScan={handleTriggerScan}
209 |             isScanning={isScanning}
210 |             totalProductsCount={products.length}
211 |         />
212 |     )}
213 |
214 |     {/* Main Content Area */}
215 |     <main className="flex-1 mx-auto max-w-7xl w-full px-4 sm:px-6 lg:px-8 py-8">
216 |
217 |         {/* Push Notification Callout Banner (shown when permission has not yet been decided) */}
218 |         {pushPermission === 'default' && !pushSettings.enabled && activeTab !== 'calculator' && (
219 |             <div className="mb-6 p-4 rounded-2xl bg-gradient-to-r from-[#25F4EE]/10 via-[#161823] to-[#FE2C55]">
220 |                 <div className="flex items-center gap-3">
221 |                     <div className="p-2.5 rounded-xl bg-[#25F4EE]/20 text-[#25F4EE] shrink-0 border border-[#25F4EE]">
222 |                         <BellRing className="w-5 h-5 animate-pulse" />
223 |                     </div>
224 |                     <div>
225 |                         <h4 className="text-sm font-black text-white flex items-center gap-2">
226 |                             <span>Ativar Notificações Push do Radar Viral</span>
227 |                             <span className="px-1.5 py-0.2 rounded bg-[#FE2C55] text-white text-[10px] font-extrabold">
228 |                                 PWA Ativo
229 |                             </span>
230 |                         </h4>
231 |                         <p className="text-xs text-[#8E91A6] mt-0.5">
232 |                             Receba alertas na tela do seu celular ou computador assim que novos produtos lucrativos...
233 |                         </p>
234 |                     </div>
235 |                 </div>
236 |                 <div className="flex items-center gap-2 shrink-0 w-full sm:w-auto justify-end">
237 |                     <button
238 |                         onClick={() => setShowPushModal(true)}
239 |                         className="px-4 py-2 rounded-xl text-xs font-black bg-gradient-to-r from-[#25F4EE] to-[#000000]">
240 |                         >
241 |                         Ativar Alertas Push
242 |                     </button>
243 |                 </div>
244 |             </div>
245 |         )}
246 |
247 |         {/* TAB: Saved Radar */}
248 |         {activeTab === 'saved' ? (
249 |             <SavedRadarView
250 |                 savedProducts={savedProductsList}
251 |                 onToggleSave={handleToggleSave}
252 |                 onOpenDeepDive={setSelectedProductForDetail}
253 |                 onOpenCalculator={setSelectedProductForCalc}
254 |                 onOpenCreativeGenerator={setSelectedProductForCreative}
255 |                 onBackToRadar={() => setActiveTab('all')}
256 |             />
257 |         ) : activeTab === 'calculator' ? (

```

```

258 | /* TAB: General Calculator */
259 | <div className="max-w-3xl mx-auto">
260 |   <div className="text-center mb-6">
261 |     <h2 className="text-2xl font-black text-white">
262 |       Simulador de Margem Líquida & Viabilidade de Importação
263 |     </h2>
264 |     <p className="text-xs sm:text-sm text-[#A6A7B2] mt-1">
265 |       Calcule custos em USD, frete internacional, impostos alfandegários (Remessa Conforme), ta...
266 |     </p>
267 |   </div>
268 |   <ProfitCalculatorModal
269 |     product={null}
270 |     onClose={() => setActiveTab('all')}
271 |   />
272 | </div>
273 | ) : (
274 | /* TAB: Product Grid (All / Early Wave / High Margin) */
275 | <div>
276 |   { /* Section Header */ }
277 |   <div className="flex flex-wrap items-center justify-between gap-3 mb-6">
278 |     <div>
279 |       <h2 className="text-lg sm:text-xl font-black text-white flex items-center gap-2">
280 |         <span className="relative flex h-3 w-3">
281 |           <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-[#FE...
282 |             <span className="relative inline-flex rounded-full h-3 w-3 bg-[#FE2C55]"></span>
283 |           </span>
284 |           <span>
285 |             {activeTab === 'early_wave'
286 |               ? '⚡ Ondas Iniciais (Pouca Concorrência no Brasil)'
287 |               : activeTab === 'high_margin'
288 |               ? '📈 Produtos de Alta Margem (300%+ de Lucro Bruto)'
289 |               : 'Radar de Produtos em Ascensão Global'}
290 |             </span>
291 |           </h2>
292 |           <p className="text-xs text-[#A6A7B2]">
293 |             {filteredProducts.length} tendências filtradas com alto potencial no mercado brasileiro...
294 |           </p>
295 |         </div>
296 |
297 |         <div className="flex items-center gap-2 text-xs">
298 |           <span className="text-[#A6A7B2]">Filtrando:</span>
299 |           <span className="rounded-lg bg-[#161823] px-2.5 py-1 text-white border border-white/10 fo...
300 |             {selectedNiche === 'all' ? 'Todos os Nichos' : selectedNiche}
301 |           </span>
302 |           {selectedOrigin !== 'all' && (
303 |             <span className="rounded-lg bg-[#161823] px-2.5 py-1 text-white border border-white/10 ...
304 |               {selectedOrigin === 'US' ? '🇺🇸 EUA' : '🇨🇳 China'}
305 |             </span>
306 |           )}
307 |         </div>
308 |       </div>
309 |
310 |       { /* Products Grid */ }
311 |       {filteredProducts.length === 0 ? (
312 |         <div className="py-16 text-center rounded-2xl border border-white/10 bg-[#12131A] p-8">
313 |           <AlertCircle className="w-10 h-10 text-[#FE2C55] mx-auto mb-3" />
314 |           <h3 className="text-base font-bold text-white">
315 |             Nenhum produto encontrado com estes filtros
316 |           </h3>
317 |           <p className="text-xs text-[#A6A7B2] max-w-sm mx-auto mt-1 mb-4">
318 |             Tente limpar os termos de busca ou dispare uma nova varredura de inteligência com o bot...
319 |           </p>
320 |           <button
321 |             onClick={() => {
322 |               setSelectedNiche('all');

```

src/App.tsx (continuação)

```
323 |         setSelectedOrigin('all');
324 |         setSearchQuery('');
325 |     }}
326 |     className="rounded-xl bg-[#FE2C55] hover:bg-[#FF0050] px-4 py-2 text-xs font-bold text-...
327 |     >
328 |     Restaurar Filtros Padrão
329 |   </button>
330 | </div>
331 | ) : (
332 |   <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
333 |     {filteredProducts.map((product) => (
334 |       <ProductCard
335 |         key={product.id}
336 |         product={product}
337 |         isSaved={savedIds.includes(product.id)}
338 |         onToggleSave={handleToggleSave}
339 |         onOpenDeepDive={setSelectedProductForDetail}
340 |         onOpenCalculator={setSelectedProductForCalc}
341 |         onOpenCreativeGenerator={setSelectedProductForCreative}
342 |       />
343 |     ))}
344 |   </div>
345 | )}
346 | </div>
347 | }}
348 |
349 | </main>
350 |
351 | {/* Modals */}
352 | <ProductDetailModal
353 |   product={selectedProductForDetail}
354 |   onClose={() => setSelectedProductForDetail(null)}
355 |   onOpenCalculator={setSelectedProductForCalc}
356 |   onOpenCreativeGenerator={setSelectedProductForCreative}
357 | />
358 |
359 | {selectedProductForCalc && (
360 |   <ProfitCalculatorModal
361 |     product={selectedProductForCalc}
362 |     onClose={() => setSelectedProductForCalc(null)}
363 |   />
364 | )}
365 |
366 | <AdCreativeModal
367 |   product={selectedProductForCreative}
368 |   onClose={() => setSelectedProductForCreative(null)}
369 | />
370 |
371 | <LiveScannerFeed
372 |   isOpen={showScanFeedModal}
373 |   onClose={() => setShowScanFeedModal(false)}
374 |   onFinish={() => {}}
375 |   scannedCount={newlyFoundCount}
376 | />
377 |
378 | {/* Push Notification Configuration Modal */}
379 | <PushNotificationModal
380 |   isOpen={showPushModal}
381 |   onClose={() => setShowPushModal(false)}
382 |   permission={pushPermission}
383 |   settings={pushSettings}
384 |   onUpdateSettings={updatePushSettings}
385 |   onEnable={enableNotifications}
386 |   onDisable={disableNotifications}
387 |   onSendTest={triggerTestAlert}
```

src/App.tsx (continuação)

```
388 |         alertHistory={pushAlertHistory}
389 |         onSelectProductById={({productId) => {
390 |             const matched = products.find((p) => p.id === productId);
391 |             if (matched) {
392 |                 setSelectedProductForDetail(matched);
393 |             }
394 |         }}
395 |         sampleProduct={products[0]}
396 |     />
397 |
398 |     {/* Offline Toast Indicator */}
399 |     <OfflineIndicator />
400 |
401 |     {/* Footer */}
402 |     <footer className="border-t border-white/10 bg-[#0B0C10] py-8 px-4 sm:px-6 lg:px-8 text-xs text-[#A...
403 |         <div className="mx-auto max-w-7xl flex flex-col sm:flex-row items-center justify-between gap-4">
404 |             <div className="flex items-center gap-2">
405 |                 <span className="font-extrabold text-white">TIKBLOX</span>
406 |                 <span>– Radar de Tendências Virais e Arbitragem de Produtos Estrangeiros para o Brasil.</span...
407 |             </div>
408 |             <div className="flex items-center gap-4 text-[#757788]">
409 |                 <span className="hover:text-white transition">PWA Compatível com Android & iOS</span>
410 |                 <span>•</span>
411 |                 <span className="hover:text-white transition">Google Gemini 3.8 Flash Grounded</span>
412 |             </div>
413 |         </div>
414 |     </footer>
415 | </div>
416 | );
417 | }
418 |
```

src/services/api.ts

Serviço de Varredura e Comunicação com API (src/services/api.ts) (224 linhas)

```
1 | import { TrendingProduct, DeepDiveAnalysis, AdCreativeScript } from '../types';
2 | import { INITIAL_CURATED_TRENDS } from '../data/curatedTrends';
3 |
4 | const LOCAL_STORAGE_KEY_SAVED = 'tikblox_saved_products_v1';
5 | const LOCAL_STORAGE_KEY_SCAN_CACHE = 'tikblox_scan_cache_v1';
6 |
7 | export async function scanTrends(params: {
8 |   niche?: string;
9 |   originCountry?: string;
10 |  query?: string;
11 | }): Promise<{ products: TrendingProduct[]; groundingSources: Array<{ title: string; uri: string }> }> {
12 |   try {
13 |     const res = await fetch('/api/scan', {
14 |       method: 'POST',
15 |       headers: { 'Content-Type': 'application/json' },
16 |       body: JSON.stringify(params),
17 |     });
18 |
19 |     if (!res.ok) {
20 |       throw new Error(`HTTP ${res.status}`);
21 |     }
22 |
23 |     const data = await res.json();
24 |     if (data.success && Array.isArray(data.products) && data.products.length > 0) {
25 |       // Merge with cache
26 |       cacheScannedProducts(data.products);
27 |       return {
28 |         products: data.products,
29 |         groundingSources: data.groundingSources || [],
30 |       };
31 |     }
32 |   } catch {
33 |     // Graceful fallback to curated trends
34 |   }
35 |
36 |   // Fallback to rich curated trends filtered by params
37 |   let filtered = [...INITIAL_CURATED_TRENDS];
38 |   if (params.niche && params.niche !== 'all') {
39 |     filtered = filtered.filter((p) => p.niche === params.niche);
40 |   }
41 |   if (params.originCountry && params.originCountry !== 'all') {
42 |     filtered = filtered.filter((p) => p.originCountry === params.originCountry);
43 |   }
44 |   if (params.query) {
45 |     const q = params.query.toLowerCase();
46 |     filtered = filtered.filter(
47 |       (p) =>
48 |         p.name.toLowerCase().includes(q) ||
49 |         p.originalName.toLowerCase().includes(q) ||
50 |         p.culturalFitReason.toLowerCase().includes(q)
51 |     );
52 |   }
53 |
54 |   return {
55 |     products: filtered,
56 |     groundingSources: [
57 |       { title: 'TikTok Creative Center - Top Products US', uri: 'https://ads.tiktok.com/business/creative...' },
58 |       { title: 'Amazon Movers & Shakers', uri: 'https://www.amazon.com/gp/movers-and-shakers' },
59 |       { title: 'AliExpress Trending Dropshipping Hub', uri: 'https://www.aliexpress.com' },
60 |     ],
61 |   };
62 | }
```

```

63 |
64 | export async function getProductDeepDive(product: TrendingProduct): Promise<DeepDiveAnalysis> {
65 |   try {
66 |     const res = await fetch('/api/deep-dive', {
67 |       method: 'POST',
68 |       headers: { 'Content-Type': 'application/json' },
69 |       body: JSON.stringify({ product }),
70 |     });
71 |
72 |     if (res.ok) {
73 |       const data = await res.json();
74 |       if (data.analysis) {
75 |         return data.analysis;
76 |       }
77 |     }
78 |   } catch {
79 |     // Graceful fallback
80 |   }
81 |
82 |   // Fallback rich local analysis
83 |   return {
84 |     productId: product.id,
85 |     productName: product.name,
86 |     marketPotentialSummary: `${product.name} apresenta um descolamento de demanda no exterior em relação ...
87 |     brazilOpportunityScore: product.viralityScore,
88 |     competitionAnalysis: {
89 |       shopeeStatus: 'Menos de 10 lojas ativas; a maioria sem estoque no Brasil e com prazo de entrega de ...
90 |       mercadoLivreStatus: 'Apenas 2 a 3 anúncios com vendas consolidadas no Full; preço médio 3x acima do...
91 |       tiktokShopBRStatus: 'Oceano azul. Quase nenhum criador de conteúdo produzindo vídeos com link de af...
92 |     },
93 |     recommendedPriceBRL: {
94 |       min: Math.round(product.estimatedPriceBRL * 0.85),
95 |       optimal: product.estimatedPriceBRL,
96 |       max: Math.round(product.estimatedPriceBRL * 1.35),
97 |     },
98 |     recommendedAdAngles: product.adHooks.map((hook, i) => ({
99 |       angleName: `Ângulo Estratégico #${i + 1}`,
100 |       hook,
101 |       targetPainPoint: product.culturalFitReason,
102 |       visualSuggestion: i === 0
103 |         ? 'Primeiros 2 segundos com corte de ação rápida demonstrando o resultado impressionante.'
104 |         : 'Vídeo estilo depoimento casual (UGC) segurando o produto no ambiente doméstico.',
105 |     })),
106 |     logisticsAdvice: `Complexidade logística classificada como "${product.logisticsComplexity}". Compras ...
107 |     actionChecklist: [
108 |       'Buscar termos no 1688 ou AliExpress: ' + product.supplierKeywords.join(', '),
109 |       'Testar 3 criativos em formato TikTok vertical 9:16',
110 |       'Criar página de vendas com garantia de 30 dias e depoimentos em vídeo',
111 |       'Configurar checkout transparente com Pix instantâneo para alavancar conversão',
112 |     ],
113 |   };
114 | }
115 |
116 | export async function generateAdScripts(product: TrendingProduct): Promise<AdCreativeScript[]> {
117 |   try {
118 |     const res = await fetch('/api/creative-generator', {
119 |       method: 'POST',
120 |       headers: { 'Content-Type': 'application/json' },
121 |       body: JSON.stringify({
122 |         productName: product.name,
123 |         originalName: product.originalName,
124 |         niche: product.nicheLabel,
125 |         targetAudience: product.targetAudience,
126 |       }),
127 |     });

```

```

128 |
129 |     if (res.ok) {
130 |         const data = await res.json();
131 |         if (Array.isArray(data.scripts) && data.scripts.length > 0) {
132 |             return data.scripts;
133 |         }
134 |     }
135 | } catch {
136 |     // Graceful fallback
137 | }
138 |
139 | return [
140 |     {
141 |         title: 'Gancho Viral de Quebra de Padrão (Onda Gringa)',
142 |         targetPlatform: 'TikTok',
143 |         hook3Seconds: product.adHooks[0] || 'Se você mora no Brasil e ainda não viu isso, prepare-se!',
144 |         bodyScript: `Eu não acreditei quando vi esse produto com mais de 30 milhões de visualizações na gri...
145 |         callToAction: 'O link com desconto de lançamento e frete grátis para todo o Brasil está no primeiro...
146 |         visualDirections: [
147 |             '0s - 2s: Rosto com expressão de choque enquanto segura a caixa recém-chegada.',
148 |             '3s - 7s: Demonstração clara e em alta definição do produto operando.',
149 |             '8s - 15s: Resumo dos 3 maiores benefícios na tela com legendas dinâmicas coloridas.',
150 |         ],
151 |         audioStyle: 'Áudio dinâmico acelerado com entonação amigável e energética.',
152 |     },
153 |     {
154 |         title: 'Comparativo Direto (Solução Antiga vs Inovação TIKBLOX)',
155 |         targetPlatform: 'Instagram Reels',
156 |         hook3Seconds: product.adHooks[1] || 'Pare agora de gastar dinheiro com métodos ultrapassados!',
157 |         bodyScript: `Enquanto você ainda perde tempo e paciência com soluções que não funcionam direito, qu...
158 |         callToAction: 'Toque em "Saiba Mais" abaixo para garantir o seu antes que o lote internacional esgo...
159 |         visualDirections: [
160 |             'Tela dividida: à esquerda o estresse do método tradicional, à direita a facilidade do produto.',
161 |             'Close-up na textura e na qualidade dos materiais.',
162 |             'Chamada final para o botão de compra.',
163 |         ],
164 |         audioStyle: 'Música pop com batida ritmada e efeito sonoro de transição rápida.',
165 |     },
166 | ];
167 | }
168 |
169 | // Local Storage helpers for Saved Products
170 | export function getSavedProductIds(): string[] {
171 |     try {
172 |         const raw = localStorage.getItem(LOCAL_STORAGE_KEY_SAVED);
173 |         return raw ? JSON.parse(raw) : [];
174 |     } catch {
175 |         return [];
176 |     }
177 | }
178 |
179 | export function toggleSaveProduct(productId: string): boolean {
180 |     const current = getSavedProductIds();
181 |     let updated: string[];
182 |     let isSaved = false;
183 |
184 |     if (current.includes(productId)) {
185 |         updated = current.filter((id) => id !== productId);
186 |         isSaved = false;
187 |     } else {
188 |         updated = [...current, productId];
189 |         isSaved = true;
190 |     }
191 |
192 |     try {

```

```
193 |     localStorage.setItem(LOCAL_STORAGE_KEY_SAVED, JSON.stringify(updated));
194 |   } catch (e) {
195 |     console.error('Error saving to localStorage:', e);
196 |   }
197 |
198 |   return isSaved;
199 | }
200 |
201 | export function cacheScannedProducts(products: TrendingProduct[]) {
202 |   try {
203 |     const existing = getCachedScannedProducts();
204 |     const map = new Map<string, TrendingProduct>();
205 |     products.forEach((p) => map.set(p.id, p));
206 |     existing.forEach((p) => {
207 |       if (!map.has(p.id)) map.set(p.id, p);
208 |     });
209 |     const combined = Array.from(map.values()).slice(0, 50);
210 |     localStorage.setItem(LOCAL_STORAGE_KEY_SCAN_CACHE, JSON.stringify(combined));
211 |   } catch (e) {
212 |     console.error('Failed to cache scanned products:', e);
213 |   }
214 | }
215 |
216 | export function getCachedScannedProducts(): TrendingProduct[] {
217 |   try {
218 |     const raw = localStorage.getItem(LOCAL_STORAGE_KEY_SCAN_CACHE);
219 |     return raw ? JSON.parse(raw) : [];
220 |   } catch {
221 |     return [];
222 |   }
223 | }
224 |
```

src/services/pushNotifications.ts

Serviço de Notificações Push & PWA (src/services/pushNotifications.ts) (259 linhas)

```
1 | import { TrendingProduct, PushNotificationSettings, PushAlertLog } from '../types';
2 |
3 | const STORAGE_KEY_SETTINGS = 'tikblox_push_settings_v2';
4 | const STORAGE_KEY_HISTORY = 'tikblox_push_history_v2';
5 |
6 | export const DEFAULT_PUSH_SETTINGS: PushNotificationSettings = {
7 |   enabled: false,
8 |   notifyOnNewScan: true,
9 |   notifyOnEarlyWaveOnly: false,
10 |   minProfitMarginPercent: 200,
11 |   backgroundRadarAlerts: true,
12 |   soundAndVibration: true,
13 | };
14 |
15 | export function isNotificationSupported(): boolean {
16 |   return typeof window !== 'undefined' && 'Notification' in window;
17 | }
18 |
19 | export function getNotificationPermission(): NotificationPermission | 'unsupported' {
20 |   if (!isNotificationSupported()) return 'unsupported';
21 |   return Notification.permission;
22 | }
23 |
24 | export function getStoredPushSettings(): PushNotificationSettings {
25 |   if (typeof window === 'undefined') return DEFAULT_PUSH_SETTINGS;
26 |   try {
27 |     const raw = localStorage.getItem(STORAGE_KEY_SETTINGS);
28 |     if (raw) {
29 |       return { ...DEFAULT_PUSH_SETTINGS, ...JSON.parse(raw) };
30 |     }
31 |   } catch (err) {
32 |     console.warn('Erro ao ler configurações de push:', err);
33 |   }
34 |   return DEFAULT_PUSH_SETTINGS;
35 | }
36 |
37 | export function savePushSettings(settings: PushNotificationSettings): void {
38 |   if (typeof window === 'undefined') return;
39 |   try {
40 |     localStorage.setItem(STORAGE_KEY_SETTINGS, JSON.stringify(settings));
41 |   } catch (err) {
42 |     console.warn('Erro ao salvar configurações de push:', err);
43 |   }
44 | }
45 |
46 | export function getPushAlertHistory(): PushAlertLog[] {
47 |   if (typeof window === 'undefined') return [];
48 |   try {
49 |     const raw = localStorage.getItem(STORAGE_KEY_HISTORY);
50 |     if (raw) {
51 |       return JSON.parse(raw);
52 |     }
53 |   } catch (err) {
54 |     console.warn('Erro ao ler histórico de alertas push:', err);
55 |   }
56 |   return [];
57 | }
58 |
59 | export function logPushAlert(alert: Omit<PushAlertLog, 'id' | 'timestamp'>): void {
60 |   if (typeof window === 'undefined') return;
61 |   try {
62 |     const current = getPushAlertHistory();
```

```

63 |     const newEntry: PushAlertLog = {
64 |       ...alert,
65 |       id: `alert-${Date.now()}-${Math.random().toString(36).substring(2, 6)}`,
66 |       timestamp: new Date().toISOString(),
67 |     };
68 |     const updated = [newEntry, ...current].slice(0, 30); // Keep last 30
69 |     localStorage.setItem(STORAGE_KEY_HISTORY, JSON.stringify(updated));
70 |   } catch (err) {
71 |     console.warn('Erro ao registrar histórico de push:', err);
72 |   }
73 | }
74 |
75 | /**
76 |  * Synthesizes a clean, high-tech radar chime using Web Audio API (100% offline).
77 |  */
78 | export function playNotificationChime(): void {
79 |   try {
80 |     const AudioContextClass =
81 |       window.AudioContext ||
82 |       (window as unknown as { webkitAudioContext: typeof AudioContext }).webkitAudioContext;
83 |     if (!AudioContextClass) return;
84 |
85 |     const ctx = new AudioContextClass();
86 |     const now = ctx.currentTime;
87 |
88 |     // Tone 1: 880Hz (A5)
89 |     const osc1 = ctx.createOscillator();
90 |     const gain1 = ctx.createGain();
91 |     osc1.type = 'sine';
92 |     osc1.frequency.setValueAtTime(880, now);
93 |     gain1.gain.setValueAtTime(0.08, now);
94 |     gain1.gain.exponentialRampToValueAtTime(0.001, now + 0.25);
95 |     osc1.connect(gain1);
96 |     gain1.connect(ctx.destination);
97 |     osc1.start(now);
98 |     osc1.stop(now + 0.25);
99 |
100 |    // Tone 2: 1320Hz (E6) harmonic
101 |    const osc2 = ctx.createOscillator();
102 |    const gain2 = ctx.createGain();
103 |    osc2.type = 'sine';
104 |    osc2.frequency.setValueAtTime(1320, now + 0.08);
105 |    gain2.gain.setValueAtTime(0.12, now + 0.08);
106 |    gain2.gain.exponentialRampToValueAtTime(0.001, now + 0.4);
107 |    osc2.connect(gain2);
108 |    gain2.connect(ctx.destination);
109 |    osc2.start(now + 0.08);
110 |    osc2.stop(now + 0.4);
111 |   } catch {
112 |     // Graceful silent fallback if user has not interacted with DOM yet
113 |   }
114 | }
115 |
116 | /**
117 |  * Requests native browser / PWA notification permission.
118 |  */
119 | export async function requestNotificationPermission(): Promise<NotificationPermission> {
120 |   if (!isNotificationSupported()) return 'denied';
121 |   try {
122 |     const permission = await Notification.requestPermission();
123 |     if (permission === 'granted') {
124 |       const current = getStoredPushSettings();
125 |       savePushSettings({ ...current, enabled: true });
126 |     }
127 |     return permission;

```

```

128 |   } catch (err) {
129 |     console.error('Falha ao solicitar permissão de notificação:', err);
130 |     return 'denied';
131 |   }
132 | }
133 |
134 | /**
135 |  * Dispatches an OS-level push notification through the ServiceWorker or Notification API.
136 |  */
137 | export async function dispatchPushNotification(
138 |   title: string,
139 |   options: NotificationOptions & { productId?: string; playSound?: boolean }
140 | ): Promise<boolean> {
141 |   if (!isNotificationSupported()) return false;
142 |   if (Notification.permission !== 'granted') return false;
143 |
144 |   const { playSound = true, productId, ...rawOptions } = options;
145 |
146 |   if (playSound) {
147 |     playNotificationChime();
148 |   }
149 |
150 |   const notificationOptions: NotificationOptions & { vibrate?: number[]; [key: string]: any } = {
151 |     icon: '/pwa-192x192.png',
152 |     badge: '/pwa-192x192.png',
153 |     vibrate: [250, 100, 250, 100, 250],
154 |     tag: rawOptions.tag || `tikblox-${Date.now()}`,
155 |     renotify: true,
156 |     data: {
157 |       url: productId ? `/?product=${encodeURIComponent(productId)}` : '/',
158 |       productId: productId || null,
159 |       timestamp: Date.now(),
160 |       ...rawOptions.data,
161 |     },
162 |     ...rawOptions,
163 |   };
164 |
165 |   try {
166 |     // Prioritize ServiceWorker registration for true PWA OS-drawer notifications
167 |     if (typeof window !== 'undefined' && 'serviceWorker' in navigator) {
168 |       const registration = await Promise.race([
169 |         navigator.serviceWorker.ready,
170 |         new Promise<null>((resolve) => setTimeout(() => resolve(null), 1500)),
171 |       ]);
172 |       if (registration && 'showNotification' in registration) {
173 |         await registration.showNotification(title, notificationOptions);
174 |         return true;
175 |       }
176 |     }
177 |
178 |     // Standard Desktop / Browser window fallback
179 |     if (typeof window !== 'undefined' && 'Notification' in window) {
180 |       new Notification(title, notificationOptions);
181 |       return true;
182 |     }
183 |     return false;
184 |   } catch (err) {
185 |     console.warn('Erro ao exibir notificação via ServiceWorker, tentando construtor padrão:', err);
186 |     try {
187 |       new Notification(title, notificationOptions);
188 |       return true;
189 |     } catch (fallbackErr) {
190 |       console.error('Falha ao disparar notificação:', fallbackErr);
191 |       return false;
192 |     }

```

```

193 |   }
194 | }
195 |
196 | /**
197 |  * Sends a viral product alert if it meets user notification filter settings.
198 |  */
199 | export async function notifyProductDetected(
200 |   product: TrendingProduct,
201 |   settings: PushNotificationSettings
202 | ): Promise<boolean> {
203 |   if (!settings.enabled || Notification.permission !== 'granted') {
204 |     return false;
205 |   }
206 |
207 |   // Filter 1: Check early wave requirement
208 |   if (settings.notifyOnEarlyWaveOnly && product.waveStage !== 'early_wave') {
209 |     return false;
210 |   }
211 |
212 |   // Filter 2: Check profit margin threshold
213 |   if (product.estimatedProfitMarginPercent < settings.minProfitMarginPercent) {
214 |     return false;
215 |   }
216 |
217 |   const originLabel = product.originCountry === 'US' ? 'EUA' : product.originCountry === 'CN' ? 'Chi...
218 |   const title = `ðŸ”ð [RADAR VIRAL] ${product.name}`;
219 |   const body = `Explodindo nos ${originLabel}! Margem de ${product.estimatedProfitMarginPercent}% e Score...
220 |
221 |   const success = await dispatchPushNotification(title, {
222 |     body,
223 |     productId: product.id,
224 |     playSound: settings.soundAndVibration,
225 |     tag: `tikblox-prod-${product.id}`,
226 |     data: {
227 |       productId: product.id,
228 |       margin: product.estimatedProfitMarginPercent,
229 |       virality: product.viralityScore,
230 |     },
231 |   });
232 |
233 |   if (success) {
234 |     logPushAlert({
235 |       productId: product.id,
236 |       productName: product.name,
237 |       viralityScore: product.viralityScore,
238 |       profitMarginPercent: product.estimatedProfitMarginPercent,
239 |       status: 'delivered',
240 |     });
241 |   }
242 |
243 |   return success;
244 | }
245 |
246 | /**
247 |  * Sends an instant welcome/confirmation test notification to verify OS setup.
248 |  */
249 | export async function sendTestNotification(): Promise<boolean> {
250 |   const title = '& Radar TIKBLOX: Notificações Push Ativas!';
251 |   const body = 'Seu PWA está configurado para monitorar tendências virais do exterior e avisar você em te...
252 |
253 |   return await dispatchPushNotification(title, {
254 |     body,
255 |     playSound: true,
256 |     tag: 'tikblox-test-notification',
257 |   });

```

```
258 | }  
259 |
```

src/hooks/usePushNotifications.ts

Hook React para Notificações (src/hooks/usePushNotifications.ts) (148 linhas)

```
1 | import { useState, useEffect, useCallback, useRef } from 'react';
2 | import {
3 |   PushNotificationSettings,
4 |   PushAlertLog,
5 |   TrendingProduct,
6 | } from '../types';
7 | import {
8 |   isNotificationSupported,
9 |   getNotificationPermission,
10 |   getStoredPushSettings,
11 |   savePushSettings,
12 |   requestNotificationPermission,
13 |   sendTestNotification,
14 |   notifyProductDetected,
15 |   getPushAlertHistory,
16 | } from '../services/pushNotifications';
17 |
18 | export function usePushNotifications(currentProducts?: TrendingProduct[]) {
19 |   const [isSupported, setIsSupported] = useState<boolean>(false);
20 |   const [permission, setPermission] = useState<NotificationPermission | 'unsupported'>('default');
21 |   const [settings, setSettings] = useState<PushNotificationSettings>(() => getStoredPushSettings());
22 |   const [history, setHistory] = useState<PushAlertLog[]>(() => getPushAlertHistory());
23 |   const notifiedProductIdsRef = useRef<Set<string>>(new Set());
24 |
25 |   // Initialize permission and support state
26 |   useEffect(() => {
27 |     const supported = isNotificationSupported();
28 |     setIsSupported(supported);
29 |     if (supported) {
30 |       setPermission(getNotificationPermission());
31 |     } else {
32 |       setPermission('unsupported');
33 |     }
34 |   }, []);
35 |
36 |   // Sync settings changes to localStorage
37 |   const updateSettings = useCallback((partial: Partial<PushNotificationSettings>) => {
38 |     setSettings((prev) => {
39 |       const updated = { ...prev, ...partial };
40 |       savePushSettings(updated);
41 |       return updated;
42 |     });
43 |   }, []);
44 |
45 |   // Toggle or Request Permission
46 |   const enableNotifications = useCallback(async (): Promise<boolean> => {
47 |     if (!isNotificationSupported()) {
48 |       return false;
49 |     }
50 |
51 |     const perm = await requestNotificationPermission();
52 |     setPermission(perm);
53 |
54 |     if (perm === 'granted') {
55 |       updateSettings({ enabled: true });
56 |       await sendTestNotification();
57 |       setHistory(getPushAlertHistory());
58 |       return true;
59 |     } else {
60 |       updateSettings({ enabled: false });
61 |       return false;
62 |     }
63 |   }, [updateSettings, setHistory, setPermission]);
```

```

63 |   }, [updateSettings]);
64 |
65 |   const disableNotifications = useCallback(() => {
66 |     updateSettings({ enabled: false });
67 |   }, [updateSettings]);
68 |
69 |   // Test notification trigger
70 |   const triggerTestAlert = useCallback(async (): Promise<boolean> => {
71 |     if (permission !== 'granted') {
72 |       const granted = await enableNotifications();
73 |       if (!granted) return false;
74 |     }
75 |     const sent = await sendTestNotification();
76 |     setHistory(getPushAlertHistory());
77 |     return sent;
78 |   }, [permission, enableNotifications]);
79 |
80 |   // Notify for a specific product
81 |   const notifyProduct = useCallback(
82 |     async (product: TrendingProduct): Promise<boolean> => {
83 |       if (!settings.enabled || permission !== 'granted') {
84 |         return false;
85 |       }
86 |       if (notifiedProductIdsRef.current.has(product.id)) {
87 |         return false;
88 |       }
89 |
90 |       const sent = await notifyProductDetected(product, settings);
91 |       if (sent) {
92 |         notifiedProductIdsRef.current.add(product.id);
93 |         setHistory(getPushAlertHistory());
94 |       }
95 |       return sent;
96 |     },
97 |     [settings, permission]
98 |   );
99 |
100 |   // Background Radar Simulator / Auto-watcher
101 |   // When user is in the app or app is running as PWA with notifications enabled,
102 |   // this simulates detecting an unexpected emergent viral product in background.
103 |   useEffect(() => {
104 |     if (!settings.enabled || !settings.backgroundRadarAlerts || permission !== 'granted' || !currentProdu...
105 |       return;
106 |   });
107 |
108 |   // Seed notified ids so we don't spam on initial boot
109 |   currentProducts.forEach((p) => {
110 |     if (!notifiedProductIdsRef.current.has(p.id)) {
111 |       // Reserve the highest margin products for alert opportunities
112 |     }
113 |   });
114 |
115 |   // Check periodically if an early-wave or high-growth item can be notified
116 |   const interval = setInterval(() => {
117 |     if (document.hidden && settings.backgroundRadarAlerts) {
118 |       // Find a standout product not yet notified
119 |       const candidate = currentProducts.find(
120 |         (p) =>
121 |           !notifiedProductIdsRef.current.has(p.id) &&
122 |           p.estimatedProfitMarginPercent >= settings.minProfitMarginPercent &&
123 |           (settings.notifyOnEarlyWaveOnly ? p.waveStage === 'early_wave' : true)
124 |       );
125 |
126 |       if (candidate) {
127 |         notifyProduct(candidate);

```

```
128 |     }
129 |   }
130 |   }, 45000); // 45 seconds check
131 |
132 |   return () => clearInterval(interval);
133 | }, [settings, permission, currentProducts, notifyProduct]);
134 |
135 | return {
136 |   isSupported,
137 |   permission,
138 |   settings,
139 |   updateSettings,
140 |   enableNotifications,
141 |   disableNotifications,
142 |   triggerTestAlert,
143 |   notifyProduct,
144 |   history,
145 |   refreshHistory: () => setHistory(getPushAlertHistory()),
146 | };
147 | }
148 |
```

src/hooks/usePWAInstall.ts

Hook de Instalação PWA (src/hooks/usePWAInstall.ts) (63 linhas)

```
1 | import { useEffect, useState } from 'react';
2 |
3 | interface BeforeInstallPromptEvent extends Event {
4 |   prompt: () => Promise<void>;
5 |   userChoice: Promise<{ outcome: 'accepted' | 'dismissed'; platform: string }>;
6 | }
7 |
8 | export function usePWAInstall() {
9 |   const [deferredPrompt, setDeferredPrompt] = useState<BeforeInstallPromptEvent | null>(null);
10 |   const [isInstalled, setIsInstalled] = useState(false);
11 |   const [isIOS, setIsIOS] = useState(false);
12 |
13 |   useEffect(() => {
14 |     // Detect standalone mode (already installed)
15 |     const isStandalone =
16 |       window.matchMedia('(display-mode: standalone)').matches ||
17 |       (window.navigator as unknown as { standalone?: boolean }).standalone === true;
18 |     setIsInstalled(isStandalone);
19 |
20 |     // Detect iOS devices
21 |     const userAgent = window.navigator.userAgent.toLowerCase();
22 |     const isIOSDevice = /iphone|ipad|ipod/.test(userAgent);
23 |     setIsIOS(isIOSDevice);
24 |
25 |     const handleBeforeInstallPrompt = (e: Event) => {
26 |       e.preventDefault();
27 |       setDeferredPrompt(e as BeforeInstallPromptEvent);
28 |     };
29 |
30 |     const handleAppInstalled = () => {
31 |       setIsInstalled(true);
32 |       setDeferredPrompt(null);
33 |     };
34 |
35 |     window.addEventListener('beforeinstallprompt', handleBeforeInstallPrompt);
36 |     window.addEventListener('appinstalled', handleAppInstalled);
37 |
38 |     return () => {
39 |       window.removeEventListener('beforeinstallprompt', handleBeforeInstallPrompt);
40 |       window.removeEventListener('appinstalled', handleAppInstalled);
41 |     };
42 |   }, []);
43 |
44 |   const install = async () => {
45 |     if (!deferredPrompt) return false;
46 |     await deferredPrompt.prompt();
47 |     const { outcome } = await deferredPrompt.userChoice;
48 |     if (outcome === 'accepted') {
49 |       setIsInstalled(true);
50 |       setDeferredPrompt(null);
51 |       return true;
52 |     }
53 |     return false;
54 |   };
55 |
56 |   return {
57 |     isInstallable: !!deferredPrompt,
58 |     isInstalled,
59 |     isIOS,
60 |     install,
61 |   };
62 | }
```


src/hooks/useOnlineStatus.ts

Hook de Detecção Online/Offline (src/hooks/useOnlineStatus.ts) (23 linhas)

```
1 | import { useEffect, useState } from 'react';
2 |
3 | export function useOnlineStatus() {
4 |   const [isOnline, setIsOnline] = useState(
5 |     typeof navigator !== 'undefined' ? navigator.onLine : true
6 |   );
7 |
8 |   useEffect(() => {
9 |     const handleOnline = () => setIsOnline(true);
10 |    const handleOffline = () => setIsOnline(false);
11 |
12 |    window.addEventListener('online', handleOnline);
13 |    window.addEventListener('offline', handleOffline);
14 |
15 |    return () => {
16 |      window.removeEventListener('online', handleOnline);
17 |      window.removeEventListener('offline', handleOffline);
18 |    };
19 |   }, []);
20 |
21 |   return isOnline;
22 | }
23 |
```

src/components/Navbar.tsx

Barra de Navegação e Controles (src/components/Navbar.tsx) (269 linhas)

```
1 | import React, { useState } from 'react';
2 | import { Bookmark, Calculator, RefreshCw, Layers, Download, Bell, BellRing, ShieldCheck } from 'lucide-re...
3 | import { PWAInstallButton } from './PWAInstallButton';
4 | import { TikBloxLogo } from './TikBloxLogo';
5 | import { BrandAssetsModal } from './BrandAssetsModal';
6 | import { FreeTierInfoModal } from './FreeTierInfoModal';
7 |
8 | interface NavbarProps {
9 |   activeTab: 'all' | 'early_wave' | 'high_margin' | 'saved' | 'calculator';
10 |   setActiveTab: (tab: 'all' | 'early_wave' | 'high_margin' | 'saved' | 'calculator') => void;
11 |   savedCount: number;
12 |   onTriggerScan: () => void;
13 |   isScanning: boolean;
14 |   onOpenPushModal?: () => void;
15 |   isPushActive?: boolean;
16 | }
17 |
18 | export const Navbar: React.FC<NavbarProps> = ({
19 |   activeTab,
20 |   setActiveTab,
21 |   savedCount,
22 |   onTriggerScan,
23 |   isScanning,
24 |   onOpenPushModal,
25 |   isPushActive = false,
26 | }) => {
27 |   const [showBrandAssetsModal, setShowBrandAssetsModal] = useState(false);
28 |   const [showFreeTierModal, setShowFreeTierModal] = useState(false);
29 |
30 |   return (
31 |     <header className="sticky top-0 z-40 w-full border-b border-white/10 bg-[#010101]/95 backdrop-blur-2x...
32 |       /* Subtle top edge chromatic glow line */
33 |       <div className="h-[1px] w-full bg-gradient-to-r from-transparent via-[#25F4EE]/60 to-[#FE2C55]/60" ...
34 |
35 |       <div className="mx-auto max-w-7xl px-4 sm:px-6 lg:px-8">
36 |         <div className="flex h-16 items-center justify-between gap-4">
37 |
38 |           /* Brand Logo & Live Radar Status */
39 |           <div className="flex items-center gap-3">
40 |             <button
41 |               onClick={() => setActiveTab('all')}
42 |               className="flex items-center text-left focus:outline-none group cursor-pointer"
43 |               title="Voltar ao início do TIKBLOX"
44 |             >
45 |               <TikBloxLogo size="md" showText={true} showBadge={true} animate={true} />
46 |             </button>
47 |           </div>
48 |
49 |           /* Center Navigation Tabs (Desktop) */
50 |           <nav className="hidden md:flex items-center gap-1 rounded-xl bg-[#12131A] p-1 border border-whi...
51 |             <button
52 |               id="nav-tab-all"
53 |               onClick={() => setActiveTab('all')}
54 |               className={`flex items-center gap-1.5 px-3 py-1.5 rounded-lg font-bold transition-all ${
55 |                 activeTab === 'all'
56 |                   ? 'bg-gradient-to-r from-[#FE2C55] to-[#FF0050] text-white shadow-md shadow-[#FE2C55]/3...'
57 |                   : 'text-[#A6A7B2] hover:text-white hover:bg-white/5'
58 |               }`}
59 |             >
60 |               <Layers className="w-3.5 h-3.5" />
61 |               <span>Todos os Produtos</span>
62 |             </button>
```

```

63 |
64 |         <button
65 |             id="nav-tab-early-wave"
66 |             onClick={() => setActiveTab('early_wave')}
67 |             className={`flex items-center gap-1.5 px-3 py-1.5 rounded-lg font-bold transition-all ${
68 |                 activeTab === 'early_wave'
69 |                     ? 'bg-emerald-500/20 text-emerald-300 border border-emerald-500/40 shadow-sm'
70 |                     : 'text-[#A6A7B2] hover:text-white hover:bg-white/5'
71 |             }`}
72 |         >
73 |             <span className="relative flex h-2 w-2">
74 |                 <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-emerald-500/50">
75 |                     <span className="relative inline-flex rounded-full h-2 w-2 bg-emerald-500"></span>
76 |                 </span>
77 |                 <span>🌊 Ondas Iniciais</span>
78 |             </button>
79 |
80 |         <button
81 |             id="nav-tab-high-margin"
82 |             onClick={() => setActiveTab('high_margin')}
83 |             className={`flex items-center gap-1.5 px-3 py-1.5 rounded-lg font-bold transition-all ${
84 |                 activeTab === 'high_margin'
85 |                     ? 'bg-amber-500/20 text-amber-300 border border-amber-500/40 shadow-sm'
86 |                     : 'text-[#A6A7B2] hover:text-white hover:bg-white/5'
87 |             }`}
88 |         >
89 |             <span>📏 Alta Margem (300%+)</span>
90 |         </button>
91 |
92 |         <button
93 |             id="nav-tab-calculator"
94 |             onClick={() => setActiveTab('calculator')}
95 |             className={`flex items-center gap-1.5 px-3 py-1.5 rounded-lg font-bold transition-all ${
96 |                 activeTab === 'calculator'
97 |                     ? 'bg-[#25F4EE]/20 text-[#25F4EE] border border-[#25F4EE]/40 shadow-sm'
98 |                     : 'text-[#A6A7B2] hover:text-white hover:bg-white/5'
99 |             }`}
100 |         >
101 |             <Calculator className="w-3.5 h-3.5" />
102 |             <span>Simulador</span>
103 |         </button>
104 |
105 |         <button
106 |             id="nav-tab-saved"
107 |             onClick={() => setActiveTab('saved')}
108 |             className={`flex items-center gap-1.5 px-3 py-1.5 rounded-lg font-bold transition-all ${
109 |                 activeTab === 'saved'
110 |                     ? 'bg-[#FE2C55]/20 text-[#FE2C55] border border-[#FE2C55]/40 shadow-sm'
111 |                     : 'text-[#A6A7B2] hover:text-white hover:bg-white/5'
112 |             }`}
113 |         >
114 |             <Bookmark className="w-3.5 h-3.5" />
115 |             <span>Radar Salvo</span>
116 |             {savedCount > 0 && (
117 |                 <span className="ml-1 rounded-full bg-[#FE2C55] px-1.5 py-0.2 text-[10px] font-extrabold ...
118 |                     {savedCount}
119 |                 </span>
120 |             )}
121 |         </button>
122 |     </nav>
123 |
124 |     {/* Right Action Controls: Scan Trigger & PWA Install */}
125 |     <div className="flex items-center gap-2.5">
126 |         <button
127 |             id="quick-scan-btn"

```

```

128 |         onClick={onTriggerScan}
129 |         disabled={isScanning}
130 |         className={`flex items-center gap-2 rounded-xl border border-[#25F4EE]/40 bg-[#161823] hove...
131 |             isScanning ? 'opacity-50 cursor-not-allowed' : ''
132 |         `}
133 |         title="Iniciar varredura em tempo real da internet"
134 |     >
135 |         <RefreshCw className={`w-3.5 h-3.5 text-[#25F4EE] ${isScanning ? 'animate-spin' : ''}`} />
136 |         <span className="hidden sm:inline">
137 |             {isScanning ? 'Varrendo...' : 'Escanear Web'}
138 |         </span>
139 |     </button>
140 |
141 |     <button
142 |         id="free-tier-info-btn"
143 |         onClick={() => setShowFreeTierModal(true)}
144 |         className="hidden lg:flex items-center gap-1.5 rounded-xl border border-emerald-500/30 bg-e...
145 |         title="API 100% Gratuita Permanente (R$ 0,00)"
146 |     >
147 |         <ShieldCheck className="w-3.5 h-3.5 text-emerald-400" />
148 |         <span>API R$ 0 Grátis</span>
149 |     </button>
150 |
151 |     <button
152 |         id="brand-assets-btn"
153 |         onClick={() => setShowBrandAssetsModal(true)}
154 |         className="flex items-center gap-1.5 rounded-xl border border-white/10 bg-[#121420] hover:b...
155 |         title="Baixar arquivos do Logo e Tipografia separados (SVG)"
156 |     >
157 |         <Download className="w-3.5 h-3.5 text-[#25F4EE]" />
158 |         <span className="hidden sm:inline">Baixar Logo</span>
159 |     </button>
160 |
161 |     {onOpenPushModal && (
162 |         <button
163 |             id="push-notifications-btn"
164 |             onClick={onOpenPushModal}
165 |             className={`relative flex items-center gap-1.5 rounded-xl border px-3 py-1.5 text-xs font...
166 |                 isPushActive
167 |                 ? 'border-emerald-500/40 bg-emerald-950/25 text-emerald-300 hover:border-emerald-400 ...
168 |                 : 'border-white/10 bg-[#121420] text-[#8E91A6] hover:border-[#FE2C55]/40 hover:text-w...
169 |             `}
170 |             title="Configurar Notificações Push do Radar PWA"
171 |         >
172 |             <div className="relative flex items-center justify-center">
173 |                 {isPushActive ? (
174 |                     <>
175 |                         <BellRing className="w-3.5 h-3.5 text-emerald-400" />
176 |                         <span className="absolute -top-1 -right-1 flex h-2 w-2">
177 |                             <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-...
178 |                             <span className="relative inline-flex rounded-full h-2 w-2 bg-emerald-400"></span>
179 |                         </span>
180 |                     </>
181 |                 ) : (
182 |                     <>
183 |                         <Bell className="w-3.5 h-3.5 text-[#FE2C55]" />
184 |                         <span className="absolute -top-0.5 -right-0.5 h-1.5 w-1.5 rounded-full bg-[#FE2C55]...
185 |                     </>
186 |                 )}
187 |             </div>
188 |             <span className="hidden sm:inline">
189 |                 {isPushActive ? 'Radar Push' : 'Push'}
190 |             </span>
191 |         </button>
192 |     )}

```

```

193 |
194 |         <PWAInstallButton />
195 |     </div>
196 | </div>
197 |
198 | {/* Mobile Navigation Row */}
199 | <div className="flex md:hidden items-center gap-1 overflow-x-auto py-2.5 scrollbar-none border-t ...
200 |     <button
201 |         onClick={() => setActiveTab('all')}
202 |         className={`px-3 py-1 rounded-lg shrink-0 font-bold transition ${
203 |             activeTab === 'all'
204 |                 ? 'bg-[#FE2C55] text-white shadow-md shadow-[#FE2C55]/30'
205 |                 : 'text-[#A6A7B2] bg-[#12131A]'
206 |             }`>
207 |         >
208 |             Todos
209 |         </button>
210 |         <button
211 |             onClick={() => setActiveTab('early_wave')}
212 |             className={`px-3 py-1 rounded-lg shrink-0 font-bold transition ${
213 |                 activeTab === 'early_wave'
214 |                     ? 'bg-emerald-500/20 text-emerald-300 border border-emerald-500/40'
215 |                     : 'text-[#A6A7B2] bg-[#12131A]'
216 |                 }`>
217 |             >
218 |                 ⚡ Ondas
219 |             </button>
220 |             <button
221 |                 onClick={() => setActiveTab('high_margin')}
222 |                 className={`px-3 py-1 rounded-lg shrink-0 font-bold transition ${
223 |                     activeTab === 'high_margin'
224 |                         ? 'bg-amber-500/20 text-amber-300 border border-amber-500/40'
225 |                         : 'text-[#A6A7B2] bg-[#12131A]'
226 |                     }`>
227 |                 >
228 |                     &i Alta Margem
229 |                 </button>
230 |                 <button
231 |                     onClick={() => setActiveTab('calculator')}
232 |                     className={`px-3 py-1 rounded-lg shrink-0 font-bold transition ${
233 |                         activeTab === 'calculator'
234 |                             ? 'bg-[#25F4EE]/20 text-[#25F4EE] border border-[#25F4EE]/40'
235 |                             : 'text-[#A6A7B2] bg-[#12131A]'
236 |                         }`>
237 |                     >
238 |                         Simulador
239 |                     </button>
240 |                     <button
241 |                         onClick={() => setActiveTab('saved')}
242 |                         className={`px-3 py-1 rounded-lg shrink-0 font-bold transition ${
243 |                             activeTab === 'saved'
244 |                                 ? 'bg-[#FE2C55]/20 text-[#FE2C55] border border-[#FE2C55]/40'
245 |                                 : 'text-[#A6A7B2] bg-[#12131A]'
246 |                             }`>
247 |                         >
248 |                             Salvos ({savedCount})
249 |                         </button>
250 |                     </div>
251 |
252 | </div>
253 |
254 | {/* Brand Assets Export Modal */}
255 | <BrandAssetsModal
256 |     isOpen={showBrandAssetsModal}
257 |     onClose={() => setShowBrandAssetsModal(false)}

```

```
258 |         />
259 |
260 |         { /* Free Tier Info Modal */ }
261 |         <FreeTierInfoModal
262 |           isOpen={showFreeTierModal}
263 |           onClose={() => setShowFreeTierModal(false)}
264 |         />
265 |       </header>
266 |     );
267 |   };
268 |
269 |
```

src/components/RadarHero.tsx

Painel Superior de Filtros e Busca (src/components/RadarHero.tsx) (178 linhas)

```
1 | import React from 'react';
2 | import { Search, Sparkles, Compass, Globe, Filter } from 'lucide-react';
3 | import { ProductNiche } from '../types';
4 |
5 | interface RadarHeroProps {
6 |   selectedNiche: ProductNiche;
7 |   setSelectedNiche: (niche: ProductNiche) => void;
8 |   selectedOrigin: string;
9 |   setSelectedOrigin: (origin: string) => void;
10 |   searchQuery: string;
11 |   setSearchQuery: (query: string) => void;
12 |   onTriggerScan: () => void;
13 |   isScanning: boolean;
14 |   totalProductsCount: number;
15 | }
16 |
17 | const NICHES: Array<{ id: ProductNiche; label: string; icon: string }> = [
18 |   { id: 'all', label: 'Todos os Nichos', icon: '🏠' },
19 |   { id: 'tech', label: 'Gadgets & Tech', icon: '&i' },
20 |   { id: 'home', label: 'Casa & Cozinha', icon: '🏠' },
21 |   { id: 'beauty', label: 'Beleza & Estética', icon: '💄' },
22 |   { id: 'fitness', label: 'Fitness & Treino', icon: '🏋️' },
23 |   { id: 'pets', label: 'Pet Inovador', icon: '🐾' },
24 |   { id: 'accessories', label: 'Moda & Acessórios', icon: '👗' },
25 |   { id: 'kids', label: 'Infantil & Bebês', icon: '👶' },
26 |   { id: 'tools', label: 'Ferramentas & Obra', icon: '🔧' },
27 |   { id: 'auto', label: 'Automotivo & Carros', icon: '🚗' },
28 |   { id: 'health', label: 'Saúde & Bem-Estar', icon: '🩺' },
29 | ];
30 |
31 | export const RadarHero: React.FC<RadarHeroProps> = ({
32 |   selectedNiche,
33 |   setSelectedNiche,
34 |   selectedOrigin,
35 |   setSelectedOrigin,
36 |   searchQuery,
37 |   setSearchQuery,
38 |   onTriggerScan,
39 |   isScanning,
40 |   totalProductsCount,
41 | }) => {
42 |   return (
43 |     <section className="relative overflow-hidden border-b border-white/10 bg-[#010101] px-4 py-8 sm:py-12...
44 |       /* TikTok Dual Ambient Glow: Cyan on top-left, Pink on top-right */
45 |       <div className="absolute top-0 left-1/4 -translate-y-1/2 -translate-x-1/2 w-[550px] h-[350px] bg-[#...
46 |       <div className="absolute top-0 right-1/4 -translate-y-1/2 translate-x-1/2 w-[550px] h-[350px] bg-[#...
47 |
48 |       /* Background radar concentric rings */
49 |       <div className="absolute top-1/2 left-1/2 -translate-x-1/2 -translate-y-1/2 pointer-events-none opa...
50 |         <div className="h-[480px] w-[480px] rounded-full border border-[#25F4EE]/30 animate-[ping_8s_line...
51 |           <div className="absolute inset-0 m-auto h-[320px] w-[320px] rounded-full border border-[#FE2C55]/...
52 |           <div className="absolute inset-0 m-auto h-[170px] w-[170px] rounded-full border border-white/20" ...
53 |         </div>
54 |
55 |       <div className="relative mx-auto max-w-7xl">
56 |
57 |         /* Top Badges & Live Status */
58 |         <div className="flex flex-wrap items-center justify-between gap-3 mb-6">
59 |           <div className="inline-flex items-center gap-2 rounded-full bg-[#161823] border border-[#25F4EE]...
60 |             <span className="relative flex h-2 w-2">
61 |               <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-[#FE2C55] ...
62 |               <span className="relative inline-flex rounded-full h-2 w-2 bg-[#FE2C55]"></span>
```

```

63 |         </span>
64 |         <span>Varredura Contínua: TikTok Shop US + Douyin China + Amazon Movers</span>
65 |     </div>
66 |
67 |     <div className="flex items-center gap-4 text-xs text-[#A6A7B2]">
68 |         <div className="flex items-center gap-1.5">
69 |             <span className="text-white font-extrabold">{totalProductsCount}</span>
70 |             <span>produtos no radar</span>
71 |         </div>
72 |         <span className="text-white/20">|</span>
73 |         <div className="flex items-center gap-1.5">
74 |             <span className="text-[#25F4EE] font-extrabold">300%+</span>
75 |             <span>margem média</span>
76 |         </div>
77 |     </div>
78 | </div>
79 |
80 | {/* Title and Value Proposition */}
81 | <div className="max-w-3xl mb-8">
82 |     <h1 className="text-2xl sm:text-4xl lg:text-5xl font-black tracking-tight text-white leading-ti...
83 |         Descubra produtos virais no exterior{' '}
84 |         <span className="bg-gradient-to-r from-white via-[#25F4EE] to-[#FE2C55] bg-clip-text text-tra...
85 |         antes de estourarem no Brasil.
86 |     </span>
87 | </h1>
88 |     <p className="mt-3 text-sm sm:text-base text-[#C5C6D0] font-normal leading-relaxed">
89 |         O TIKBLOX escaneia a internet em tempo real identificando produtos com tração explosiva nos E...
90 |     </p>
91 | </div>
92 |
93 | {/* Action & Filter Console */}
94 | <div className="rounded-2xl border border-white/10 bg-[#12131A]/90 backdrop-blur-xl p-4 sm:p-5 sh...
95 |     <div className="flex flex-col lg:flex-row items-stretch lg:items-center gap-3">
96 |
97 |         {/* Search Input */}
98 |         <div className="relative flex-1">
99 |             <Search className="absolute left-3.5 top-1/2 -translate-y-1/2 h-4 w-4 text-[#A6A7B2]" />
100 |             <input
101 |                 id="search-products-input"
102 |                 type="text"
103 |                 placeholder="Buscar por nome do produto, dor resolvida ou público (ex: projetor, postura,...
104 |                 value={searchQuery}
105 |                 onChange={(e) => setSearchQuery(e.target.value)}
106 |                 className="w-full rounded-xl border border-white/10 bg-[#0B0C10] py-2.5 pl-10 pr-4 text-x...
107 |             />
108 |             {searchQuery && (
109 |                 <button
110 |                     onClick={() => setSearchQuery('')}
111 |                     className="absolute right-3 top-1/2 -translate-y-1/2 text-xs text-[#A6A7B2] hover:text-...
112 |                 >
113 |                     Limpar
114 |                 </button>
115 |             )}
116 |         </div>
117 |
118 |         {/* Origin Country Selector */}
119 |         <div className="flex items-center gap-2">
120 |             <div className="relative">
121 |                 <select
122 |                     id="origin-country-select"
123 |                     value={selectedOrigin}
124 |                     onChange={(e) => setSelectedOrigin(e.target.value)}
125 |                     aria-label="Filtrar por país ou plataforma de origem"
126 |                     className="rounded-xl border border-white/10 bg-[#0B0C10] px-3 py-2.5 text-xs text-whit...
127 |                 >

```

```

128 |         <option value="all">ð<ß Todas as Origens</option>
129 |         <option value="US">ð<Ýúð<Ýø EUA (TikTok Shop / Amazon)</option>
130 |         <option value="CN">ð<Ýèð<Ýó China (Douyin / 1688 / AliExpress)</option>
131 |     </select>
132 | </div>
133 |
134 |     { /* Big Scan Button - TikTok Primary Glow */ }
135 |     <button
136 |       id="hero-trigger-scan-btn"
137 |       onClick={onTriggerScan}
138 |       disabled={isScanning}
139 |       className={`flex items-center justify-center gap-2 rounded-xl bg-gradient-to-r from-[#FE2...
140 |         isScanning ? 'opacity-60 cursor-not-allowed' : ''
141 |       }`}
142 |     >
143 |       <Sparkles className={`w-4 h-4 ${isScanning ? 'animate-spin' : ''}`} />
144 |       <span>{isScanning ? 'Varrendo a Web com IA...' : 'Disparar Varredura Global'}</span>
145 |     </button>
146 | </div>
147 |
148 | </div>
149 |
150 | { /* Niches Carousel / Pills */ }
151 | <div className="mt-4 pt-3 border-t border-white/10 flex items-center gap-2 overflow-x-auto scro...
152 |   <span className="text-[11px] font-bold text-[#A6A7B2] shrink-0 flex items-center gap-1">
153 |     <Filter className="w-3 h-3 text-[#25F4EE]" />
154 |     <span>Nicho:</span>
155 |   </span>
156 |   {NICHES.map((niche) => (
157 |     <button
158 |       key={niche.id}
159 |       onClick={() => setSelectedNiche(niche.id)}
160 |       className={`flex items-center gap-1.5 rounded-lg px-3 py-1 text-xs font-bold shrink-0 tra...
161 |         selectedNiche === niche.id
162 |           ? 'bg-gradient-to-r from-[#FE2C55] to-[#FF0050] text-white shadow-md shadow-[#FE2C55]...'
163 |           : 'bg-[#161823] text-[#A6A7B2] hover:bg-[#202230] hover:text-white border border-whit...
164 |       }`}
165 |     >
166 |       <span>{niche.icon}</span>
167 |       <span>{niche.label}</span>
168 |     </button>
169 |   ))}
170 | </div>
171 |
172 | </div>
173 |
174 | </div>
175 | </section>
176 | );
177 | };
178 |

```

src/components/ProductCard.tsx

Card de Produto Viral e Métricas (src/components/ProductCard.tsx) (259 linhas)

```
1 | import React from 'react';
2 | import {
3 |   TrendingUp,
4 |   DollarSign,
5 |   Globe2,
6 |   Bookmark,
7 |   BookmarkCheck,
8 |   Zap,
9 |   Activity,
10 |   Tv,
11 |   Wind,
12 |   ShieldCheck,
13 |   Sparkles,
14 |   Scissors,
15 |   Calculator,
16 |   Video,
17 |   ArrowRight,
18 | } from 'lucide-react';
19 | import { TrendingProduct } from '../types';
20 |
21 | interface ProductCardProps {
22 |   product: TrendingProduct;
23 |   isSaved: boolean;
24 |   onToggleSave: (id: string) => void;
25 |   onOpenDeepDive: (product: TrendingProduct) => void;
26 |   onOpenCalculator: (product: TrendingProduct) => void;
27 |   onOpenCreativeGenerator: (product: TrendingProduct) => void;
28 | }
29 |
30 | export const ProductCard: React.FC<ProductCardProps> = ({
31 |   product,
32 |   isSaved,
33 |   onToggleSave,
34 |   onOpenDeepDive,
35 |   onOpenCalculator,
36 |   onOpenCreativeGenerator,
37 | }) => {
38 |   // Render icon based on iconType
39 |   const renderProductIcon = () => {
40 |     const iconClass = 'w-6 h-6 text-[#25F4EE] group-hover:text-[#FE2C55] transition-colors';
41 |     switch (product.iconType) {
42 |       case 'Wind':
43 |         return <Wind className={iconClass} />;
44 |       case 'Tv':
45 |         return <Tv className={iconClass} />;
46 |       case 'ShieldCheck':
47 |         return <ShieldCheck className={iconClass} />;
48 |       case 'Scissors':
49 |         return <Scissors className={iconClass} />;
50 |       case 'Activity':
51 |         return <Activity className={iconClass} />;
52 |       case 'Sparkles':
53 |         return <Sparkles className={iconClass} />;
54 |       default:
55 |         return <Zap className={iconClass} />;
56 |     }
57 |   };
58 |
59 |   // Wave status styling
60 |   const getWaveBadge = () => {
61 |     switch (product.waveStage) {
62 |       case 'early_wave':
```

```

63 |         return (
64 |             <span className="inline-flex items-center gap-1.5 rounded-full bg-emerald-500/15 border border-...
65 |                 <span className="relative flex h-1.5 w-1.5">
66 |                     <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-emerald-40...
67 |                     <span className="relative inline-flex rounded-full h-1.5 w-1.5 bg-emerald-500"></span>
68 |                 </span>
69 |                 <span>Onda Inicial (Oceano Azul)</span>
70 |             </span>
71 |         );
72 |     case 'rising_wave':
73 |         return (
74 |             <span className="inline-flex items-center gap-1.5 rounded-full bg-[#25F4EE]/15 border border-[#...
75 |                 <span>&#10138; Onda Crescente (Alta Tração)</span>
76 |             </span>
77 |         );
78 |     case 'peak_wave':
79 |     default:
80 |         return (
81 |             <span className="inline-flex items-center gap-1.5 rounded-full bg-[#FE2C55]/15 border border-[#...
82 |                 <span>&#10138; Ápice de Demanda (Volume Alto)</span>
83 |             </span>
84 |         );
85 |     }
86 | };
87 |
88 | // Flag emoji based on country
89 | const getCountryFlag = (country: string) => {
90 |     switch (country) {
91 |         case 'US':
92 |             return '&#10138;&#10138; EUA';
93 |         case 'CN':
94 |             return '&#10138;&#10138; China';
95 |         case 'KR':
96 |             return '&#10138;&#10138; Coreia';
97 |         case 'JP':
98 |             return '&#10138;&#10138; Japão';
99 |         default:
100 |             return '&#10138;&#10138; Global';
101 |     }
102 | };
103 |
104 | return (
105 |     <div
106 |         id={`product-card-${product.id}`}
107 |         className="group relative flex flex-col justify-between rounded-2xl border border-white/10 bg-[#121...
108 |     >
109 |         <div>
110 |             {/* Top Header: Wave Badge, Score & Bookmark */}
111 |             <div className="flex items-center justify-between gap-2 mb-3">
112 |                 <div className="flex flex-wrap items-center gap-2">
113 |                     {getWaveBadge()}
114 |                     <span className="text-[11px] font-medium text-[#A6A7B2]">
115 |                         {getCountryFlag(product.originCountry)}
116 |                     </span>
117 |                 </div>
118 |
119 |                 <div className="flex items-center gap-2">
120 |                     {/* Virality Score Gauge */}
121 |                     <div
122 |                         className="flex items-center gap-1 rounded-lg bg-[#161823] border border-white/10 px-2 py-0...
123 |                         title="Pontuação de Viralidade TIKBLOX (0-100)"
124 |                     >
125 |                         <TrendingUp className="w-3 h-3 text-[#FE2C55]" />
126 |                         <span>{product.viralityScore}</span>
127 |                     </div>

```

```

128 |
129 |     {/* Bookmark button */}
130 |     <button
131 |       onClick={() => onToggleSave(product.id)}
132 |       className={`p-1.5 rounded-lg border transition ${
133 |         isSaved
134 |           ? 'bg-[#FE2C55]/20 border-[#FE2C55]/50 text-[#FE2C55]'
135 |           : 'bg-[#161823] border-white/10 text-[#A6A7B2] hover:text-white hover:border-white/20'
136 |       }`}
137 |       title={isSaved ? 'Remover do Radar Salvo' : 'Salvar no meu Radar'}
138 |     >
139 |       {isSaved ? (
140 |         <BookmarkCheck className="w-4 h-4 text-[#FE2C55]" />
141 |       ) : (
142 |         <Bookmark className="w-4 h-4" />
143 |       )}
144 |     </button>
145 |   </div>
146 | </div>
147 |
148 |   {/* Product Title & Original Reference */}
149 |   <div className="flex items-start gap-3 mb-3">
150 |     <div className="flex h-11 w-11 shrink-0 items-center justify-center rounded-xl bg-[#161823] bor...
151 |       {renderProductIcon()}
152 |     </div>
153 |     <div className="flex-1 min-w-0">
154 |       <h3 className="text-base font-bold text-white leading-snug group-hover:text-[#25F4EE] transit...
155 |         {product.name}
156 |       </h3>
157 |       <p className="text-xs text-[#A6A7B2] italic truncate mt-0.5">
158 |         Origem: {product.originalName}
159 |       </p>
160 |     </div>
161 |   </div>
162 |
163 |   {/* Financial Arbitrage Box (Cost USD vs Price BRL & Margin) */}
164 |   <div className="rounded-xl border border-white/10 bg-[#0B0C10] p-3 mb-3.5">
165 |     <div className="grid grid-cols-3 gap-2 text-center">
166 |       <div>
167 |         <span className="block text-[10px] text-[#A6A7B2] uppercase font-semibold">
168 |           Custo Exterior
169 |         </span>
170 |         <span className="text-xs sm:text-sm font-bold text-[#E1E2EC]">
171 |           ${product.estimatedCostUSD.toFixed(2)} USD
172 |         </span>
173 |       </div>
174 |       <div>
175 |         <span className="block text-[10px] text-[#A6A7B2] uppercase font-semibold">
176 |           Preço no Brasil
177 |         </span>
178 |         <span className="text-xs sm:text-sm font-extrabold text-white">
179 |           R$ {product.estimatedPriceBRL.toFixed(2)}
180 |         </span>
181 |       </div>
182 |       <div>
183 |         <span className="block text-[10px] text-emerald-400 uppercase font-semibold">
184 |           Margem Estimada
185 |         </span>
186 |         <span className="text-xs sm:text-sm font-black text-emerald-400">
187 |           +{product.estimatedProfitMarginPercent}%
188 |         </span>
189 |       </div>
190 |     </div>
191 |   </div>
192 |

```

```

193 |     { /* Market Validation: Cultural Fit in Brazil */ }
194 |     <div className="mb-3.5">
195 |       <span className="text-[11px] font-bold text-[#A6A7B2] uppercase tracking-wider block mb-1">
196 |         Por que vende muito no Brasil:
197 |       </span>
198 |       <p className="text-xs text-[#C5C6D0] line-clamp-2 leading-relaxed bg-[#161823] rounded-lg p-2.5...">
199 |         {product.culturalFitReason}
200 |       </p>
201 |     </div>
202 |
203 |     { /* Saturation in Brazil & Platform Origin */ }
204 |     <div className="flex flex-wrap items-center justify-between gap-2 text-[11px] text-[#A6A7B2] mb-4...">
205 |       <div className="flex items-center gap-1.5">
206 |         <Globe2 className="w-3.5 h-3.5 text-[#25F4EE]" />
207 |         <span className="text-white font-medium">{product.originPlatform}</span>
208 |       </div>
209 |       <div className="flex items-center gap-1.5">
210 |         <span>Saturação BR:</span>
211 |         <span
212 |           className={`font-bold ${
213 |             product.saturationInBrazil === 'Muito Baixa'
214 |               ? 'text-emerald-400'
215 |               : product.saturationInBrazil === 'Baixa'
216 |               ? 'text-[#25F4EE]'
217 |               : 'text-amber-400'
218 |             }`}
219 |         >
220 |           {product.saturationInBrazil}
221 |         </span>
222 |       </div>
223 |     </div>
224 |   </div>
225 |
226 |   { /* Action Buttons Row */ }
227 |   <div className="space-y-2">
228 |     <button
229 |       onClick={() => onOpenDeepDive(product)}
230 |       className="w-full flex items-center justify-center gap-2 rounded-xl bg-gradient-to-r from-[#FE2...">
231 |       >
232 |       <span>Ver Raio-X Completo da Onda</span>
233 |       <ArrowRight className="w-3.5 h-3.5" />
234 |     </button>
235 |
236 |     <div className="grid grid-cols-2 gap-2">
237 |       <button
238 |         onClick={() => onOpenCreativeGenerator(product)}
239 |         className="flex items-center justify-center gap-1.5 rounded-lg border border-white/10 bg-[#16...">
240 |         title="Gerar scripts de vídeo para TikTok e Reels"
241 |       >
242 |         <Video className="w-3 h-3 text-[#25F4EE]" />
243 |         <span>Criativos IA</span>
244 |       </button>
245 |
246 |       <button
247 |         onClick={() => onOpenCalculator(product)}
248 |         className="flex items-center justify-center gap-1.5 rounded-lg border border-white/10 bg-[#16...">
249 |         title="Simular impostos e margem líquida real"
250 |       >
251 |         <Calculator className="w-3 h-3 text-emerald-400" />
252 |         <span>Simular Lucro</span>
253 |       </button>
254 |     </div>
255 |   </div>
256 | </div>
257 | );

```

```
258 | };  
259 |
```

src/components/ProductDetailModal.tsx

Modal de Raio-X e Roteiros de Vídeo (src/components/ProductDetailModal.tsx) (404 linhas)

```
1 | import React, { useEffect, useState } from 'react';
2 | import {
3 |   X,
4 |   TrendingUp,
5 |   ExternalLink,
6 |   Copy,
7 |   Check,
8 |   ShieldCheck,
9 |   Package,
10 | Calculator,
11 | Video,
12 | Layers,
13 | Sparkles,
14 | ShoppingBag,
15 | } from 'lucide-react';
16 | import { TrendingProduct, DeepDiveAnalysis } from '../types';
17 | import { getProductDeepDive } from '../services/api';
18 |
19 | interface ProductDetailModalProps {
20 |   product: TrendingProduct | null;
21 |   onClose: () => void;
22 |   onOpenCalculator: (product: TrendingProduct) => void;
23 |   onOpenCreativeGenerator: (product: TrendingProduct) => void;
24 | }
25 |
26 | export const ProductDetailModal: React.FC<ProductDetailModalProps> = ({
27 |   product,
28 |   onClose,
29 |   onOpenCalculator,
30 |   onOpenCreativeGenerator,
31 | }) => {
32 |   const [analysis, setAnalysis] = useState<DeepDiveAnalysis | null>(null);
33 |   const [isLoading, setIsLoading] = useState(false);
34 |   const [copiedIndex, setCopiedIndex] = useState<number | null>(null);
35 |
36 |   useEffect(() => {
37 |     if (!product) return;
38 |     let isMounted = true;
39 |     setIsLoading(true);
40 |
41 |     getProductDeepDive(product).then((data) => {
42 |       if (isMounted) {
43 |         setAnalysis(data);
44 |         setIsLoading(false);
45 |       }
46 |     });
47 |
48 |     return () => {
49 |       isMounted = false;
50 |     };
51 |   }, [product]);
52 |
53 |   if (!product) return null;
54 |
55 |   const handleCopyHook = (text: string, index: number) => {
56 |     navigator.clipboard.writeText(text);
57 |     setCopiedIndex(index);
58 |     setTimeout(() => setCopiedIndex(null), 2000);
59 |   };
60 |
61 |   // Verification & supplier search URLs
62 |   const aliexpressSearchUrl = `https://www.aliexpress.com/wholesale?SearchText=${encodeURIComponent(
```

src/components/ProductDetailModal.tsx (continuação)

```
63 |     product.supplierKeywords[0] || product.originalName
64 |   )}`;
65 |   const googleTrendsUrl = `https://trends.google.com/trends/explore?geo=BR&q=${encodeURIComponent(
66 |     product.name.split(' ').slice(0, 3).join(' ')
67 |   )}`;
68 |   const cjDropshippingUrl = `https://cjdropsipping.com/search/${encodeURIComponent(
69 |     product.supplierKeywords[0] || product.originalName
70 |   )}.html`;
71 |   const mercadoLivreUrl = `https://lista.mercadolivre.com.br/${encodeURIComponent(
72 |     product.name.split(' ').slice(0, 3).join(' ')
73 |   )}`;
74 |   const shopeeBrUrl = `https://shopee.com.br/search?keyword=${encodeURIComponent(
75 |     product.name.split(' ').slice(0, 3).join(' ')
76 |   )}`;
77 |   const tiktokViralUrl = `https://www.tiktok.com/search?q=${encodeURIComponent(
78 |     product.originalName
79 |   )}`;
80 |
81 |   return (
82 |     <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/80 backdrop-blur-md p-3 ...
83 |       <div className="relative w-full max-w-3xl rounded-2xl border border-slate-800 bg-slate-900 shadow-2...
84 |
85 |         {/* Modal Header */}
86 |         <div className="flex items-center justify-between border-b border-slate-800 bg-slate-950/60 p-4 s...
87 |           <div>
88 |             <div className="flex items-center gap-2 mb-1">
89 |               <span className="rounded-full bg-cyan-500/20 border border-cyan-500/30 px-2.5 py-0.5 text-[...
90 |                 Raio-X de Inteligência TIKBLOX
91 |               </span>
92 |               <span className="text-xs text-slate-400 font-medium">
93 |                 {product.nicheLabel}
94 |               </span>
95 |             </div>
96 |             <h2 className="text-lg sm:text-xl font-black text-white leading-tight">
97 |               {product.name}
98 |             </h2>
99 |             <p className="text-xs text-slate-400 italic">
100 |               Nome de busca internacional: {product.originalName}
101 |             </p>
102 |           </div>
103 |
104 |           <button
105 |             onClick={onClose}
106 |             className="rounded-xl p-2 text-slate-400 hover:text-white hover:bg-slate-800 transition"
107 |             title="Fechar"
108 |           >
109 |             <X className="w-5 h-5" />
110 |           </button>
111 |         </div>
112 |
113 |         {/* Modal Scrollable Body */}
114 |         <div className="flex-1 overflow-y-auto p-4 sm:p-6 space-y-6">
115 |
116 |           {/* Arbitrage Snapshot Grid */}
117 |           <div className="grid grid-cols-2 sm:grid-cols-4 gap-3">
118 |             <div className="rounded-xl border border-slate-800 bg-slate-950/70 p-3">
119 |               <span className="text-[11px] text-slate-400 block mb-0.5">Custo Fornecedor</span>
120 |               <span className="text-base font-bold text-white">${product.estimatedCostUSD.toFixed(2)} USD...
121 |               <span className="text-[10px] text-slate-500 block">H R$ {(product.estimatedCostUSD * 5.6).t...
122 |             </div>
123 |
124 |             <div className="rounded-xl border border-slate-800 bg-slate-950/70 p-3">
125 |               <span className="text-[11px] text-slate-400 block mb-0.5">Preço Médio no BR</span>
126 |               <span className="text-base font-bold text-cyan-300">R$ {product.estimatedPriceBRL.toFixed(2...
127 |               <span className="text-[10px] text-slate-500 block">Preço de saída validado</span>
```

```

128 |         </div>
129 |
130 |         <div className="rounded-xl border border-emerald-500/30 bg-emerald-950/20 p-3">
131 |             <span className="text-[11px] text-emerald-300 block mb-0.5">Margem Bruta</span>
132 |             <span className="text-base font-black text-emerald-400">+{product.estimatedProfitMarginPerc...
133 |             <span className="text-[10px] text-emerald-400/80 block">Lucro bruto expressivo</span>
134 |         </div>
135 |
136 |         <div className="rounded-xl border border-slate-800 bg-slate-950/70 p-3">
137 |             <span className="text-[11px] text-slate-400 block mb-0.5">Score de Viralidade</span>
138 |             <div className="flex items-center gap-1.5">
139 |                 <TrendingUp className="w-4 h-4 text-cyan-400" />
140 |                 <span className="text-base font-bold text-white">{product.viralityScore}/100</span>
141 |             </div>
142 |             <span className="text-[10px] text-cyan-400 block">{product.trendingMetrics.searchVolumeBR}<...
143 |         </div>
144 |     </div>
145 |
146 |     {/ * Diagnosis on Brazilian Competition */}
147 |     <div className="rounded-xl border border-slate-800 bg-slate-950/50 p-4">
148 |         <h3 className="text-sm font-bold text-white flex items-center gap-2 mb-3">
149 |             <ShoppingBag className="w-4 h-4 text-cyan-400" />
150 |             <span>Diagnóstico de Mercado no Brasil (Oceano Azul)</span>
151 |         </h3>
152 |
153 |         <div className="grid grid-cols-1 sm:grid-cols-3 gap-3 text-xs">
154 |             <div className="p-3 rounded-lg bg-slate-900/80 border border-slate-800 flex flex-col justif...
155 |                 <div>
156 |                     <div className="flex items-center justify-between mb-1">
157 |                         <span className="text-[11px] font-bold text-amber-400">Shopee Brasil</span>
158 |                         <a
159 |                             href={shopeeBrUrl}
160 |                             target="_blank"
161 |                             rel="noopener noreferrer"
162 |                             className="text-[10px] text-amber-400 hover:text-amber-300 underline flex items-cen...
163 |                             title="Checar concorrentes na Shopee Brasil agora"
164 |                         >
165 |                             <span>Checar</span>
166 |                             <ExternalLink className="w-2.5 h-2.5" />
167 |                         </a>
168 |                     </div>
169 |                     <p className="text-slate-300 text-[11px]">
170 |                         {analysis?.competitionAnalysis.shopeeStatus || 'Poucos anúncios internacionais com pr...
171 |                     </p>
172 |                 </div>
173 |             </div>
174 |
175 |             <div className="p-3 rounded-lg bg-slate-900/80 border border-slate-800 flex flex-col justif...
176 |                 <div>
177 |                     <div className="flex items-center justify-between mb-1">
178 |                         <span className="text-[11px] font-bold text-yellow-400">Mercado Livre Full</span>
179 |                         <a
180 |                             href={mercadoLivreUrl}
181 |                             target="_blank"
182 |                             rel="noopener noreferrer"
183 |                             className="text-[10px] text-yellow-400 hover:text-yellow-300 underline flex items-c...
184 |                             title="Checar concorrentes no Mercado Livre agora"
185 |                         >
186 |                             <span>Checar</span>
187 |                             <ExternalLink className="w-2.5 h-2.5" />
188 |                         </a>
189 |                     </div>
190 |                     <p className="text-slate-300 text-[11px]">
191 |                         {analysis?.competitionAnalysis.mercadoLivreStatus || 'Raros vendedores com estoque na...
192 |                     </p>

```

```

193 |     </div>
194 |   </div>
195 |
196 |   <div className="p-3 rounded-lg bg-slate-900/80 border border-slate-800 flex flex-col justif...
197 |     <div>
198 |       <div className="flex items-center justify-between mb-1">
199 |         <span className="text-[11px] font-bold text-pink-400">TikTok Viral</span>
200 |         <a
201 |           href={tiktokViralUrl}
202 |           target="_blank"
203 |           rel="noopener noreferrer"
204 |           className="text-[10px] text-pink-400 hover:text-pink-300 underline flex items-cente...
205 |           title="Checar videos virais no TikTok"
206 |         >
207 |           <span>Checar</span>
208 |           <ExternalLink className="w-2.5 h-2.5" />
209 |         </a>
210 |       </div>
211 |       <p className="text-slate-300 text-[11px]">
212 |         {analysis?.competitionAnalysis.tiktokShopBRStatus || 'Ponto cego total: vídeos virali...
213 |       </p>
214 |     </div>
215 |   </div>
216 | </div>
217 |
218 |   <div className="mt-3 p-3 rounded-lg bg-cyan-950/30 border border-cyan-500/20 text-xs text-cya...
219 |     <strong className="text-cyan-300">Entendimento cultural: </strong>
220 |     {product.culturalFitReason}
221 |   </div>
222 | </div>
223 |
224 | {/* Ad Hooks for TikTok/Reels */}
225 | <div>
226 |   <div className="flex items-center justify-between mb-3">
227 |     <h3 className="text-sm font-bold text-white flex items-center gap-2">
228 |       <Sparkles className="w-4 h-4 text-cyan-400" />
229 |       <span>Ganchos de Anúncio Testados (Primeiros 3 Segundos)</span>
230 |     </h3>
231 |     <button
232 |       onClick={() => onOpenCreativeGenerator(product)}
233 |       className="text-xs font-semibold text-cyan-400 hover:text-cyan-300 flex items-center gap-...
234 |     >
235 |       <span>Ver Roteiros Completos</span>
236 |       <ExternalLink className="w-3 h-3" />
237 |     </button>
238 |   </div>
239 |
240 |   <div className="space-y-2">
241 |     {product.adHooks.map((hook, idx) => (
242 |       <div
243 |         key={idx}
244 |         className="flex items-center justify-between gap-3 p-3 rounded-xl bg-slate-950/70 borde...
245 |       >
246 |         <p className="text-slate-200 italic font-medium">
247 |           "{hook}"
248 |         </p>
249 |         <button
250 |           onClick={() => handleCopyHook(hook, idx)}
251 |           className="p-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 hover:text...
252 |           title="Copiar gancho"
253 |         >
254 |           {copiedIndex === idx ? (
255 |             <Check className="w-4 h-4 text-emerald-400" />
256 |           ) : (
257 |             <Copy className="w-4 h-4" />

```

```

258 |         }}
259 |       </button>
260 |     </div>
261 |   )}}
262 | </div>
263 | </div>
264 |
265 | {/* Direct Supplier Search Links */}
266 | <div className="rounded-xl border border-slate-800 bg-slate-950/50 p-4">
267 |   <h3 className="text-sm font-bold text-white flex items-center gap-2 mb-3">
268 |     <Package className="w-4 h-4 text-cyan-400" />
269 |     <span>Onde Encontrar Fornecedores Rápidos</span>
270 |   </h3>
271 |
272 |   <div className="flex flex-wrap gap-2 text-xs">
273 |     <a
274 |       href={aliexpressSearchUrl}
275 |       target="_blank"
276 |       rel="noopener noreferrer"
277 |       className="flex items-center gap-1.5 rounded-lg border border-orange-500/30 bg-orange-500/10"
278 |     >
279 |       <span>AliExpress (Busca Direta)</span>
280 |       <ExternalLink className="w-3 h-3" />
281 |     </a>
282 |
283 |     <a
284 |       href={cjDropshippingUrl}
285 |       target="_blank"
286 |       rel="noopener noreferrer"
287 |       className="flex items-center gap-1.5 rounded-lg border border-blue-500/30 bg-blue-500/10 ..."
288 |     >
289 |       <span>CJ Dropshipping</span>
290 |       <ExternalLink className="w-3 h-3" />
291 |     </a>
292 |
293 |     <a
294 |       href={googleTrendsUrl}
295 |       target="_blank"
296 |       rel="noopener noreferrer"
297 |       className="flex items-center gap-1.5 rounded-lg border border-emerald-500/30 bg-emerald-500/10 ..."
298 |     >
299 |       <span>Google Trends Brasil</span>
300 |       <ExternalLink className="w-3 h-3" />
301 |     </a>
302 |   </div>
303 |
304 |   <div className="mt-3 text-[11px] text-slate-400 flex items-center gap-2">
305 |     <ShieldCheck className="w-4 h-4 text-emerald-400 shrink-0" />
306 |     <span>
307 |       <strong>Logística & Despacho:</strong> {product.logisticsComplexity}. Sugerido fornecedor...
308 |     </span>
309 |   </div>
310 | </div>
311 |
312 | {/* Action Checklist */}
313 | <div className="rounded-xl border border-slate-800 bg-slate-950/50 p-4">
314 |   <h3 className="text-sm font-bold text-white flex items-center gap-2 mb-2">
315 |     <Layers className="w-4 h-4 text-cyan-400" />
316 |     <span>Plano de Ação para Surfar Essa Onda</span>
317 |   </h3>
318 |   <ul className="space-y-1.5 text-xs text-slate-300">
319 |     {analysis?.actionChecklist?.map((action, i) => (
320 |       <li key={i} className="flex items-start gap-2">
321 |         <span className="text-cyan-400 font-bold">'</span>
322 |         <span>{action}</span>

```

```

323 |         </li>
324 |       )) || (
325 |         <>
326 |           <li className="flex items-start gap-2">
327 |             <span className="text-cyan-400 font-bold">' </span>
328 |             <span>Validar fornecedor com envio rápido no AliExpress ou CJ Dropshipping.</span>
329 |           </li>
330 |           <li className="flex items-start gap-2">
331 |             <span className="text-cyan-400 font-bold">' </span>
332 |             <span>Criar landing page de produto único com prova social internacional.</span>
333 |           </li>
334 |           <li className="flex items-start gap-2">
335 |             <span className="text-cyan-400 font-bold">' </span>
336 |             <span>Testar os 3 primeiros ganchos em campanhas no TikTok Ads ou Reels.</span>
337 |           </li>
338 |         </>
339 |       )}
340 |     </ul>
341 |   </div>
342 |
343 |   {/* Grounding web sources if any */}
344 |   {product.groundingSources && product.groundingSources.length > 0 && (
345 |     <div className="pt-2 border-t border-slate-800 text-xs">
346 |       <span className="text-[11px] font-semibold text-slate-400 block mb-2">
347 |         Fontes Web e Referências Rastreadas:
348 |       </span>
349 |       <div className="flex flex-wrap gap-2">
350 |         {product.groundingSources.map((source, idx) => (
351 |           <a
352 |             key={idx}
353 |             href={source.uri}
354 |             target="_blank"
355 |             rel="noopener noreferrer"
356 |             className="inline-flex items-center gap-1 rounded bg-slate-800 px-2 py-1 text-[10px] ..."
357 |           >
358 |             <span>{source.title}</span>
359 |             <ExternalLink className="w-2.5 h-2.5" />
360 |           </a>
361 |         ))}
362 |       </div>
363 |     </div>
364 |   )}
365 |
366 | </div>
367 |
368 | {/* Modal Footer Controls */}
369 | <div className="border-t border-slate-800 bg-slate-950/80 p-4 flex flex-col sm:flex-row items-cen...
370 |   <div className="text-xs text-slate-400 text-center sm:text-left">
371 |     <span>Margem líquida média estimada: </span>
372 |     <strong className="text-emerald-400 font-bold">+{product.estimatedProfitMarginPercent}%</stro...
373 |   </div>
374 |
375 |   <div className="flex items-center gap-2 w-full sm:w-auto">
376 |     <button
377 |       onClick={() => {
378 |         onClose();
379 |         onOpenCreativeGenerator(product);
380 |       }}
381 |       className="flex-1 sm:flex-initial flex items-center justify-center gap-1.5 rounded-xl borde...
382 |     >
383 |       <Video className="w-4 h-4 text-cyan-400" />
384 |       <span>Gerar Roteiros UGC</span>
385 |     </button>
386 |
387 |     <button

```

```
388 |         onClick={() => {
389 |             onClose();
390 |             onOpenCalculator(product);
391 |         }}
392 |         className="flex-1 sm:flex-initial flex items-center justify-center gap-1.5 rounded-xl bg-gr...
393 |     >
394 |         <Calculator className="w-4 h-4 text-slate-950" />
395 |         <span>Simular Lucro Líquido</span>
396 |     </button>
397 | </div>
398 | </div>
399 |
400 | </div>
401 | </div>
402 | );
403 | };
404 |
```

src/components/ProfitCalculatorModal.tsx

Calculadora e Simulador de Lucro (src/components/ProfitCalculatorModal.tsx) (296 linhas)

```
1 | import React, { useState, useMemo } from 'react';
2 | import { X, Calculator, DollarSign, ArrowRight, ShieldCheck, HelpCircle } from 'lucide-react';
3 | import { TrendingProduct } from '../types';
4 |
5 | interface ProfitCalculatorModalProps {
6 |   product: TrendingProduct | null;
7 |   onClose: () => void;
8 | }
9 |
10 | export const ProfitCalculatorModal: React.FC<ProfitCalculatorModalProps> = ({
11 |   product,
12 |   onClose,
13 | }) => {
14 |   // Configurable parameters
15 |   const [costUSD, setCostUSD] = useState<number>(product ? product.estimatedCostUSD : 10.0);
16 |   const [usdRate, setUsdRate] = useState<number>(5.65);
17 |   const [shippingBRL, setShippingBRL] = useState<number>(18.0);
18 |   const [sellingPriceBRL, setSellingPriceBRL] = useState<number>(
19 |     product ? product.estimatedPriceBRL : 179.9
20 |   );
21 |   const [applyRemessaConforme, setApplyRemessaConforme] = useState<boolean>(true);
22 |   const [gatewayPercent, setGatewayPercent] = useState<number>(4.99);
23 |   const [gatewayFixedBRL, setGatewayFixedBRL] = useState<number>(1.0);
24 |   const [adSpendCpaBRL, setAdSpendCpaBRL] = useState<number>(35.0);
25 |
26 |   // Financial calculations
27 |   const calculation = useMemo(() => {
28 |     const rawCostBRL = costUSD * usdRate;
29 |
30 |     // Remessa Conforme tax estimation (approx 20% federal import tax for <=$50 + 17% ICMS, or standard)
31 |     let importTaxBRL = 0;
32 |     if (applyRemessaConforme) {
33 |       if (costUSD <= 50) {
34 |         // approx 20% federal + 17% ICMS = ~44% effective on CIF
35 |         importTaxBRL = (rawCostBRL + shippingBRL) * 0.35;
36 |       } else {
37 |         // 60% above $50 + ICMS
38 |         importTaxBRL = (rawCostBRL + shippingBRL) * 0.6;
39 |       }
40 |     }
41 |
42 |     const totalProductCostBRL = rawCostBRL + shippingBRL + importTaxBRL;
43 |     const gatewayFeeBRL = (sellingPriceBRL * (gatewayPercent / 100)) + gatewayFixedBRL;
44 |     const totalExpensesBRL = totalProductCostBRL + gatewayFeeBRL + adSpendCpaBRL;
45 |     const netProfitBRL = sellingPriceBRL - totalExpensesBRL;
46 |     const netMarginPercent = sellingPriceBRL > 0 ? (netProfitBRL / sellingPriceBRL) * 100 : 0;
47 |     const breakEvenRoas = (sellingPriceBRL - totalProductCostBRL - gatewayFeeBRL) > 0
48 |       ? sellingPriceBRL / (sellingPriceBRL - totalProductCostBRL - gatewayFeeBRL)
49 |       : 0;
50 |
51 |     return {
52 |       rawCostBRL,
53 |       importTaxBRL,
54 |       totalProductCostBRL,
55 |       gatewayFeeBRL,
56 |       totalExpensesBRL,
57 |       netProfitBRL,
58 |       netMarginPercent,
59 |       breakEvenRoas,
60 |     };
61 |   }, [costUSD, usdRate, shippingBRL, sellingPriceBRL, applyRemessaConforme, gatewayPercent, gatewayFixedB...
62 |
```

```

63 |   return (
64 |     <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/80 backdrop-blur-md p-3 ...
65 |       <div className="relative w-full max-w-2xl rounded-2xl border border-slate-800 bg-slate-900 shadow-2...
66 |
67 |       {/* Header */}
68 |       <div className="flex items-center justify-between border-b border-slate-800 bg-slate-950/60 p-4 s...
69 |         <div className="flex items-center gap-3">
70 |           <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-emerald-500/20 borde...
71 |             <Calculator className="h-5 w-5 text-emerald-400" />
72 |           </div>
73 |           <div>
74 |             <h2 className="text-base sm:text-lg font-bold text-white">
75 |               Simulador de Viabilidade & Lucro Líquido
76 |             </h2>
77 |             <p className="text-xs text-slate-400">
78 |               {product ? product.name : 'Simule importação, taxas e margem real em BRL'}
79 |             </p>
80 |           </div>
81 |         </div>
82 |
83 |         <button
84 |           onClick={onClose}
85 |           className="rounded-xl p-2 text-slate-400 hover:text-white hover:bg-slate-800 transition"
86 |           title="Fechar"
87 |         >
88 |           <X className="w-5 h-5" />
89 |         </button>
90 |       </div>
91 |
92 |       {/* Scrollable Content */}
93 |       <div className="flex-1 overflow-y-auto p-4 sm:p-6 space-y-5">
94 |
95 |         {/* Inputs Section */}
96 |         <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
97 |
98 |           {/* Cost in USD */}
99 |           <div>
100 |             <label className="block text-xs font-semibold text-slate-300 mb-1">
101 |               Custo no Fornecedor (USD)
102 |             </label>
103 |             <div className="relative">
104 |               <span className="absolute left-3 top-1/2 -translate-y-1/2 text-xs text-slate-400">$/span...
105 |               <input
106 |                 type="number"
107 |                 step="0.1"
108 |                 min="0.5"
109 |                 value={costUSD}
110 |                 onChange={(e) => setCostUSD(parseFloat(e.target.value) || 0)}
111 |                 className="w-full rounded-xl border border-slate-700 bg-slate-950/80 py-2 pl-7 pr-3 tex...
112 |               </div>
113 |             </div>
114 |           </div>
115 |
116 |           {/* Dólar Rate */}
117 |           <div>
118 |             <label className="block text-xs font-semibold text-slate-300 mb-1">
119 |               Cotação Dólar (BRL)
120 |             </label>
121 |             <div className="relative">
122 |               <span className="absolute left-3 top-1/2 -translate-y-1/2 text-xs text-slate-400">R$/spa...
123 |               <input
124 |                 type="number"
125 |                 step="0.05"
126 |                 value={usdRate}
127 |                 onChange={(e) => setUsdRate(parseFloat(e.target.value) || 1)}

```

```

128 |         className="w-full rounded-xl border border-slate-700 bg-slate-950/80 py-2 pl-8 pr-3 tex...
129 |       />
130 |     </div>
131 |   </div>
132 |
133 |   {/* International Shipping */}
134 |   <div>
135 |     <label className="block text-xs font-semibold text-slate-300 mb-1">
136 |       Frete Internacional Estimado (BRL)
137 |     </label>
138 |     <div className="relative">
139 |       <span className="absolute left-3 top-1/2 -translate-y-1/2 text-xs text-slate-400">R$</spa...
140 |       <input
141 |         type="number"
142 |         step="1"
143 |         value={shippingBRL}
144 |         onChange={(e) => setShippingBRL(parseFloat(e.target.value) || 0)}
145 |         className="w-full rounded-xl border border-slate-700 bg-slate-950/80 py-2 pl-8 pr-3 tex...
146 |       />
147 |     </div>
148 |   </div>
149 |
150 |   {/* Selling Price in Brazil */}
151 |   <div>
152 |     <label className="block text-xs font-semibold text-emerald-400 mb-1">
153 |       Preço de Venda Pretendido no BR (BRL)
154 |     </label>
155 |     <div className="relative">
156 |       <span className="absolute left-3 top-1/2 -translate-y-1/2 text-xs text-emerald-400 font-b...
157 |       <input
158 |         type="number"
159 |         step="1"
160 |         value={sellingPriceBRL}
161 |         onChange={(e) => setSellingPriceBRL(parseFloat(e.target.value) || 0)}
162 |         className="w-full rounded-xl border border-emerald-500/50 bg-slate-950/90 py-2 pl-8 pr-...
163 |       />
164 |     </div>
165 |   </div>
166 |
167 |   {/* Estimated CPA (Ad Spend per purchase) */}
168 |   <div>
169 |     <label className="block text-xs font-semibold text-slate-300 mb-1">
170 |       CPA Tráfego Pago / Anúncios (BRL)
171 |     </label>
172 |     <div className="relative">
173 |       <span className="absolute left-3 top-1/2 -translate-y-1/2 text-xs text-slate-400">R$</spa...
174 |       <input
175 |         type="number"
176 |         step="1"
177 |         value={adSpendCpaBRL}
178 |         onChange={(e) => setAdSpendCpaBRL(parseFloat(e.target.value) || 0)}
179 |         className="w-full rounded-xl border border-slate-700 bg-slate-950/80 py-2 pl-8 pr-3 tex...
180 |       />
181 |     </div>
182 |     <span className="text-[10px] text-slate-500">Média de R$ 25 a R$ 45 no TikTok Ads</span>
183 |   </div>
184 |
185 |   {/* Remessa Conforme Switch */}
186 |   <div className="flex flex-col justify-center">
187 |     <label className="flex items-center gap-2 cursor-pointer pt-2">
188 |       <input
189 |         type="checkbox"
190 |         checked={applyRemessaConforme}
191 |         onChange={(e) => setApplyRemessaConforme(e.target.checked)}
192 |         className="h-4 w-4 rounded border-slate-700 bg-slate-950 text-cyan-500 focus:ring-cyan-...

```

```

193 |         />
194 |         <span className="text-xs font-medium text-slate-200">
195 |             Incluir Imposto de Importação / Remessa Conforme
196 |         </span>
197 |     </label>
198 |     <span className="text-[10px] text-slate-500 pl-6 mt-1">
199 |         Calcula alíquota com despacho aduaneiro seguro.
200 |     </span>
201 | </div>
202 |
203 | </div>
204 |
205 | {/* Results Summary Cards */}
206 | <div className="rounded-2xl border border-slate-800 bg-slate-950/80 p-4 sm:p-5">
207 |     <h3 className="text-xs font-bold uppercase tracking-wider text-slate-400 mb-3">
208 |         Resultado da Operação Unitária
209 |     </h3>
210 |
211 |     <div className="grid grid-cols-2 sm:grid-cols-3 gap-3 mb-4">
212 |         <div className="p-3 rounded-xl bg-slate-900 border border-slate-800">
213 |             <span className="text-[11px] text-slate-400 block mb-0.5">Custo Mercadoria + Frete + Taxa...
214 |             <span className="text-sm sm:text-base font-bold text-slate-200">
215 |                 R$ {calculation.totalProductCostBRL.toFixed(2)}
216 |             </span>
217 |         </div>
218 |
219 |         <div className="p-3 rounded-xl bg-slate-900 border border-slate-800">
220 |             <span className="text-[11px] text-slate-400 block mb-0.5">Taxa de Gateway (4.99%)</span>
221 |             <span className="text-sm sm:text-base font-bold text-slate-300">
222 |                 R$ {calculation.gatewayFeeBRL.toFixed(2)}
223 |             </span>
224 |         </div>
225 |
226 |         <div className="p-3 rounded-xl bg-slate-900 border border-slate-800">
227 |             <span className="text-[11px] text-slate-400 block mb-0.5">ROAS de Equilíbrio (Mínimo)</sp...
228 |             <span className="text-sm sm:text-base font-bold text-cyan-400">
229 |                 {calculation.breakEvenRoas.toFixed(2)}x
230 |             </span>
231 |         </div>
232 |     </div>
233 |
234 | {/* Net Profit Big Banner */}
235 | <div className={`p-4 rounded-xl border flex flex-col sm:flex-row items-center justify-between...
236 |     calculation.netProfitBRL > 0
237 |     ? 'bg-emerald-950/30 border-emerald-500/40 text-emerald-300'
238 |     : 'bg-rose-950/30 border-rose-500/40 text-rose-300'
239 | }`>
240 |     <div>
241 |         <span className="text-xs font-semibold block uppercase">
242 |             Lucro Líquido Real por Venda
243 |         </span>
244 |         <span className="text-2xl sm:text-3xl font-black">
245 |             R$ {calculation.netProfitBRL.toFixed(2)}
246 |         </span>
247 |     </div>
248 |
249 |     <div className="text-right">
250 |         <span className="text-xs font-semibold block uppercase">
251 |             Margem Líquida
252 |         </span>
253 |         <span className="text-xl sm:text-2xl font-extrabold">
254 |             {calculation.netMarginPercent.toFixed(1)}%
255 |         </span>
256 |     </div>
257 | </div>

```

```

258 |
259 |     {/* Scaling projections */}
260 |     <div className="mt-4 pt-3 border-t border-slate-800 text-xs">
261 |       <span className="text-slate-400 font-semibold block mb-2">Projeção Mensal de Lucro:</span>
262 |       <div className="grid grid-cols-3 gap-2 text-center">
263 |         <div className="p-2 rounded-lg bg-slate-900/60 border border-slate-800/80">
264 |           <span className="text-[10px] text-slate-500 block">30 vendas/mês</span>
265 |           <span className="font-bold text-white">R$ {(calculation.netProfitBRL * 30).toFixed(2)}<...
266 |         </div>
267 |         <div className="p-2 rounded-lg bg-slate-900/60 border border-slate-800/80">
268 |           <span className="text-[10px] text-slate-500 block">100 vendas/mês</span>
269 |           <span className="font-bold text-emerald-400">R$ {(calculation.netProfitBRL * 100).toFix...
270 |         </div>
271 |         <div className="p-2 rounded-lg bg-slate-900/60 border border-slate-800/80">
272 |           <span className="text-[10px] text-slate-500 block">300 vendas/mês</span>
273 |           <span className="font-bold text-cyan-400">R$ {(calculation.netProfitBRL * 300).toFixed(...
274 |         </div>
275 |       </div>
276 |     </div>
277 |
278 |   </div>
279 |
280 | </div>
281 |
282 |   {/* Footer */}
283 |   <div className="border-t border-slate-800 bg-slate-950/80 p-4 flex justify-end">
284 |     <button
285 |       onClick={onClose}
286 |       className="rounded-xl bg-cyan-500 hover:bg-cyan-400 px-5 py-2 text-xs font-bold text-slate-95...
287 |     >
288 |       Concluir Simulação
289 |     </button>
290 |   </div>
291 |
292 | </div>
293 | </div>
294 | );
295 | };
296 |

```

src/components/LiveScannerFeed.tsx

Feed em Tempo Real de Produtos (src/components/LiveScannerFeed.tsx) (134 linhas)

```
1 | import React, { useEffect, useState } from 'react';
2 | import { Radio, CheckCircle, Sparkles, X, Globe, ShieldAlert } from 'lucide-react';
3 |
4 | interface LiveScannerFeedProps {
5 |   isOpen: boolean;
6 |   onClose: () => void;
7 |   onFinish: () => void;
8 |   scannedCount: number;
9 | }
10 |
11 | const SCAN_STEPS = [
12 |   '🌀🌐 Conectando aos nós globais de telemetria de e-commerce...',
13 |   '🌀🔍 Minerando feeds virais do TikTok Shop US e Amazon Movers & Shakers...',
14 |   '🌀📦 Varrendo lançamentos no Douyin, Taobao e catálogo de fábricas 1688...',
15 |   '🌀📊 Cruzando índice de concorrência ativa na Shopee e Mercado Livre Brasil...',
16 |   '🌀🌐 Computando spread de preço internacional x preço de venda no Brasil...',
17 |   '🌀✅ Validação de fit cultural e margens superiores a 250% concluída!',
18 | ];
19 |
20 | export const LiveScannerFeed: React.FC<LiveScannerFeedProps> = ({
21 |   isOpen,
22 |   onClose,
23 |   onFinish,
24 |   scannedCount,
25 | }) => {
26 |   const [currentStepIndex, setCurrentStepIndex] = useState(0);
27 |   const [isCompleted, setIsCompleted] = useState(false);
28 |
29 |   useEffect(() => {
30 |     if (!isOpen) {
31 |       setCurrentStepIndex(0);
32 |       setIsCompleted(false);
33 |       return;
34 |     }
35 |
36 |     const interval = setInterval(() => {
37 |       setCurrentStepIndex((prev) => {
38 |         if (prev < SCAN_STEPS.length - 1) {
39 |           return prev + 1;
40 |         } else {
41 |           clearInterval(interval);
42 |           setIsCompleted(true);
43 |           return prev;
44 |         }
45 |       });
46 |     }, 900);
47 |
48 |     return () => clearInterval(interval);
49 |   }, [isOpen]);
50 |
51 |   if (!isOpen) return null;
52 |
53 |   return (
54 |     <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/80 backdrop-blur-md p-4 ...
55 |       <div className="relative w-full max-w-lg rounded-2xl border border-cyan-500/40 bg-slate-950 p-6 sha...
56 |
57 |         {/* Animated Radar Sweep Header */}
58 |         <div className="flex items-center justify-between border-b border-slate-800 pb-4 mb-5">
59 |           <div className="flex items-center gap-3">
60 |             <div className="relative flex h-10 w-10 items-center justify-center rounded-xl bg-cyan-500/20...
61 |               <Radio className="h-5 w-5 text-cyan-400 animate-spin" />
62 |             </div>
```

```

63 |         <div>
64 |             <h3 className="text-base font-bold text-white flex items-center gap-2">
65 |                 <span>Scanner em Tempo Real TIKBLOX</span>
66 |             </h3>
67 |             <p className="text-xs text-cyan-400">Varredura de tendências internacionais</p>
68 |         </div>
69 |     </div>
70 |
71 |     <button
72 |         onClick={onClose}
73 |         className="rounded-lg p-1.5 text-slate-400 hover:text-white hover:bg-slate-800"
74 |     >
75 |         <X className="w-4 h-4" />
76 |     </button>
77 | </div>
78 |
79 | {/* Radar Terminal Log */}
80 | <div className="space-y-3 rounded-xl bg-slate-900/90 border border-slate-800/80 p-4 font-mono tex...
81 |     {SCAN_STEPS.slice(0, currentStepIndex + 1).map((step, idx) => (
82 |         <div
83 |             key={idx}
84 |             className={`flex items-start gap-2 ${
85 |                 idx === currentStepIndex && !isCompleted
86 |                 ? 'text-cyan-300 font-semibold animate-pulse'
87 |                 : 'text-slate-300'
88 |             }`}
89 |         >
90 |             <span className="text-cyan-500 mt-0.5">'o</span>
91 |             <span>{step}</span>
92 |         </div>
93 |     )]}
94 | </div>
95 |
96 | {/* Progress Bar */}
97 | <div className="w-full bg-slate-800 rounded-full h-2 mb-5 overflow-hidden">
98 |     <div
99 |         className="bg-gradient-to-r from-cyan-500 to-emerald-400 h-full transition-all duration-700 e...
100 |         style={{ width: `${((currentStepIndex + 1) / SCAN_STEPS.length) * 100}%` }}
101 |     />
102 | </div>
103 |
104 | {/* Completion Message */}
105 | {isCompleted ? (
106 |     <div className="rounded-xl bg-emerald-950/40 border border-emerald-500/40 p-4 text-center">
107 |         <CheckCircle className="w-8 h-8 text-emerald-400 mx-auto mb-2" />
108 |         <h4 className="text-sm font-bold text-white">
109 |             Varredura Finalizada com Sucesso!
110 |         </h4>
111 |         <p className="text-xs text-slate-300 mt-1">
112 |             O radar identificou novas oportunidades com alto spread de lucro e baixa concorrência no Br...
113 |         </p>
114 |         <button
115 |             onClick={() => {
116 |                 onFinish();
117 |                 onClose();
118 |             }}
119 |             className="mt-4 w-full rounded-xl bg-gradient-to-r from-cyan-500 to-emerald-400 py-2 text-x...
120 |         >
121 |             Explorar Produtos Descobertos
122 |         </button>
123 |     </div>
124 | ) : (
125 |     <div className="text-center text-xs text-slate-500 italic">
126 |         Conectando com Google Search e feeds internacionais de produtos virais...
127 |     </div>

```

```
128 |         }}  
129 |  
130 |         </div>  
131 |     </div>  
132 | );  
133 | };  
134 |
```

src/components/SavedRadarView.tsx

Área de Produtos Salvos (src/components/SavedRadarView.tsx) (83 linhas)

```
1 | import React from 'react';
2 | import { Bookmark, Sparkles, Layers } from 'lucide-react';
3 | import { TrendingProduct } from '../types';
4 | import { ProductCard } from './ProductCard';
5 |
6 | interface SavedRadarViewProps {
7 |   savedProducts: TrendingProduct[];
8 |   onToggleSave: (id: string) => void;
9 |   onOpenDeepDive: (product: TrendingProduct) => void;
10 |   onOpenCalculator: (product: TrendingProduct) => void;
11 |   onOpenCreativeGenerator: (product: TrendingProduct) => void;
12 |   onBackToRadar: () => void;
13 | }
14 |
15 | export const SavedRadarView: React.FC<SavedRadarViewProps> = ({
16 |   savedProducts,
17 |   onToggleSave,
18 |   onOpenDeepDive,
19 |   onOpenCalculator,
20 |   onOpenCreativeGenerator,
21 |   onBackToRadar,
22 | }) => {
23 |   if (savedProducts.length === 0) {
24 |     return (
25 |       <div className="py-16 text-center max-w-md mx-auto px-4">
26 |         <div className="w-16 h-16 rounded-2xl bg-cyan-500/10 border border-cyan-500/20 flex items-center ...
27 |           <Bookmark className="w-8 h-8 text-cyan-400" />
28 |         </div>
29 |         <h3 className="text-lg font-bold text-white mb-2">
30 |           Seu Radar Pessoal está Vazio
31 |         </h3>
32 |         <p className="text-xs text-slate-400 leading-relaxed mb-6">
33 |           Navegue pelas tendências e clique no ícone de marcador para salvar os produtos que você deseja ...
34 |         </p>
35 |         <button
36 |           onClick={onBackToRadar}
37 |           className="inline-flex items-center gap-2 rounded-xl bg-cyan-500 hover:bg-cyan-400 px-5 py-2.5 ...
38 |         >
39 |           <Layers className="w-4 h-4" />
40 |           <span>Ver Produtos em Alta</span>
41 |         </button>
42 |       </div>
43 |     );
44 |   }
45 |
46 |   return (
47 |     <div className="space-y-6">
48 |       <div className="flex items-center justify-between">
49 |         <div>
50 |           <h2 className="text-xl font-black text-white flex items-center gap-2">
51 |             <Bookmark className="w-5 h-5 text-cyan-400" />
52 |             <span>Meu Radar Pessoal ({savedProducts.length})</span>
53 |           </h2>
54 |           <p className="text-xs text-slate-400">
55 |             Produtos salvos para validação de fornecedores e lançamento de criativos.
56 |           </p>
57 |         </div>
58 |
59 |         <button
60 |           onClick={onBackToRadar}
61 |           className="text-xs font-semibold text-cyan-400 hover:text-cyan-300"
62 |         >
```

```
63 |         !• Voltar para Todos os Produtos
64 |       </button>
65 |     </div>
66 |
67 |     <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
68 |       {savedProducts.map((product) => (
69 |         <ProductCard
70 |           key={product.id}
71 |           product={product}
72 |           isSaved={true}
73 |           onToggleSave={onToggleSave}
74 |           onOpenDeepDive={onOpenDeepDive}
75 |           onOpenCalculator={onOpenCalculator}
76 |           onOpenCreativeGenerator={onOpenCreativeGenerator}
77 |         />
78 |       ))}
79 |     </div>
80 |   </div>
81 | );
82 | };
83 |
```

src/components/PushNotificationModal.tsx

Modal de Configurações de Notificações (src/components/PushNotificationModal.tsx) (414 linhas)

```
1 | import React, { useState } from 'react';
2 | import {
3 |   Bell,
4 |   BellOff,
5 |   BellRing,
6 |   CheckCircle2,
7 |   AlertTriangle,
8 |   Sparkles,
9 |   Sliders,
10 | Volume2,
11 | VolumeX,
12 | Radio,
13 | ExternalLink,
14 | X,
15 | Flame,
16 | ShieldCheck,
17 | } from 'lucide-react';
18 | import { PushNotificationSettings, PushAlertLog, TrendingProduct } from '../types';
19 |
20 | interface PushNotificationModalProps {
21 |   isOpen: boolean;
22 |   onClose: () => void;
23 |   permission: NotificationPermission | 'unsupported';
24 |   settings: PushNotificationSettings;
25 |   onUpdateSettings: (partial: Partial<PushNotificationSettings>) => void;
26 |   onEnable: () => Promise<boolean>;
27 |   onDisable: () => void;
28 |   onSendTest: () => Promise<boolean>;
29 |   alertHistory: PushAlertLog[];
30 |   onSelectProductById?: (productId: string) => void;
31 |   sampleProduct?: TrendingProduct;
32 | }
33 |
34 | export const PushNotificationModal: React.FC<PushNotificationModalProps> = ({
35 |   isOpen,
36 |   onClose,
37 |   permission,
38 |   settings,
39 |   onUpdateSettings,
40 |   onEnable,
41 |   onDisable,
42 |   onSendTest,
43 |   alertHistory,
44 |   onSelectProductById,
45 |   sampleProduct,
46 | }) => {
47 |   const [testSent, setTestSent] = useState(false);
48 |   const [isLoading, setIsLoading] = useState(false);
49 |
50 |   if (!isOpen) return null;
51 |
52 |   const isGranted = permission === 'granted' && settings.enabled;
53 |   const isDenied = permission === 'denied';
54 |   const isDefault = permission === 'default' || (permission === 'granted' && !settings.enabled);
55 |
56 |   const handleToggleActive = async () => {
57 |     setIsLoading(true);
58 |     try {
59 |       if (isGranted) {
60 |         onDisable();
61 |       } else {
62 |         await onEnable();
```

```

63 |     }
64 |   } finally {
65 |     setIsLoading(false);
66 |   }
67 | };
68 |
69 | const handleTestClick = async () => {
70 |   setTestSent(false);
71 |   setIsLoading(true);
72 |   try {
73 |     const ok = await onSendTest();
74 |     if (ok) {
75 |       setTestSent(true);
76 |       setTimeout(() => setTestSent(false), 4000);
77 |     }
78 |   } finally {
79 |     setIsLoading(false);
80 |   }
81 | };
82 |
83 | return (
84 |   <div className="fixed inset-0 z-50 flex items-center justify-center p-3 sm:p-4 bg-black/80 backdrop-b...
85 |     <div className="relative w-full max-w-2xl max-h-[92vh] overflow-y-auto rounded-3xl border border-wh...
86 |
87 |       {/* Close Button */}
88 |       <button
89 |         onClick={onClose}
90 |         className="absolute top-4 right-4 p-2 rounded-full bg-white/5 hover:bg-white/10 text-white/70 h...
91 |         title="Fechar"
92 |       >
93 |         <X className="w-5 h-5" />
94 |       </button>
95 |
96 |       {/* Modal Header */}
97 |       <div className="flex items-center gap-3.5 mb-5">
98 |         <div className="relative flex items-center justify-center w-12 h-12 rounded-2xl bg-gradient-to-...
99 |           <BellRing className="w-6 h-6 text-[#25F4EE] animate-pulse" />
100 |          <span className="absolute -top-1 -right-1 w-3.5 h-3.5 rounded-full bg-[#FE2C55] border-2 bord...
101 |        </div>
102 |        <div>
103 |          <h2 className="text-xl sm:text-2xl font-black tracking-tight flex items-center gap-2">
104 |            <span>Radar Push PWA</span>
105 |            <span className="px-2 py-0.5 rounded-full bg-[#25F4EE]/10 border border-[#25F4EE]/30 text-[...
106 |              Tempo Real
107 |            </span>
108 |          </h2>
109 |          <p className="text-xs sm:text-sm text-[#8E91A6]">
110 |            Alertas instantâneos no seu dispositivo quando novos produtos virais explodirem nos EUA e C...
111 |          </p>
112 |        </div>
113 |      </div>
114 |
115 |      {/* Status Card Banner */}
116 |      <div
117 |        className={`p-4 rounded-2xl border transition-all ${
118 |          isGranted
119 |            ? 'bg-emerald-950/25 border-emerald-500/40 text-emerald-300'
120 |            : isDenied
121 |            ? 'bg-rose-950/25 border-rose-500/40 text-rose-300'
122 |            : 'bg-amber-950/25 border-amber-500/40 text-amber-300'
123 |        }`}
124 |      >
125 |        <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4">
126 |          <div className="flex items-start gap-3">
127 |            {isGranted ? (

```

```

128 |         <CheckCircle2 className="w-6 h-6 text-emerald-400 shrink-0 mt-0.5" />
129 |       ) : isDenied ? (
130 |         <AlertTriangle className="w-6 h-6 text-rose-400 shrink-0 mt-0.5" />
131 |       ) : (
132 |         <Radio className="w-6 h-6 text-amber-400 shrink-0 mt-0.5 animate-pulse" />
133 |       )}
134 |     <div>
135 |       <h4 className="font-black text-sm sm:text-base text-white">
136 |         {isGranted
137 |           ? 'Radar Push Ativado com Sucesso'
138 |           : isDenied
139 |           ? 'Permissão Bloqueada no Navegador'
140 |           : 'Alertas em Espera (Permissão Pendente)'}
141 |       </h4>
142 |       <p className="text-xs text-white/75 mt-0.5 leading-relaxed">
143 |         {isGranted
144 |           ? 'Seu PWA está pronto para emitir notificações no sistema operacional (mesmo com o a...'
145 |           : isDenied
146 |           ? 'Você bloqueou as notificações deste site. Clique no cadeado ⓘ na barra de endereç...'
147 |           : 'Ative para ser alertado sobre as ondas virais de produtos antes da saturação no Br...'
148 |         }
149 |       </p>
150 |     </div>
151 |
152 |     {/ Main Toggle Button */}
153 |     {!isDenied && (
154 |       <button
155 |         onClick={handleToggleActive}
156 |         disabled={isLoading}
157 |         className={`px-4 py-2 rounded-xl text-xs font-black shadow-lg transition active:scale-95 ...
158 |           isGranted
159 |             ? 'bg-white/10 hover:bg-white/15 text-white/90 border border-white/15'
160 |             : 'bg-gradient-to-r from-[#25F4EE] to-[#00CED1] hover:from-[#4BF6F1] hover:to-[#25F4E...
161 |         }`}
162 |       >
163 |         {isGranted ? (
164 |           <>
165 |             <BellOff className="w-4 h-4 text-[#FE2C55]" />
166 |             <span>Desativar</span>
167 |           </>
168 |         ) : (
169 |           <>
170 |             <Bell className="w-4 h-4 text-[#05060A]" />
171 |             <span>Ativar Alertas Push</span>
172 |           </>
173 |         )}
174 |       </button>
175 |     )}
176 |   </div>
177 | </div>
178 |
179 | {/ Quick Action: Test Button */}
180 | <div className="mt-4 flex flex-wrap items-center justify-between gap-3 p-3.5 rounded-2xl bg-[#141...
181 |   <div className="flex items-center gap-2">
182 |     <Sparkles className="w-4 h-4 text-[#25F4EE]" />
183 |     <span className="text-xs font-bold text-white">Testar no seu sistema:</span>
184 |     <span className="text-xs text-[#8E91A6]">
185 |       Dispara uma notificação nativa de teste com som e vibração.
186 |     </span>
187 |   </div>
188 |   <button
189 |     onClick={handleTestClick}
190 |     disabled={isLoading || isDenied}
191 |     className="px-3.5 py-1.5 rounded-xl bg-white/10 hover:bg-white/15 active:scale-95 text-white ...
192 |   >

```

```

193 |         <BellRing className="w-3.5 h-3.5 text-[#25F4EE]" />
194 |         <span>Disparar Notificação Teste</span>
195 |     </button>
196 | </div>
197 |
198 | {testSent && (
199 |     <div className="mt-2.5 p-2.5 rounded-xl bg-emerald-500/10 border border-emerald-500/30 text-eme...
200 |         <CheckCircle2 className="w-4 h-4 text-emerald-400 shrink-0" />
201 |         <span>Notificação de teste disparada! Verifique a central de notificações do seu aparelho/PC...
202 |     </div>
203 | )}
204 |
205 | {/* Notification Settings Form */}
206 | <div className="mt-5 space-y-4">
207 |     <div className="flex items-center gap-2 text-xs font-black uppercase tracking-wider text-[#8E91A6]">
208 |         <Sliders className="w-3.5 h-3.5 text-[#FE2C55]" />
209 |         <span>Regras de Disparo do Radar</span>
210 |     </div>
211 |
212 |     <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">
213 |
214 |         {/* Setting 1: Notify on scan */}
215 |         <label className="flex items-start gap-3 p-3 rounded-xl bg-[#121422] border border-white/10 h...
216 |             <input
217 |                 type="checkbox"
218 |                 checked={settings.notifyOnNewScan}
219 |                 onChange={(e) => onUpdateSettings({ notifyOnNewScan: e.target.checked })}
220 |                 className="mt-1 accent-[#25F4EE] w-4 h-4 rounded cursor-pointer"
221 |             />
222 |             <div>
223 |                 <span className="text-xs font-bold text-white block">Novas Varreduras do Radar</span>
224 |                 <span className="text-[11px] text-[#8E91A6] leading-tight block mt-0.5">
225 |                     Avisar imediatamente quando a varredura detectar novos produtos virais.
226 |                 </span>
227 |             </div>
228 |         </label>
229 |
230 |         {/* Setting 2: Early Wave Only */}
231 |         <label className="flex items-start gap-3 p-3 rounded-xl bg-[#121422] border border-white/10 h...
232 |             <input
233 |                 type="checkbox"
234 |                 checked={settings.notifyOnEarlyWaveOnly}
235 |                 onChange={(e) => onUpdateSettings({ notifyOnEarlyWaveOnly: e.target.checked })}
236 |                 className="mt-1 accent-[#FE2C55] w-4 h-4 rounded cursor-pointer"
237 |             />
238 |             <div>
239 |                 <span className="text-xs font-bold text-white block flex items-center gap-1.5">
240 |                     <span>Apenas Early Wave</span>
241 |                     <span className="px-1.5 py-0.2 rounded bg-amber-500/20 text-amber-300 text-[10px] font-...
242 |                         Oceano Azul
243 |                     </span>
244 |                 </span>
245 |                 <span className="text-[11px] text-[#8E91A6] leading-tight block mt-0.5">
246 |                     Filtrar apenas produtos no início da ascensão antes da concorrência no Brasil.
247 |                 </span>
248 |             </div>
249 |         </label>
250 |
251 |         {/* Setting 3: Sound & Vibration */}
252 |         <label className="flex items-start gap-3 p-3 rounded-xl bg-[#121422] border border-white/10 h...
253 |             <input
254 |                 type="checkbox"
255 |                 checked={settings.soundAndVibration}
256 |                 onChange={(e) => onUpdateSettings({ soundAndVibration: e.target.checked })}
257 |                 className="mt-1 accent-[#25F4EE] w-4 h-4 rounded cursor-pointer"

```

```

258 |         />
259 |         <div>
260 |             <span className="text-xs font-bold text-white block flex items-center gap-1.5">
261 |                 {settings.soundAndVibration ? (
262 |                     <Volume2 className="w-3.5 h-3.5 text-[#25F4EE]" />
263 |                 ) : (
264 |                     <VolumeX className="w-3.5 h-3.5 text-gray-400" />
265 |                 )}
266 |             <span>Som & Vibração Tátil</span>
267 |         </span>
268 |         <span className="text-[11px] text-[#8E91A6] leading-tight block mt-0.5">
269 |             Tocar o chime de radar sintetizado em alta resolução ao receber o push.
270 |         </span>
271 |     </div>
272 | </label>
273 |
274 | {/* Setting 4: Background Radar Watcher */}
275 | <label className="flex items-start gap-3 p-3 rounded-xl bg-[#121422] border border-white/10 h...
276 |     <input
277 |         type="checkbox"
278 |         checked={settings.backgroundRadarAlerts}
279 |         onChange={(e) => onUpdateSettings({ backgroundRadarAlerts: e.target.checked })}
280 |         className="mt-1 accent-[#FE2C55] w-4 h-4 rounded cursor-pointer"
281 |     />
282 |     <div>
283 |         <span className="text-xs font-bold text-white block flex items-center gap-1.5">
284 |             <Radio className="w-3.5 h-3.5 text-[#FE2C55]" />
285 |             <span>Radar em Segundo Plano</span>
286 |         </span>
287 |         <span className="text-[11px] text-[#8E91A6] leading-tight block mt-0.5">
288 |             Monitorar tendências e picos virais mesmo com o app minimizado no celular.
289 |         </span>
290 |     </div>
291 | </label>
292 | </div>
293 |
294 | {/* Threshold Margin */}
295 | <div className="p-3.5 rounded-xl bg-[#121422] border border-white/10">
296 |     <div className="flex items-center justify-between mb-2">
297 |         <span className="text-xs font-bold text-white flex items-center gap-1.5">
298 |             <Flame className="w-3.5 h-3.5 text-[#FE2C55]" />
299 |             <span>Margem de Lucro Mínima para Notificar:</span>
300 |         </span>
301 |         <span className="text-xs font-black text-[#25F4EE] bg-[#25F4EE]/10 px-2 py-0.5 rounded-md b...
302 |             +{settings.minProfitMarginPercent}%
303 |         </span>
304 |     </div>
305 |     <div className="grid grid-cols-4 gap-2">
306 |         {[150, 200, 250, 300].map((marginVal) => (
307 |             <button
308 |                 key={marginVal}
309 |                 type="button"
310 |                 onClick={() => onUpdateSettings({ minProfitMarginPercent: marginVal })}
311 |                 className={`py-1.5 rounded-lg text-xs font-bold transition cursor-pointer ${
312 |                     settings.minProfitMarginPercent === marginVal
313 |                         ? 'bg-[#25F4EE] text-[#010101] shadow-md shadow-[#25F4EE]/20'
314 |                         : 'bg-white/5 hover:bg-white/10 text-white/70 border border-white/10'
315 |                     }`}
316 |             >
317 |                 +{marginVal}%
318 |             </button>
319 |         )]}
320 |     </div>
321 | </div>
322 | </div>

```

```

323 |
324 |     {/* Visual Preview of the Push Notification Card */}
325 |     <div className="mt-5 p-4 rounded-2xl bg-[#090A11] border border-white/15">
326 |       <div className="flex items-center justify-between text-[11px] font-bold text-[#8E91A6] mb-2.5">
327 |         <span className="flex items-center gap-1.5">
328 |           <ShieldCheck className="w-3.5 h-3.5 text-[#25F4EE]" />
329 |           <span>Simulação da Notificação no Sistema Operacional:</span>
330 |         </span>
331 |         <span className="text-[10px] text-white/50">Android / iOS / Windows</span>
332 |       </div>
333 |
334 |       <div className="p-3.5 rounded-xl bg-[#141724] border border-white/10 shadow-lg flex items-start...
335 |         
340 |         <div className="flex-1 min-w-0">
341 |           <div className="flex items-center justify-between gap-2">
342 |             <span className="text-xs font-black text-white truncate">
343 |               Ø=" [RADAR VIRAL] {sampleProduct?.name || 'Copos Térmicos Tumbler Gravados a Laser'}
344 |             </span>
345 |             <span className="text-[10px] text-white/40 shrink-0">agora</span>
346 |           </div>
347 |           <p className="text-[11px] text-white/75 mt-0.5 line-clamp-2 leading-tight">
348 |             Explodindo nos EUA Ø<YúØ<YØ! Margem estimada de {sampleProduct?.estimatedProfitMarginPercent ...
349 |           </p>
350 |           <div className="mt-2 flex items-center gap-2">
351 |             <span className="text-[10px] font-bold text-[#25F4EE] bg-[#25F4EE]/10 px-2 py-0.5 rounded...
352 |               Ø=Y% Ver Produto no Radar
353 |             </span>
354 |             <span className="text-[10px] text-white/50">Dispensar</span>
355 |           </div>
356 |         </div>
357 |       </div>
358 |     </div>
359 |
360 |     {/* History of triggered alerts */}
361 |     {alertHistory.length > 0 && (
362 |       <div className="mt-5">
363 |         <div className="text-xs font-black uppercase tracking-wider text-[#8E91A6] mb-2 flex items-ce...
364 |           <span>Histórico de Alertas Recentes ({alertHistory.length})</span>
365 |         </div>
366 |         <div className="space-y-1.5 max-h-36 overflow-y-auto pr-1">
367 |           {alertHistory.slice(0, 5).map((log) => (
368 |             <div
369 |               key={log.id}
370 |               className="flex items-center justify-between p-2.5 rounded-xl bg-[#121422] border borde...
371 |             >
372 |               <div className="flex items-center gap-2 truncate">
373 |                 <span className="w-2 h-2 rounded-full bg-emerald-400 shrink-0" />
374 |                 <span className="font-bold text-white truncate">{log.productName}</span>
375 |               </div>
376 |               <div className="flex items-center gap-3 shrink-0">
377 |                 <span className="text-[11px] font-extrabold text-[#25F4EE]">
378 |                   +{log.profitMarginPercent}%
379 |                 </span>
380 |                 {onSelectProductById && (
381 |                   <button
382 |                     onClick={() => {
383 |                       onSelectProductById(log.productId);
384 |                       onClose();
385 |                     }}
386 |                     className="text-[11px] text-white/70 hover:text-white underline flex items-center...
387 |                   >

```

```
388 |         <span>Abrir</span>
389 |         <ExternalLink className="w-3 h-3" />
390 |       </button>
391 |     )}
392 |   </div>
393 | </div>
394 |   )}
395 | </div>
396 | </div>
397 | )}
398 |
399 | { /* Modal Footer */}
400 | <div className="mt-6 pt-4 border-t border-white/10 flex items-center justify-between text-xs text...
401 |   <span>TIKBLOX PWA Web Push • 100% Nativo</span>
402 |   <button
403 |     onClick={onClose}
404 |     className="px-4 py-2 rounded-xl bg-white/10 hover:bg-white/15 text-white font-bold transition...
405 |   >
406 |     Concluir
407 |   </button>
408 | </div>
409 |
410 | </div>
411 | </div>
412 | );
413 | };
414 |
```

src/components/FreeTierInfoModal.tsx

Modal de Arquitetura 100% Gratuita (src/components/FreeTierInfoModal.tsx) (147 linhas)

```
1 | import React from 'react';
2 | import { X, CheckCircle2, ShieldCheck, Sparkles, Zap, KeyRound, ExternalLink, Cpu } from 'lucide-react';
3 |
4 | interface FreeTierInfoModalProps {
5 |   isOpen: boolean;
6 |   onClose: () => void;
7 | }
8 |
9 | export function FreeTierInfoModal({ isOpen, onClose }: FreeTierInfoModalProps) {
10 |   if (!isOpen) return null;
11 |
12 |   return (
13 |     <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-black/80 backdrop-blur-sm ...
14 |       <div
15 |         className="relative w-full max-w-xl rounded-2xl bg-[#0F111A] border border-white/10 shadow-2xl p-...
16 |         onClick={(e) => e.stopPropagation()}
17 |       >
18 |         {/* Header */}
19 |         <div className="flex items-center justify-between pb-4 border-b border-white/10 shrink-0">
20 |           <div className="flex items-center gap-2.5">
21 |             <div className="p-2 rounded-xl bg-emerald-500/10 text-emerald-400 border border-emerald-500/2...
22 |               <ShieldCheck className="w-5 h-5" />
23 |             </div>
24 |             <div>
25 |               <h3 className="text-base font-black text-white flex items-center gap-2">
26 |                 <span>API 100% Gratuita Permanente</span>
27 |                 <span className="px-2 py-0.5 rounded-full text-[10px] font-extrabold bg-emerald-500/20 te...
28 |                   R$ 0,00 / Sem Cartão
29 |                 </span>
30 |               </h3>
31 |               <p className="text-xs text-[#8E91A6]">
32 |                 0 TIKBLOX foi arquitetado para nunca cobrar mensalidades ou exigir cartões.
33 |               </p>
34 |             </div>
35 |           </div>
36 |           <button
37 |             onClick={onClose}
38 |             className="p-1.5 rounded-lg text-[#8E91A6] hover:text-white hover:bg-white/5 transition"
39 |             aria-label="Fechar"
40 |           >
41 |             <X className="w-5 h-5" />
42 |           </button>
43 |         </div>
44 |
45 |         {/* Content */}
46 |         <div className="py-4 space-y-4 overflow-y-auto pr-1 text-xs">
47 |           {/* Banner de Garantia */}
48 |           <div className="p-3.5 rounded-xl bg-emerald-950/20 border border-emerald-500/30 flex items-star...
49 |             <CheckCircle2 className="w-5 h-5 text-emerald-400 shrink-0 mt-0.5" />
50 |             <div>
51 |               <div className="font-bold text-white text-sm">Garantia de Custo Zero Perpétuo</div>
52 |               <div className="text-[#8E91A6] mt-0.5 leading-relaxed">
53 |                 Todas as integrações do TIKBLOX utilizam planos gratuitos permanentes oficiais ou process...
54 |               </div>
55 |             </div>
56 |           </div>
57 |
58 |           {/* Opções Gratuitas Disponíveis */}
59 |           <div className="space-y-2.5">
60 |             <div className="text-xs font-black uppercase tracking-wider text-[#7E8299]">
61 |               3 Modos 100% Gratuitos Suportados
62 |             </div>
```

```

63 |
64 |     {/* Opção 1: Motor Autônomo Local */}
65 |     <div className="p-3.5 rounded-xl bg-[#161823] border border-white/5 hover:border-[#25F4EE]/30...
66 |         <div className="flex items-center justify-between mb-1.5">
67 |             <div className="flex items-center gap-2 font-bold text-white">
68 |                 <CPU className="w-4 h-4 text-[#25F4EE]" />
69 |                 <span>1. Motor Autônomo Local (Sem Chave)</span>
70 |             </div>
71 |             <span className="text-[10px] font-bold text-emerald-400 bg-emerald-500/10 px-2 py-0.5 rou...
72 |                 Padrão Ativo
73 |             </span>
74 |         </div>
75 |         <p className="text-[#8E91A6] leading-relaxed">
76 |             Funciona imediatamente sem precisar criar conta nem configurar chaves. O radar calcula ma...
77 |         </p>
78 |     </div>
79 |
80 |     {/* Opção 2: Gemini 2.5 Flash Free Tier */}
81 |     <div className="p-3.5 rounded-xl bg-[#161823] border border-white/5 hover:border-[#25F4EE]/30...
82 |         <div className="flex items-center justify-between mb-1.5">
83 |             <div className="flex items-center gap-2 font-bold text-white">
84 |                 <Sparkles className="w-4 h-4 text-[#FE2C55]" />
85 |                 <span>2. Google Gemini 2.5 Flash (Free Tier)</span>
86 |             </div>
87 |             <span className="text-[10px] font-bold text-[#FE2C55] bg-[#FE2C55]/10 px-2 py-0.5 rounded...
88 |                 15 req/min Grátis
89 |             </span>
90 |         </div>
91 |         <p className="text-[#8E91A6] leading-relaxed mb-2">
92 |             Plano gratuito permanente do Google com busca web em tempo real. Não pede cartão de crédi...
93 |         </p>
94 |         <a
95 |             href="https://aistudio.google.com/app/apikey"
96 |             target="_blank"
97 |             rel="noreferrer"
98 |             className="inline-flex items-center gap-1.5 text-[#25F4EE] hover:underline font-semibold ...
99 |         >
100 |             <KeyRound className="w-3.5 h-3.5" />
101 |             Obter Chave Gratuita no Google AI Studio
102 |             <ExternalLink className="w-3 h-3" />
103 |         </a>
104 |     </div>
105 |
106 |     {/* Opção 3: Groq Cloud Llama 3.3 Free Tier */}
107 |     <div className="p-3.5 rounded-xl bg-[#161823] border border-white/5 hover:border-[#25F4EE]/30...
108 |         <div className="flex items-center justify-between mb-1.5">
109 |             <div className="flex items-center gap-2 font-bold text-white">
110 |                 <Zap className="w-4 h-4 text-amber-400" />
111 |                 <span>3. Groq Cloud (Llama 3.3 70B Free Tier)</span>
112 |             </div>
113 |             <span className="text-[10px] font-bold text-amber-400 bg-amber-500/10 px-2 py-0.5 rounded...
114 |                 Ultra Rápido
115 |             </span>
116 |         </div>
117 |         <p className="text-[#8E91A6] leading-relaxed mb-2">
118 |             Modelo open-source rodando em servidores LPUs de alta velocidade. 100% gratuito sem cartã...
119 |         </p>
120 |         <a
121 |             href="https://console.groq.com/keys"
122 |             target="_blank"
123 |             rel="noreferrer"
124 |             className="inline-flex items-center gap-1.5 text-amber-400 hover:underline font-semibold ...
125 |         >
126 |             <KeyRound className="w-3.5 h-3.5" />
127 |             Obter Chave Gratuita no Groq Console

```

```
128 |         <ExternalLink className="w-3 h-3" />
129 |       </a>
130 |     </div>
131 |   </div>
132 | </div>
133 |
134 |   {/* Footer */}
135 |   <div className="pt-3 border-t border-white/10 flex justify-end shrink-0">
136 |     <button
137 |       onClick={onClose}
138 |       className="px-4 py-2 rounded-xl text-xs font-black bg-white/10 hover:bg-white/15 text-white t...
139 |     >
140 |       Entendido
141 |     </button>
142 |   </div>
143 | </div>
144 | </div>
145 | );
146 | }
147 |
```

src/components/BrandAssetsModal.tsx

Modal de Identidade Visual TikBlox (src/components/BrandAssetsModal.tsx) (209 linhas)

```
1 | import React, { useState } from 'react';
2 | import { Download, Copy, Check, X, Layers, Type, Sparkles, ExternalLink } from 'lucide-react';
3 |
4 | interface BrandAssetsModalProps {
5 |   isOpen: boolean;
6 |   onClose: () => void;
7 | }
8 |
9 | export const BrandAssetsModal: React.FC<BrandAssetsModalProps> = ({ isOpen, onClose }) => {
10 |   const [copiedKey, setCopiedKey] = useState<string | null>(null);
11 |
12 |   if (!isOpen) return null;
13 |
14 |   const assets = [
15 |     {
16 |       id: 'symbol',
17 |       title: 'Símbolo 3D Blox (Com Container Squircle)',
18 |       description: 'Ideal para foto de perfil, ícone de app, avatar, favicon e redes sociais.',
19 |       filename: 'tikblox-symbol.svg',
20 |       path: '/tikblox-symbol.svg',
21 |       preview: '/tikblox-symbol.svg',
22 |       isSymbol: true,
23 |     },
24 |     {
25 |       id: 'symbol-transparent',
26 |       title: 'Símbolo 3D Blox (Fundo Transparente)',
27 |       description: 'Apenas os cubos 3D isométricos com efeito cromático para aplicar em qualquer layout o...',
28 |       filename: 'tikblox-symbol-transparent.svg',
29 |       path: '/tikblox-symbol-transparent.svg',
30 |       preview: '/tikblox-symbol-transparent.svg',
31 |       isSymbol: true,
32 |     },
33 |     {
34 |       id: 'typography',
35 |       title: 'Tipografia TIKBLOX (Com Fundo Escuro)',
36 |       description: 'Logotype com efeito cromático vetorial e submarca "RADAR BR • SPY VIRAL".',
37 |       filename: 'tikblox-typography.svg',
38 |       path: '/tikblox-typography.svg',
39 |       preview: '/tikblox-typography.svg',
40 |       isSymbol: false,
41 |     },
42 |     {
43 |       id: 'typography-transparent',
44 |       title: 'Tipografia TIKBLOX (Fundo Transparente)',
45 |       description: 'Tipografia vetorial isolada em fundo transparente pronta para vídeos, banners e apres...',
46 |       filename: 'tikblox-typography-transparent.svg',
47 |       path: '/tikblox-typography-transparent.svg',
48 |       preview: '/tikblox-typography-transparent.svg',
49 |       isSymbol: false,
50 |     },
51 |     {
52 |       id: 'full-logo',
53 |       title: 'Logo Completo (Símbolo + Tipografia)',
54 |       description: 'Lockup horizontal completo com todos os elementos integrados em alta resolução.',
55 |       filename: 'tikblox-full-logo.svg',
56 |       path: '/tikblox-full-logo.svg',
57 |       preview: '/tikblox-full-logo.svg',
58 |       isSymbol: false,
59 |     },
60 |   ];
61 |
62 |   const handleDownload = (path: string, filename: string) => {
```

src/components/BrandAssetsModal.tsx (continuação)

```
63 |     const link = document.createElement('a');
64 |     link.href = path;
65 |     link.download = filename;
66 |     document.body.appendChild(link);
67 |     link.click();
68 |     document.body.removeChild(link);
69 | };
70 |
71 | const handleCopyLink = async (path: string, id: string) => {
72 |   try {
73 |     const url = `${window.location.origin}${path}`;
74 |     await navigator.clipboard.writeText(url);
75 |     setCopiedKey(id);
76 |     setTimeout(() => setCopiedKey(null), 2500);
77 |   } catch {
78 |     // Fallback
79 |     setCopiedKey(id);
80 |     setTimeout(() => setCopiedKey(null), 2500);
81 |   }
82 | };
83 |
84 | return (
85 |   <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-black/80 backdrop-blur-md ...
86 |     <div className="relative w-full max-w-3xl max-h-[90vh] overflow-y-auto rounded-3xl border border-wh...
87 |
88 |       {/* Close Button */}
89 |       <button
90 |         onClick={onClose}
91 |         className="absolute top-5 right-5 p-2 rounded-full bg-white/5 hover:bg-white/10 text-white/70 h...
92 |         title="Fechar"
93 |       >
94 |         <X className="w-5 h-5" />
95 |       </button>
96 |
97 |       {/* Modal Header */}
98 |       <div className="flex items-center gap-3 mb-2">
99 |         <div className="flex items-center justify-center w-10 h-10 rounded-xl bg-gradient-to-tr from-[#...
100 |           <Sparkles className="w-5 h-5 text-[#25F4EE]" />
101 |         </div>
102 |         <div>
103 |           <h2 className="text-xl sm:text-2xl font-black tracking-tight">Kit de Marca TIKBLOX</h2>
104 |           <p className="text-xs sm:text-sm text-[#8E91A6]">
105 |             Arquivos vetoriais em SVG de alta resolução separados para download direto.
106 |           </p>
107 |         </div>
108 |       </div>
109 |
110 |       {/* Assets Grid */}
111 |       <div className="mt-6 space-y-5">
112 |         {assets.map((asset) => (
113 |           <div
114 |             key={asset.id}
115 |             className="flex flex-col sm:flex-row items-center gap-5 p-4 sm:p-5 rounded-2xl border borde...
116 |           >
117 |             {/* Asset Preview Thumbnail */}
118 |             <div className="relative flex items-center justify-center w-full sm:w-48 h-32 rounded-xl bg...
119 |               {/* Checkerboard background for transparent items */}
120 |               {asset.id.includes('transparent') && (
121 |                 <div
122 |                   className="absolute inset-0 opacity-15 pointer-events-none"
123 |                   style={{
124 |                     backgroundImage:
125 |                       'linear-gradient(45deg, #333 25%, transparent 25%), linear-gradient(-45deg, #333 ...
126 |                     backgroundImageSize: '16px 16px',
127 |                     backgroundPosition: '0 0, 0 8px, 8px -8px, -8px -8px, -8px 0px',
```

```

128 |         }}
129 |       />
130 |     )}
131 |     <img
132 |       src={asset.preview}
133 |       alt={asset.title}
134 |       className="max-h-24 max-w-full object-contain relative z-10 drop-shadow-md"
135 |     />
136 |   </div>
137 |
138 |   {/* Asset Info & Download Action */}
139 |   <div className="flex-1 w-full text-left">
140 |     <div className="flex items-center gap-2 mb-1">
141 |       {asset.isSymbol ? (
142 |         <Layers className="w-4 h-4 text-[#25F4EE]" />
143 |       ) : (
144 |         <Type className="w-4 h-4 text-[#FE2C55]" />
145 |       )}
146 |       <h3 className="font-bold text-sm sm:text-base text-white">{asset.title}</h3>
147 |     </div>
148 |     <p className="text-xs text-[#8E91A6] mb-3 leading-relaxed">
149 |       {asset.description}
150 |     </p>
151 |
152 |     {/* Buttons Row */}
153 |     <div className="flex flex-wrap items-center gap-2">
154 |       <button
155 |         onClick={() => handleDownload(asset.path, asset.filename)}
156 |         className="flex items-center gap-1.5 px-3.5 py-1.5 rounded-lg bg-gradient-to-r from-[...
157 |       >
158 |         <Download className="w-3.5 h-3.5" />
159 |         <span>Baixar SVG</span>
160 |       </button>
161 |
162 |       <a
163 |         href={asset.path}
164 |         target="_blank"
165 |         rel="noopener noreferrer"
166 |         className="flex items-center gap-1.5 px-3 py-1.5 rounded-lg bg-white/5 hover:bg-white...
167 |       >
168 |         <ExternalLink className="w-3.5 h-3.5" />
169 |         <span>Abrir</span>
170 |       </a>
171 |
172 |       <button
173 |         onClick={() => handleCopyLink(asset.path, asset.id)}
174 |         className="flex items-center gap-1.5 px-3 py-1.5 rounded-lg bg-white/5 hover:bg-white...
175 |       >
176 |         {copiedKey === asset.id ? (
177 |           <>
178 |             <Check className="w-3.5 h-3.5 text-emerald-400" />
179 |             <span className="text-emerald-400">URL Copiada!</span>
180 |           </>
181 |         ) : (
182 |           <>
183 |             <Copy className="w-3.5 h-3.5" />
184 |             <span>Copiar Link</span>
185 |           </>
186 |         )}
187 |       </button>
188 |     </div>
189 |   </div>
190 | </div>
191 |   )}}
192 | </div>

```

```
193 |  
194 |     {/* Modal Footer Note */}  
195 |     <div className="mt-6 pt-4 border-t border-white/10 flex flex-col sm:flex-row items-center justify...  
196 |         <span>Formatos: Vetor SVG escalável sem perda de qualidade (100% editável no Figma, Illustrator...  
197 |         <button  
198 |             onClick={onClose}  
199 |             className="px-4 py-1.5 rounded-lg bg-white/10 hover:bg-white/15 text-white font-bold transiti...  
200 |         >  
201 |             Concluir  
202 |         </button>  
203 |     </div>  
204 |  
205 | </div>  
206 | </div>  
207 | );  
208 | };  
209 |
```

src/components/PWAInstallButton.tsx

Botão de Instalar App (src/components/PWAInstallButton.tsx) (113 linhas)

```
1 | import React, { useState } from 'react';
2 | import { Download, Share, PlusSquare, X, CheckCircle2 } from 'lucide-react';
3 | import { usePWAInstall } from '../hooks/usePWAInstall';
4 |
5 | export const PWAInstallButton: React.FC = () => {
6 |   const { isInstallable, isInstalled, isIOS, install } = usePWAInstall();
7 |   const [showIOSGuide, setShowIOSGuide] = useState(false);
8 |
9 |   // If already running as an installed PWA, show a subtle "Instalado" badge or null
10 |   if (isInstalled) {
11 |     return (
12 |       <div className="hidden sm:flex items-center gap-1.5 px-3 py-1 rounded-full bg-emerald-500/10 border...
13 |         <CheckCircle2 className="w-3.5 h-3.5" />
14 |         <span>TIKBLOX Instalado</span>
15 |       </div>
16 |     );
17 |   }
18 |
19 |   // Chromium / Android / Desktop flow
20 |   if (isInstallable) {
21 |     return (
22 |       <button
23 |         id="pwa-install-btn"
24 |         onClick={install}
25 |         className="flex items-center gap-2 rounded-xl bg-gradient-to-r from-cyan-500 to-blue-600 px-3.5 p...
26 |         title="Instalar TIKBLOX no dispositivo"
27 |       >
28 |         <Download className="w-4 h-4" />
29 |         <span>Instalar App</span>
30 |       </button>
31 |     );
32 |   }
33 |
34 |   // iOS Safari flow
35 |   if (isIOS) {
36 |     return (
37 |       <>
38 |         <button
39 |           id="pwa-ios-install-btn"
40 |           onClick={() => setShowIOSGuide(true)}
41 |           className="flex items-center gap-2 rounded-xl bg-slate-800 border border-slate-700 hover:border...
42 |         >
43 |           <Download className="w-3.5 h-3.5 text-cyan-400" />
44 |           <span>Instalar no iPhone</span>
45 |         </button>
46 |
47 |         {showIOSGuide && (
48 |           <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/70 backdrop-blur-s...
49 |             <div className="w-full max-w-sm rounded-2xl bg-slate-900 border border-slate-800 p-6 shadow-2...
50 |               <button
51 |                 onClick={() => setShowIOSGuide(false)}
52 |                 className="absolute top-4 right-4 text-slate-400 hover:text-white p-1 rounded-lg hover:bg...
53 |               >
54 |                 <X className="w-5 h-5" />
55 |               </button>
56 |
57 |               <div className="flex items-center gap-3 mb-4">
58 |                 <div className="w-10 h-10 rounded-xl bg-cyan-500/20 flex items-center justify-center bord...
59 |                   <Download className="w-5 h-5 text-cyan-400" />
60 |                 </div>
61 |                 <div>
62 |                   <h3 className="text-base font-bold text-white">Instalar TIKBLOX no iOS</h3>
```

```

63 |         <p className="text-xs text-slate-400">Tenha o radar na sua tela de início</p>
64 |     </div>
65 | </div>
66 |
67 |     <div className="space-y-3 text-xs text-slate-300">
68 |         <div className="flex items-start gap-3 p-3 rounded-xl bg-slate-800/60 border border-slate...
69 |             <Share className="w-5 h-5 text-cyan-400 shrink-0 mt-0.5" />
70 |             <div>
71 |                 <strong className="text-white block font-medium">1. Toque no botão Compartilhar</stro...
72 |                 <span>No rodapé do seu navegador Safari do iPhone.</span>
73 |             </div>
74 |         </div>
75 |
76 |         <div className="flex items-start gap-3 p-3 rounded-xl bg-slate-800/60 border border-slate...
77 |             <PlusSquare className="w-5 h-5 text-cyan-400 shrink-0 mt-0.5" />
78 |             <div>
79 |                 <strong className="text-white block font-medium">2. Adicionar à Tela de Início</stron...
80 |                 <span>Role a lista para baixo e selecione esta opção.</span>
81 |             </div>
82 |         </div>
83 |     </div>
84 |
85 |     <button
86 |         onClick={() => setShowIOSGuide(false)}
87 |         className="mt-5 w-full rounded-xl bg-cyan-500 hover:bg-cyan-400 text-slate-950 font-semib...
88 |     >
89 |         Entendi
90 |     </button>
91 | </div>
92 | </div>
93 |     )}
94 | </>
95 | );
96 | }
97 |
98 | // Generic install helper button if beforeinstallprompt is not currently ready
99 | return (
100 |     <button
101 |         id="pwa-install-generic-btn"
102 |         onClick={() => {
103 |             alert('Para instalar o TIKBLOX como PWA: abra o menu do navegador (três pontinhos no topo) e cliq...
104 |         }}
105 |         className="flex items-center gap-1.5 rounded-xl bg-slate-800/80 hover:bg-slate-800 border border-sl...
106 |         title="Instalar como aplicativo PWA"
107 |     >
108 |         <Download className="w-3.5 h-3.5 text-cyan-400" />
109 |         <span className="hidden sm:inline">PWA App</span>
110 |     </button>
111 | );
112 | };
113 |

```

src/components/OfflineIndicator.tsx

Banner de Status de Conexão (src/components/OfflineIndicator.tsx) (20 linhas)

```
1 | import React from 'react';
2 | import { WifiOff } from 'lucide-react';
3 | import { useOnlineStatus } from '../hooks/useOnlineStatus';
4 |
5 | export const OfflineIndicator: React.FC = () => {
6 |   const isOnline = useOnlineStatus();
7 |
8 |   if (isOnline) return null;
9 |
10 |   return (
11 |     <div
12 |       id="offline-banner"
13 |       className="fixed bottom-4 left-4 z-50 flex items-center gap-2.5 rounded-xl bg-amber-500/90 backdrop...
14 |     >
15 |       <WifiOff className="w-4 h-4 text-slate-950" />
16 |       <span>Modo Offline – O radar TIKBLOX está usando dados salvos no cache local.</span>
17 |     </div>
18 |   );
19 | };
20 |
```

src/components/TikBloxLogo.tsx

Logomarca e Símbolo TikBlox (src/components/TikBloxLogo.tsx) (302 linhas)

```
1 | import React from 'react';
2 |
3 | interface TikBloxLogoProps {
4 |   size?: 'sm' | 'md' | 'lg' | 'xl';
5 |   showText?: boolean;
6 |   showBadge?: boolean;
7 |   className?: string;
8 |   animate?: boolean;
9 | }
10 |
11 | export const TikBloxLogo: React.FC<TikBloxLogoProps> = ({
12 |   size = 'md',
13 |   showText = true,
14 |   showBadge = true,
15 |   className = '',
16 |   animate = false,
17 | }) => {
18 |   // Dimensions and scaling based on size
19 |   const config = {
20 |     sm: {
21 |       box: 'h-9 w-9 rounded-xl',
22 |       svg: 24,
23 |       title: 'text-base',
24 |       badge: 'text-[8px] px-1.5 py-0.5',
25 |       sub: 'text-[8px]',
26 |     },
27 |     md: {
28 |       box: 'h-11 w-11 rounded-[14px]',
29 |       svg: 28,
30 |       title: 'text-xl',
31 |       badge: 'text-[9px] px-2 py-0.5',
32 |       sub: 'text-[9px]',
33 |     },
34 |     lg: {
35 |       box: 'h-14 w-14 rounded-2xl',
36 |       svg: 36,
37 |       title: 'text-2xl',
38 |       badge: 'text-[10px] px-2.5 py-0.5',
39 |       sub: 'text-[10px]',
40 |     },
41 |     xl: {
42 |       box: 'h-18 w-18 rounded-3xl',
43 |       svg: 48,
44 |       title: 'text-3xl',
45 |       badge: 'text-xs px-3 py-1',
46 |       sub: 'text-xs',
47 |     },
48 |   }[size];
49 |
50 |   return (
51 |     <div className={`flex items-center gap-3 select-none ${className}`}>
52 |       /* 3D Isometric Blox Emblem Container */
53 |       <div
54 |         className={`relative flex items-center justify-center shrink-0 ${config.box} bg-gradient-to-br fr...
55 |       >
56 |         /* Soft atmospheric chromatic glow */
57 |         <div className="absolute -top-3 -left-3 w-10 h-10 rounded-full bg-[#25F4EE]/25 blur-md pointer-ev...
58 |         <div className="absolute -bottom-3 -right-3 w-10 h-10 rounded-full bg-[#FE2C55]/25 blur-md pointe...
59 |
60 |         /* Diagonal metallic glass sheen */
61 |         <div className="absolute inset-0 bg-gradient-to-tr from-transparent via-white/8 to-transparent po...
62 |
```

```

63 |     {/
64 |         CLEAN 3D ISOMETRIC BLOX (QUADRADINHOS 3D)
65 |         - No lightning bolt in the middle: clean, pure geometric 3D cubes
66 |         - Apex Cube (Top): Gleaming Frost White with dual chromatic reflections
67 |         - Left Cube: Electric Cyan (#25F4EE)
68 |         - Right Cube: Neon Pink (#FE2C55)
69 |         - Signature TikTok chromatic aberration offset
70 |     */
71 |     <svg
72 |         width={config.svg}
73 |         height={config.svg}
74 |         viewBox="0 0 60 60"
75 |         fill="none"
76 |         xmlns="http://www.w3.org/2000/svg"
77 |         className={`relative z-10 ${animate ? 'group-hover:scale-110' : ''} transition-transform durati...
78 |     >
79 |         <defs>
80 |             {/
81 |                 <linearGradient id="topRoofGrad" x1="0%" y1="0%" x2="100%" y2="100%">
82 |                     <stop offset="0%" stopColor="#FFFFFF" />
83 |                     <stop offset="50%" stopColor="#F0FDFA" />
84 |                     <stop offset="100%" stopColor="#CCFBF1" />
85 |                 </linearGradient>
86 |
87 |                 {/
88 |                     <linearGradient id="leftRoofGrad" x1="0%" y1="0%" x2="100%" y2="100%">
89 |                         <stop offset="0%" stopColor="#E6FFFE" />
90 |                         <stop offset="100%" stopColor="#7EF7F3" />
91 |                     </linearGradient>
92 |
93 |                     {/
94 |                         <linearGradient id="rightRoofGrad" x1="0%" y1="0%" x2="100%" y2="100%">
95 |                             <stop offset="0%" stopColor="#FFF1F3" />
96 |                             <stop offset="100%" stopColor="#FFA4B6" />
97 |                         </linearGradient>
98 |
99 |                         {/
100 |                             <linearGradient id="cyanLightFace" x1="0%" y1="0%" x2="100%" y2="100%">
101 |                                 <stop offset="0%" stopColor="#25F4EE" />
102 |                                 <stop offset="100%" stopColor="#08B6BC" />
103 |                             </linearGradient>
104 |                             <linearGradient id="cyanDarkFace" x1="0%" y1="0%" x2="100%" y2="100%">
105 |                                 <stop offset="0%" stopColor="#038085" />
106 |                                 <stop offset="100%" stopColor="#014A4E" />
107 |                             </linearGradient>
108 |
109 |                             {/
110 |                                 <linearGradient id="pinkLightFace" x1="0%" y1="0%" x2="100%" y2="100%">
111 |                                     <stop offset="0%" stopColor="#FF385C" />
112 |                                     <stop offset="100%" stopColor="#FE2C55" />
113 |                                 </linearGradient>
114 |                                 <linearGradient id="pinkDarkFace" x1="0%" y1="0%" x2="100%" y2="100%">
115 |                                     <stop offset="0%" stopColor="#B30E30" />
116 |                                     <stop offset="100%" stopColor="#66051A" />
117 |                                 </linearGradient>
118 |                             </defs>
119 |
120 |                             {/
121 |                                 {/ LAYER 1: CHROMATIC OFFSET (CYAN #25F4EE, SHIFTED TOP-LEFT)
122 |                                 {/
123 |                                 <g transform="translate(-2, -1.5)" fill="#25F4EE" opacity="0.8">
124 |                                     {/
125 |                                         <polygon points="30,6 41.5,12.5 41.5,25.5 30,32 18.5,25.5 18.5,12.5" />
126 |                                         {/
127 |                                         <polygon points="18.5,25 29.5,31.2 29.5,43.7 18.5,50 7.5,43.7 7.5,31.2" />

```

```

128 |         {/* Right Cube Silhouette */}
129 |         <polygon points="41.5,25 52.5,31.2 52.5,43.7 41.5,50 30.5,43.7 30.5,31.2" />
130 |     </g>
131 |
132 |     {/* ===== */}
133 |     {/* LAYER 2: CHROMATIC OFFSET (PINK #FE2C55, SHIFTED BOTTOM-RIGHT) */}
134 |     {/* ===== */}
135 |     <g transform="translate(2, 1.5)" fill="#FE2C55" opacity="0.8">
136 |         {/* Top Cube Silhouette */}
137 |         <polygon points="30,6 41.5,12.5 41.5,25.5 30,32 18.5,25.5 18.5,12.5" />
138 |         {/* Left Cube Silhouette */}
139 |         <polygon points="18.5,25 29.5,31.2 29.5,43.7 18.5,50 7.5,43.7 7.5,31.2" />
140 |         {/* Right Cube Silhouette */}
141 |         <polygon points="41.5,25 52.5,31.2 52.5,43.7 41.5,50 30.5,43.7 30.5,31.2" />
142 |     </g>
143 |
144 |     {/* ===== */}
145 |     {/* LAYER 3: CRISP 3D ISOMETRIC CUBES (CENTERED & BEAUTIFUL) */}
146 |     {/* ===== */}
147 |     <g transform="translate(0, 0)">
148 |
149 |         {/* --- CUBE 1: TOP APEX BLOX --- */}
150 |         {/* Top Face */}
151 |         <polygon
152 |             points="30,6 41.5,12.5 30,19 18.5,12.5"
153 |             fill="url(#topRoofGrad)"
154 |             stroke="FFFFFF"
155 |             strokeWidth="0.6"
156 |         />
157 |         {/* Left Face (Cyan illuminated) */}
158 |         <polygon
159 |             points="18.5,12.5 30,19 30,32 18.5,25.5"
160 |             fill="url(#cyanLightFace)"
161 |         />
162 |         {/* Right Face (Pink illuminated) */}
163 |         <polygon
164 |             points="30,19 41.5,12.5 41.5,25.5 30,32"
165 |             fill="url(#pinkLightFace)"
166 |         />
167 |         {/* Bevel highlight */}
168 |         <line x1="30" y1="19" x2="30" y2="32" stroke="FFFFFF" strokeWidth="0.6" opacity="0.7" />
169 |
170 |         {/* --- CUBE 2: LEFT BLOX (ELECTRIC CYAN) --- */}
171 |         {/* Top Face */}
172 |         <polygon
173 |             points="18.5,25 29.5,31.2 18.5,37.5 7.5,31.2"
174 |             fill="url(#leftRoofGrad)"
175 |             stroke="FFFFFF"
176 |             strokeWidth="0.5"
177 |         />
178 |         {/* Left Face (Vibrant Cyan) */}
179 |         <polygon
180 |             points="7.5,31.2 18.5,37.5 18.5,50 7.5,43.7"
181 |             fill="url(#cyanLightFace)"
182 |         />
183 |         {/* Right Face (Shadowed Teal) */}
184 |         <polygon
185 |             points="18.5,37.5 29.5,31.2 29.5,43.7 18.5,50"
186 |             fill="url(#cyanDarkFace)"
187 |         />
188 |         {/* Edge line */}
189 |         <line x1="18.5" y1="37.5" x2="18.5" y2="50" stroke="#25F4EE" strokeWidth="0.5" opacity="0.8" ...
190 |
191 |         {/* --- CUBE 3: RIGHT BLOX (NEON PINK) --- */}
192 |         {/* Top Face */}

```

```

193 |         <polygon
194 |           points="41.5,25 52.5,31.2 41.5,37.5 30.5,31.2"
195 |           fill="url(#rightRoofGrad)"
196 |           stroke="FFFFFF"
197 |           strokeWidth="0.5"
198 |         />
199 |         {/ * Left Face (Shadowed Ruby) */}
200 |         <polygon
201 |           points="30.5,31.2 41.5,37.5 41.5,50 30.5,43.7"
202 |           fill="url(#pinkDarkFace)"
203 |         />
204 |         {/ * Right Face (Radiant Pink) */}
205 |         <polygon
206 |           points="41.5,37.5 52.5,31.2 52.5,43.7 41.5,50"
207 |           fill="url(#pinkLightFace)"
208 |         />
209 |         {/ * Edge line */}
210 |         <line x1="41.5" y1="37.5" x2="41.5" y2="50" stroke="#FE2C55" strokeWidth="0.5" opacity="0.8" ...
211 |
212 |         {/ * Clean Floating Mini-Blox / Gem in Bottom-Center gap (Adds high-end tech craft) */}
213 |         <g transform="translate(0, 0)">
214 |           <polygon
215 |             points="30,35.5 35,38.3 30,41.2 25,38.3"
216 |             fill="FFFFFF"
217 |             stroke="FFFFFF"
218 |             strokeWidth="0.4"
219 |           />
220 |           <polygon
221 |             points="25,38.3 30,41.2 30,46.5 25,43.6"
222 |             fill="#25F4EE"
223 |           />
224 |           <polygon
225 |             points="30,41.2 35,38.3 35,43.6 30,46.5"
226 |             fill="#FE2C55"
227 |           />
228 |         </g>
229 |       </svg>
230 |     </div>
231 |
232 |     {/ * Brand Typography & Status Lockup */}
233 |     {showText && (
234 |       <div className="flex flex-col justify-center">
235 |         <div className="flex items-center gap-2">
236 |
237 |           {/ * Custom Logotype: TIK in Architecturally Crisp White, BLOX with Vector Glitch & Neon Aura ...
238 |           <div className="relative flex items-center font-black tracking-[-0.03em] leading-none select-...
239 |
240 |             {/ * TIK: Pure White Display Letterforms */}
241 |             <span className={`font-black text-white ${config.title} drop-shadow-[0_2px_12px_rgba(255,25...
242 |               TIK
243 |             </span>
244 |
245 |             {/ * BLOX: Real Layered Chromatic Glitch Vector Display */}
246 |             <span className={`relative font-black ${config.title} ml-0.5`}>
247 |               {/ * Cyan Shift Layer (-1.8px, -1.2px) */}
248 |               <span
249 |                 aria-hidden="true"
250 |                 className="absolute -left-[1.8px] -top-[1.2px] text-[#25F4EE] opacity-90 select-none po...
251 |               >
252 |                 BLOX
253 |             </span>
254 |
255 |             {/ * Pink Shift Layer (+1.8px, +1.2px) */}
256 |             <span
257 |

```

```

258 |         aria-hidden="true"
259 |         className="absolute left-[1.8px] top-[1.2px] text-[#FE2C55] opacity-90 select-none poin...
260 |     >
261 |         BLOX
262 |     </span>
263 |
264 |     {/* Foreground Crisp White Lettering */}
265 |     <span className="relative z-10 text-white drop-shadow-[0_2px_8px_rgba(0,0,0,0.8)]">
266 |         BLOX
267 |     </span>
268 | </span>
269 | </div>
270 |
271 | {/* Futuristic Live Status Badge: RADAR BR */}
272 | {showBadge && (
273 |     <div
274 |         className={`flex items-center gap-1.5 rounded-full bg-[#12131A] border border-[#25F4EE]/3...
275 |     >
276 |         <span className="relative flex h-1.5 w-1.5">
277 |             <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-[#FE2C...
278 |             <span className="relative inline-flex rounded-full h-1.5 w-1.5 bg-[#25F4EE]" />
279 |         </span>
280 |         <span className="font-extrabold text-[9px] tracking-wider text-white">
281 |             RADAR <span className="text-[#25F4EE]">BR</span>
282 |         </span>
283 |     </div>
284 | )}
285 | </div>
286 |
287 | {/* Elevated Brand Subtitle / Tagline */}
288 | <div className="flex items-center gap-1.5 mt-0.5">
289 |     <span className={`font-black tracking-[0.2em] uppercase text-[#8B8D9E] group-ho...
290 |         <span className="text-[#25F4EE]">SPY VIRAL</span>
291 |         <span className="text-white/20 text-[7px]">•</span>
292 |         <span className="text-[#C5C6D2]">ARBITRAGEM</span>
293 |         <span className="hidden sm:inline text-white/20 text-[7px]">•</span>
294 |         <span className="hidden sm:inline text-[#FE2C55]">DROPSHIPPING</span>
295 |     </span>
296 | </div>
297 | </div>
298 |     )}
299 | </div>
300 | );
301 | };
302 |

```

src/data/curatedTrends.ts

Base Curada de Produtos Internacionais (src/data/curatedTrends.ts) (8 linhas)

```
1 | import { TrendingProduct } from '../types';
2 | import { ALL_NICHE_TRENDS, NICHE_TRENDS_MAP } from './niches';
3 |
4 | export { ALL_NICHE_TRENDS, NICHE_TRENDS_MAP };
5 |
6 | // Comprehensive 300 trends (30 per niche across all 10 niches: Tech, Home, Beauty, Fitness, Pets, Access...
7 | export const INITIAL_CURATED_TRENDS: TrendingProduct[] = ALL_NICHE_TRENDS;
8 |
```

src/data/niches/index.ts

Índice de Nichos e Categorias (src/data/niches/index.ts) (60 linhas)

```
1 | import { TrendingProduct, ProductNiche } from '../types';
2 | import { techTrends } from './tech';
3 | import { homeTrends } from './home';
4 | import { beautyTrends } from './beauty';
5 | import { fitnessTrends } from './fitness';
6 | import { petTrends } from './pets';
7 | import { accessoriesTrends } from './accessories';
8 | import { kidsTrends } from './kids';
9 | import { toolsTrends } from './tools';
10 | import { autoTrends } from './auto';
11 | import { healthTrends } from './health';
12 |
13 | export {
14 |   techTrends,
15 |   homeTrends,
16 |   beautyTrends,
17 |   fitnessTrends,
18 |   petTrends,
19 |   accessoriesTrends,
20 |   kidsTrends,
21 |   toolsTrends,
22 |   autoTrends,
23 |   healthTrends
24 | };
25 |
26 | export const NICHE_TRENDS_MAP: Record<Exclude<ProductNiche, 'all'>, TrendingProduct[]> = {
27 |   tech: techTrends,
28 |   home: homeTrends,
29 |   beauty: beautyTrends,
30 |   fitness: fitnessTrends,
31 |   pets: petTrends,
32 |   accessories: accessoriesTrends,
33 |   kids: kidsTrends,
34 |   tools: toolsTrends,
35 |   auto: autoTrends,
36 |   health: healthTrends,
37 | };
38 |
39 | // All 300 curated trends across all 10 niches (30 items per niche)
40 | export const ALL_NICHE_TRENDS: TrendingProduct[] = [
41 |   ...techTrends,
42 |   ...homeTrends,
43 |   ...beautyTrends,
44 |   ...fitnessTrends,
45 |   ...petTrends,
46 |   ...accessoriesTrends,
47 |   ...kidsTrends,
48 |   ...toolsTrends,
49 |   ...autoTrends,
50 |   ...healthTrends,
51 | ];
52 |
53 | export function getNicheProducts(niche?: string): TrendingProduct[] {
54 |   if (!niche || niche === 'all') {
55 |     return ALL_NICHE_TRENDS;
56 |   }
57 |   const key = niche as Exclude<ProductNiche, 'all'>;
58 |   return NICHE_TRENDS_MAP[key] || ALL_NICHE_TRENDS;
59 | }
60 |
```


